Sharp Query Thresholds for Adaptivity under Complete Equivalence Promises

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Abstract

We study zero-error classification between complete equivalence-partition hypotheses using priced pair-equality queries. Each hypothesis retains every partition satisfying its equalities and inequalities, with unrestricted class count. Adaptive policies are compared with one global query order permitting causal permanent skips; both buy the same information and pay for their actual purchased set. For deterministic worst-input cost, a computer-assisted proof gives equality on all complete sources with at most six queries, and a positive-additive separation at seven. Under private randomization the exact minimum is four even for coverage and monotone submodular fees, with strictly increasing witnesses and explicit robust neighborhoods. On every complete source with at most four queries, additive charges plus at most one pairwise shared setup give equality. Over complete four-query sources, two positive setups are necessary and sufficient for a strict gap, also with strictly positive individual charges. If the quadratic pair-coverage fee must satisfy gross substitutes (GS), the exact positive-pair minimum is instead three, with a full-class priority lower bound and a relative robust neighborhood. For additive prices, randomized equality holds on every complete source with at most four queries and every nonnegative real price vector, leaving the first additive separation between five and seven queries. On the fixed seven-query source, the sharp deterministic ratios are **5/4** for additive and **4/3** for monotone submodular fees; unrestricted monotone ratios are unbounded. Exact finite certificates and analytic arguments support the stated quantifiers. Practical academic-evaluation gains and exact literature priority remain unestablished.

Keywords: query complexity, adaptivity, ordered decision diagrams, equivalence constraints, submodular costs

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1 Introduction

Suppose that observations are equality tests between records, and that the unknown state is an equivalence partition. Two hypotheses specify which records must agree and which must differ. The task is to identify the hypothesis while buying as little information as possible. An adaptive policy can change its next query after an answer. A priority policy fixes one global order in advance, but may permanently skip queries using the answers already observed. Both policies know the same hypotheses and fees. We ask when adaptivity first improves their optimal worst-input costs.

The hypotheses in this paper are *complete*: each contains every partition satisfying its stated SAME and DIFFERENT constraints. Completeness restricts the permitted answer tables; it does not mean that the available queries reveal the whole partition. For example, with queries $(a, b), (b, c), (a, c)$, the hypothesis $a = b$ admits exactly the words $(1, 0, 0)$ and $(1, 1, 1)$. The alternative hypothesis $b = c$, $a \neq b$ admits $(0, 1, 0)$. Unlisted cross-block merges remain possible. This small illustration explains the input contract, rather than giving an adaptivity separation. Arbitrarily retaining selected compatible worlds would define a different promise class.

The model isolates classification between relational hypotheses, rather than reconstruction of an entire partition. It can describe a controlled identity-checking task in which the constraints and query answers have precise meanings. Applying it to academic evidence would additionally require validating those meanings and the completeness assumption. Our results do not establish a natural evidence-acquisition advantage or authenticate a historical scientific execution.

1.1 The boundaries depend on randomness and fees

Write $D_{\mathcal{C}}$ for optimal deterministic worst-input cost and $R_{\mathcal{C}}$ for optimal worst-input expected cost with input-independent private randomness, where $\mathcal{C} \in \{\mathcal{A}, \mathcal{P}\}$. The fee is charged on the actual purchased set. It is finite, nonnegative, normalized at the empty set, and monotone unless a narrower class is stated. Table 1 distinguishes the source and fee quantifiers.

The first contribution is the deterministic six/seven boundary (Theorem 3). Completeness is essential: a four-query Horn promise outside this source class already separates the deterministic policies. The proof combines a complete source cover, analytic reductions and exact finite certificates, rather than extrapolation from sampled fees.

The second contribution separates private randomness from the fee class. The minimum drops to four even with positive query charges and pairwise shared setups (Theorem 26); the witness is strictly increasing and strictly submodular on incomparable sets. The general three-query lower bound is Theorem 5. At four, however, additive prices force equality on every complete source, including zero prices and ties

Policies	Fees and source scope	Boundary conclusion
Deterministic	Complete sources; monotone fees	Equality for every menu of size at most six; a positive-additive gap at seven.
Private random	Arbitrary partial tasks; monotone fees	Equality at sizes at most three.
Private random	Complete sources; monotone, submodular or coverage fees	Minimum four, already with positive individual charges and pairwise shared setups.
Private random	Complete sources; nonnegative additive prices	Equality at sizes at most four for every real price vector; the minimum size permitting a gap lies between five and seven.
Private random	Complete four-query sources; individual charges and pairwise shared setups	Minimum positive pair support two; at most one pair gives equality on every complete source, also at smaller menus.
Private random	Complete four-query sources; quadratic pair-coverage fees satisfying GS	Minimum positive pair support three; at most two pairs give equality on every complete source, also at smaller menus.

Table 1 Menu-size and pair-resource boundaries. Equality quantifies over every complete source and fee in the stated class, except the arbitrary-partial-task row; upper witnesses are existential. The last two rows count positive pair weights over complete four-query sources, with and without the GS restriction. The exact randomized additive menu minimum remains open.

(Theorem 7). This all-price exclusion uses full original adaptive and causal-skip priority families. The complete seven-query positive-additive witness gives the five-to-seven interval in Corollary 8; its exact additive threshold remains open. Theorem 30 further gives an all-source cost-structure boundary: at most one pairwise shared setup never separates the values on a complete source of size at most four. Two positive pair setups suffice at four, even with positive individual charges. This strengthens the sourcewise certificate in Theorem 28. With a GS restriction on the quadratic pair-coverage fee, the exact minimum is three instead (Theorem 31). The lower excludes all at-most-two-pair GS fees on every complete four-query source; a complete seven-world example gives the upper. Its priority value is certified directly against all global orders and causal continuations, with a relative GS neighborhood in Corollary 32. The adaptive value in that example is only upper-bounded.

A separate fixed-source analysis determines the sharp deterministic ratios on the seven-query witness: 5/4 for additive prices, 4/3 for monotone submodular fees, and unbounded ratios for unrestricted monotone fees. These ratio claims concern that source, not every menu of size seven. The specified eleven-query extension, record-acquisition comparisons, completion bounds and execution-evidence cases are supporting results with separate assumptions and evidence.

1.2 Closest prior comparisons

Priced information and certificate-competitive evaluation are established [1, 2]. Ordered diagrams permit omitted variables, and Greiner et al.'s linear strategies allow causal irrelevant-test skips [3]. Neither skipping, private worst-input randomization nor generic adaptivity separation is proposed as a new concept here.

The total read-once AND–OR comparison is especially relevant. Amano [4] studies general and recursive directional randomized evaluation. Ito and Suzuki [5] prove that depth-first and directional randomized values coincide for every such total tree at unit query costs. Directional seeds are legal priority policies, giving $R_{\mathcal{A}} \leq R_{\mathcal{P}} \leq R_{\mathrm{dir}} = R_{\mathrm{DF}}$ on that full-domain problem. Thus their upper bounds can transfer to our priority class; their depth-first lower bounds alone need not. The two larger policy sets are not nested.

Transferring that equality sharply to a relational promise still requires preserving its optimum and fees. On one complete four-query source at unit prices, Appendix U proves promise value $5/2$ but minimum full-cube value 3 over *all* compatible total functions (Proposition 25). For one compatible AND–OR completion, the hard-child product places mass $3/16$ outside the promise. These fixed-source, unit-cost obstructions leave other reductions open.

Specified-distribution objectives require a further quantifier comparison. Stochastic Boolean evaluation [6], the tight k-of-n stochastic gap [7], and correlated function certification [8] supply important prior context. A fixed-prior separation alone does not establish a gap between two separately minimized worst-input values, although a valid fixed-prior lower bound against the full priority class remains a valid minimax lower bound. Completion and representation-size work also has established methods whose transfer needs the correct cost and comparator. Appendix Q and Appendix N.4 give the detailed comparisons. Coverage and shared-machine costs are also established in batch testing [9]. That independent series-diagnosis setting charges the sum of performed batch costs in expectation. Our correlated complete promise, worst-input private objective and final-union fee require a separate source-, cost- and comparator-preserving argument. Within this established fee family, Theorem 28 counts positive pair groups on one fixed source. Shared-cost modeling and the standard proof tools are credited to existing work. The exact literature priority of our model-specific boundary remains open; no new generic minimax or completion framework is claimed. The GS restriction does not make generic adaptivity separation new: additive fees are already GS, and the product-bit k-of-n result of Nielsen et al. [7, Theorem 1] has a tight gap of two. The candidate increment is the exact pair-support boundary on complete four-query sources under our full comparator contract. The GS/S3 geometry is standard [10]. Adaptive submodularity of a posterior utility [11] imposes different conditions and supplies approximation guarantees, rather than this exact fee-based equality. Appendix Y gives the objective and transfer comparisons, including a specifically delimited Pandora route obstruction. None establishes priority of the exact new boundary.

1.3 Organization and evidence

Section 2 specifies the source and both policy classes. Section 3 proves the deterministic threshold. Section 4 states the randomized boundary; Appendix V proves the coverage/submodular strengthening. Appendix W gives the sourcewise pair-resource minimum and its sparse robust neighborhood. Appendix Y proves the GS-restricted minimum, full-class priority bound and relative robustness. Section 5 gives the continuous-price four-query exclusion. Section 6 states the fixed-source fee-class bounds. The appendices retain the complete proofs, finite-proof dependencies and applicability controls. Exact certificate replay supports the stated mathematical obligations; it is not human review, a formal proof kernel, or evidence of practical superiority.

2 Model and complete policy recursion

2.1 Complete partition fibres

Let R be a finite record set. For each label $y \in \{A, B\}$, let C_y be a conjunction of $\mathsf{SAME}(r, s)$ and $\mathsf{DIFFERENT}(r, s)$ constraints, and let Ω_y contain *all* equivalence partitions satisfying C_y. Both fibres are nonempty. The number of equivalence classes is unrestricted.

A physical menu $Q = \{q_0, \ldots, q_{m-1}\}$ assigns each coordinate a pair of records. Its answer is 1 exactly when that pair is equivalent. Write $W_y \subseteq \{0, 1\}^m$ for the projection of Ω_y onto the menu. Assume $W_A \cap W_B = \varnothing$. Distinct partitions can have the same answer vector. Distinct physical coordinates keep their separate cost identities even when their answers are equivalent.

Unlisted equalities do not imply inequalities: unconstrained blocks may merge. Arbitrary pruning of W_y, bounding the number of classes, or imposing a distribution changes the model.

For example, nonnegative query charges and shared setup fees give $F(S) = \sum_{q \in S} c_q + \sum_p a_p \mathbf{1}\{S \cap Q_p \neq \varnothing\}$. This illustrates a monotone submodular cost without assuming platform measurements. For pair groups, we also study the stronger gross substitutes (GS) condition on F. The exchange formulation and standard quadratic criterion appear in Appendix Y: the minimum pair weight in each triangle must occur at least twice. This is a property of the charged set function, without an independent-reward or adaptive-submodular-utility assumption on observations.

2.2 Policies and purchased-set costs

A deterministic adaptive policy purchases an unpurchased query using only purchased answers and stops at a pure feasible state. A *priority policy* fixes a permutation π of Q first. At each position it can TAKE the query, permanently SKIP it, or stop, using only observed answers. Skipping reveals nothing and cannot be undone. The order may depend on the instance and its costs, but not on unobserved answers.

Global order indexes coordinates, not paid-test depth: causal skips may make the kth purchase differ across inputs. Prefix policies prohibit intermediate skips. Appendix N.4 compares these classes and full-cube linear strategies.

Let $F : 2^Q \to \mathbb{R}_{\geq 0}$ satisfy $F(\varnothing) = 0$ and $F(S) \leq F(T)$ for $S \subseteq T$. For a policy T, let $P_T(w)$ be its final purchased set in world w, and define

$$C_F(T) = \max_{w \in W_A \cup W_B} F(P_T(w)).$$

The optimal adaptive and priority values are A_F and P_F. Policy-class inclusion gives $\mathsf{A}_F \leq \mathsf{P}_F$. For additive prices $\kappa \geq 0$, with $F(S) = \sum_{q \in S} \kappa_q$, write $\mathsf{A}^*(\kappa)$ and $\mathsf{P}^*(\kappa)$.

The main deterministic objective excludes noisy observations, randomization, allowed errors, expected costs, and charges depending on answer values or acquisition order. Appendix M defines a separate zero-error randomized extension with expected purchased-set cost; no randomized conclusion is implicit in the deterministic results. Repeating a physical coordinate is unnecessary with reliable answers. Computational costs of constructing a policy are separate from its query costs.

2.3 Envelopes and dominance

The envelope $E(T)$ is the family of inclusion-maximal sets among $\{P_T(w) : w \in W_A \cup W_B\}$. It is an antichain, with $C_F(T) = \max_{S \in E(T)} F(S)$. For envelopes write $E_1 \preceq E_2$ if every member of E_1 is contained in some member of E_2; equivalently, $\downarrow E_1 \subseteq \downarrow E_2$.

Lemma 1 (Cost-independent dominance) *If $E_1 \preceq E_2$, then E_1 costs no more than E_2 for every monotone F. This relation is necessary as well for domination at every nonnegative additive price vector.*

Proof Containment proves sufficiency. If $S \in E_1$ is contained in no member of E_2, give the coordinates in S unit prices and the others zero prices. The first envelope costs at least $|S|$; the second costs at most $|S| - 1$. □

The recursion combines every query and feasible child-envelope choice, then retains envelopes minimal under $\preceq$. A fixed-order recursion also permits SKIP. Induction on remaining coordinates proves completeness; adjoining a common query and uniting branch families preserve dominance. Appendix E specifies the recurrence, empty-family conventions and paid-context costs.

Lemma 2 (Universal equality criterion) *Equality $\mathsf{A}_F = \mathsf{P}_F$ holds for every admissible monotone F if and only if each minimal adaptive envelope has a realizable priority envelope dominated by it.*

Proof Every adaptive envelope has a minimal adaptive envelope below it, since the family is finite. A priority certificate below that envelope costs no more, proving $\mathsf{P}_F \leq \mathsf{A}_F$. Conversely, suppose a minimal adaptive envelope E has no priority envelope below it. Set $F(S) = 0$ on $\downarrow E$ and $F(S) = 1$ elsewhere. This is normalized and monotone. The adaptive witness costs

zero, while every priority policy has a path outside $\downarrow E$ and costs one. This converse concerns arbitrary monotone costs, not additive costs alone. □

Incompatible branch orders in one adaptive tree do not rule out a different optimal priority tree.

2.4 Private randomness and the objective

For an arbitrary partial Boolean task $f : D \to \{0,1\}$ and a finite normalized nonnegative monotone fee F, define

$$R_{\mathcal{C}}(f,F) = \min_{\mu\in\Delta(\mathcal{C})} \max_{w\in D} \mathbb{E}_{T\sim\mu} F(P_T(w)), \qquad \mathcal{C} \in \{\mathcal{A},\mathcal{P}\}.$$

Every supported tree must be correct on all of D. The input is independent of the seed. For $\mathcal{P}$ each seed fixes one global order, shared by both answer branches; permanent SKIPs are causal. Different seeds may choose different orders. When $D = W_0 \cup W_1$, we also write $R_{\mathcal{C}}(W;F) = \mathsf{R}_{\mathcal{C}}(W;F)$. For additive $F_c(S) = \sum_{q\in S} c_q$ we write $R_{\mathcal{C}}(W;c)$. The deterministic values are $D_{\mathcal{A}} = \mathsf{A}_F$ and $D_{\mathcal{P}} = \mathsf{P}_F$. Neither the cost of an expected purchased set nor a per-seed hard budget is being minimized.

3 The first deterministic separation

Theorem 3 (Computer-assisted six/seven boundary) *For complete partition fibres, if $|Q| \le 6$, then $\mathsf{A}_F = \mathsf{P}_F$ for every normalized nonnegative monotone F. A seven-query complete-fibre instance has a strict gap with strictly positive additive prices. Thus seven is the smallest menu size permitting a strict gap for each of the monotone, monotone submodular, nonnegative additive, and strictly positive additive cost classes.*

The order may depend on F; one order optimal for all costs is not asserted. The theorem minimizes queries, not records or classes. We separate the structural reduction, the finite premise, and the explicit upper-size witness.

3.1 A finite catalogue and its structural meaning

Collapse prior SAME blocks on one side. Retain blocks containing query endpoints and their induced DIFFERENT constraints. Every consistent endpoint partition extends to omitted blocks by assigning each a fresh class. Clone shared endpoints and express their identities through prior SAME constraints. At most $2m$ named endpoint positions therefore suffice, without changing physical query coordinates.

After contraction, the menu is a labeled multigraph $G = (V,Q)$, possibly with loops and parallel edges; let D be the forbidden terminal pairs. A set $X \subseteq Q$ of SAME answers is feasible exactly when

$$\mathrm{cl}_{M(G)}(X) = X \quad \text{and no pair in } D \text{ is connected in } (V,X). \tag{1}$$

Transitivity and the negative constraints prove necessity. Assigning each component of (V, X) a separate class proves sufficiency, with omitted blocks assigned fresh classes. This is an equality of answer supports, not a bijection of hidden partitions.

Equivalently, add an auxiliary edge for each forbidden pair and fix its SAME answer to zero. The legal supports are fixed-coordinate sections of the augmented graphic-flat relation. Auxiliary edges are constraints, not purchasable queries. Complementing edge answers gives demand-separated multicuts [12, 13]. All transformations preserve the original purchased sets and F.

Each circuit yields Horn implications from its other edges to the last edge; each forbidden terminal path gives a negative Horn clause. The source class is strictly narrower than arbitrary Horn relations. Indeed, suppose nonconstant coordinates satisfy $x_i = 1 \Rightarrow x_j = 1$. Merge just the feasible endpoint blocks of query i, keeping the other blocks separate. Query j can be one in this atomic world only when it joins the same pair of blocks. Hence the reverse implication also holds. A strict one-way positive implication between nonconstant coordinates is impossible.

For comparison, the four-query Horn supports $A = \{0, 2, 3, 6, 7, 11\}$ and $B = \{4, 5, 14, 15\}$, with prices $(1, 2, 2, 2)$ and integer-bit coordinates, have adaptive cost 5 and priority cost 6 for every one of the 24 orders. The first support has $x_0 \Rightarrow x_1$ without the converse. This exact finite example lies outside the complete-source class for every record count, so it does not refute Theorem 3.

The four-query comparison shows why completeness matters: the six-query equality is specific to graphic-flat sections, rather than to general Horn promises or ordered diagrams. Complete-source and 2-CNF relations are also incomparable when visible coordinates are preserved (Appendix F).

3.2 The checked finite premise

For each endpoint identification, enumerate its unrestricted partition-answer support. Intersect it with every distinct collection of forbidden-pair filters, retaining distinct nonempty supports. Iterative intersection, deduplicated by the entire resulting support, covers all conjunctions. For a fixed endpoint identification and realizable X, the components of (V, X) form the finest partition realizing X. A forbidden-pair filter admits X exactly when its endpoints are in different components. This one partition therefore satisfies all filters that individually admit X, justifying their intersection. Pair disjoint supports, then quotient by *simultaneous* coordinate permutations and label exchange. The sides cannot be relabeled independently, and arbitrary answer-bit flips are not used.

Any disjoint pair of catalogue supports has a common physical realization. Take $2m$ distinct records $r_{i,0}, r_{i,1}$ and queries $q_i = (r_{i,0}, r_{i,1})$. Under each label impose the endpoint identifications and forbidden pairs generating that support. These SAME and DIFFERENT constraints may differ by label; the physical coordinates and their cost identities remain fixed.

A coordinate permutation σ transports cost by $F_\sigma(T) = F(\sigma^{-1}(T))$. This preserves purchased-set costs and the four cost classes in Theorem 3. A representative certificate therefore covers its full orbit. Table 2 states the completed finite premise.

Layer	Count
Six-query endpoint topologies	198,091
Distinct nonempty complete one-sided supports	546,735
Disjoint support-pair orbits	2,860,696
Singleton-side orbits	51,048
Remaining orbits with a shared constant coordinate	61,923
Remaining duplicate/complement gate orbits	129,250
Residual orbits with explicit strategy certificates	2,618,475
Uncovered cases	0

Table 2 Complete six-query classification. The last four case groups are disjoint. Counts are accompanied by membership and complete-certificate checks.

Lemma 4 (Finite certificate premise) *Let $\mathcal{C}_m$ be exactly the family of nonempty supports from endpoint identifications and forbidden-pair conjunctions on m labelled query coordinates. For every disjoint pair in $\mathcal{C}_5$, and every disjoint pair in $\mathcal{C}_6$ with neither a singleton side nor a shared constant coordinate, each minimal adaptive envelope E has a correct priority policy T, compatible with one global permutation, with $E(T) \preceq E$.*

The finite check reconstructs every adaptive frontier and verifies supplied policies for legality, order compatibility and containment. It need not enumerate the whole priority frontier. This premise has no prices; Lemma 2 supplies the all-cost consequence. The orbit audit checks source membership, canonicality, uniqueness, weights and an independent Burnside count. Counts alone are insufficient. Duplicate/complement coordinates receive explicit certificates, not a cost-independent representative substitution.

The two analytic groups require short proofs. If one side is $\{a\}$, take an adaptive tree's query sequence on a. Every opposite-label world differs somewhere on that sequence. Buy it in that fixed order, stopping with the opposite label at the first difference. Each purchased set is a prefix of the set on a, so its worst monotone cost is no larger than that of the original tree. This argument works for arbitrary disjoint supports and any menu size.

If a query has a shared constant answer on both sides, remove its nodes from any policy. Purchased sets shrink and the projected supports remain disjoint. Deletion preserves complete fibres by retaining the same records and constraints and deleting only an observation. On a six-query instance the remaining menu has at most five queries. The independently checked five-query foundation covers $9,439$ supports, $34,601$ pair orbits, and $54,021$ priority-envelope witnesses. Apply it and insert the removed coordinate back as a permanent skip. If primed values denote deletion, then

$$\mathsf{A}_F \le \mathsf{P}_F \le \mathsf{P}'_{F'} = \mathsf{A}'_{F'} \le \mathsf{A}_F, \qquad F'(S) = F(S).$$

Smaller menus are padded with shared constant queries, with $F^+(S) = F(S \cap Q)$, then projected back. Positive dummy prices also suffice when proving the strictly positive additive statement, because they need not be bought.

Together, catalogue completeness, the analytic groups, full adaptive-frontier reconstruction and Lemma 2 establish the lower-size assertion. This is a computer-assisted finite premise, not a proof-assistant-kernel theorem. The executable evidence and all dependency layers remain necessary.

3.3 A complete seven-query source

With coordinates $0, \dots, 6$, use the templates

$$A = (\alpha, 1, \alpha, \beta, \alpha, \gamma, \gamma), \qquad B = (\mu, \mu, 0, \nu, \omega, \omega, \nu).$$

Each side retains all eight assignments to its three free bits; these are combinatorial freedoms, not probabilistic independence. A common vector would force $q_1 = 1$, then $\mu = 1$, $q_0 = 1$ and $\alpha = 1$, contradicting $q_2 = 0$.

For a complete realization, take records $0, \dots, 13$ and $q_i = (2i, 2i + 1)$. Under A, impose the SAME blocks

$$\{0, 4, 8\},\ \{1, 5, 9\},\ \{2, 3\},\ \{10, 12\},\ \{11, 13\},$$

with $6, 7$ unconstrained singleton blocks and no DIFFERENT constraints. Under B, impose

$$\{0, 2\},\ \{1, 3\},\ \{6, 12\},\ \{7, 13\},\ \{8, 10\},\ \{9, 11\},$$

with $4, 5$ singleton blocks and $\mathsf{DIFFERENT}(4, 5)$. Unlisted cross-block merges remain legal. Every satisfying partition has the stated template. Conversely, merge the relevant block pair for each free bit equal to one, and give the remaining blocks distinct fresh classes. Every assignment is realized, so the supports are exact complete-fibre projections.

The complete frontiers in Appendix B give, for $t > 0$ and $\varepsilon \geq 0$,

$$\kappa = (2t, 2t, 3t, \varepsilon, t, t, t), \qquad \mathsf{A}^* = \min(5t, 4t + \varepsilon), \quad \mathsf{P}^* = 5t.$$

Taking $t = 1$, $\varepsilon = 1/2$ gives strictly positive prices with values $9/2$ and 5, completing the upper-size assertion of Theorem 3. Section A.4 gives a direct proof of strict separation: for $0 < \varepsilon < 1$, it establishes $\mathsf{A}^* \leq 4 + \varepsilon < 5 = \mathsf{P}^*$ by incompatible necessary orders across two probe answers. The complete frontiers there additionally determine the exact adaptive value and the all-price ratio.

A readable pair of policies.

Let $N_i(L, R)$ mean buy q_i and follow L on answer zero or R on answer one; leaves A, B output the label. The following saved adaptive witness begins with the inexpensive probe:

$$\begin{aligned} L &= N_0\big(N_1(B, A),\ N_5(N_4(B, A), N_6(B, A))\big), \\ R &= N_4\big(N_5(N_6(A, B), A),\ N_2(B, A)\big), \\ T &= N_3(L, R). \end{aligned}$$

Its envelope is $\{11, 28, 57, 105, 120\}$ and its cost at $(2, 2, 3, 1/2, 1, 1, 1)$ is $9/2$. A priority policy in order $0, 1, 2, 3, 4, 5, 6$ is $N_0(N_1(B, A), N_2(B, A))$, skipping q_1 after $q_0 = 1$. Its envelope $\{3, 5\}$ costs 5. These policies show achievable values and the role of the probe; completeness of the frontiers, not a conflict between these particular trees, proves optimality.

4 Randomization changes the first separation

In this section F is any finite normalized nonnegative monotone purchased-set fee. The first statement even permits an arbitrary partial Boolean domain D and the objective $R_{\mathcal{C}}(f, F)$ just defined.

Theorem 5 (Generic three-query equality) *For every partial Boolean task with $|Q| \leq 3$ and every fee in the stated class, $R_{\mathcal{A}}(f, F) = R_{\mathcal{P}}(f, F)$.*

The result concerns a worst-input expectation; it does not assert equality for each specified prior or with the cheapest-certificate lower bound. Its analytic reductions and disjoint finite proof cover are in Appendix N. Complete sources satisfy the stronger certificate identity of Theorem 18.

Theorem 6 (Four-query randomized boundary; general monotone fees) *In the complete equivalence-partition fibre model, the smallest indexed pair-query menu permitting $R_{\mathcal{A}}(F) < R_{\mathcal{P}}(F)$ is four for normalized nonnegative monotone final-set fees. The same minimum holds when every nonempty set has positive fee and every proper inclusion strictly increases the fee.*

Appendix M.5 gives the original general-monotone witness and perturbations approaching unit additive fees; those particular fees remain nonsubmodular. The distinct coverage construction in Theorem 26 uses positive individual and pairwise setup charges, with exact values $125/7$ and $161/9$. Corollary 27 preserves a gap of at least $1/63$ in an explicit full-dimensional submodular fee neighborhood. The deterministic values on this same four-query source are equal by Theorem 3. Mixing therefore changes the comparison: input-specific path costs are averaged before maximizing over inputs. Pure additive fees instead force the equality proved next.

4.1 Two pair-support boundaries

Theorem 30 proves the unrestricted positive-pair minimum two on complete four-query sources. Theorem 31 raises that minimum to three when the quadratic pair-coverage fee must be GS. The latter does not follow merely from the former: two disjoint positive pair groups are GS and require a separate universal exclusion. Its proof covers six balanced hierarchy cones per price order on each of the eleven residual sources, alongside the monotone majority. An explicit equal-weight triangle supplies a strict upper witness, with $R_{\mathcal{P}} = 1025/47$ and $R_{\mathcal{A}} \leq 324871/14899$. A direct full-class Bellman recurrence proves the priority lower bound, retaining correctness on every legal world even when its prior mass is zero.

5 Randomized equality at four queries with additive costs

The fee class changes the randomized boundary. Here we prove an exclusion at four queries for every nonnegative additive price vector. This is stronger than agreement on a finite set of prices. It does not extend the exclusion to arbitrary partial Boolean promises or to all submodular final-set fees.

For $\mathcal{C} \in \{\mathcal{A}, \mathcal{P}\}$, define

$$R_{\mathcal{C}}(W; c) = \min_{\mu} \max_{w \in W} \sum_{T} \mu_T \sum_{q \in P_T(w)} c_q, \qquad c \in \mathbb{R}^{Q}_{\geq 0}.$$

The private seed is independent of w. For $\mathcal{P}$ each seed may choose its own global permutation; its two answer branches share that permutation and may permanently skip queries using the observations so far. A lower-bound prior used in a certificate is not a population assumption. The maximum is over the original correctness promise, even at inputs receiving zero probability in that prior.

Theorem 7 (Four-query additive equality, computer-assisted) *Let W_0, W_1 be the nonempty, disjoint projections of two complete partition fibres with an unrestricted number of classes, on at most four indexed pair-equality queries. For every finite nonnegative real additive price vector c,*

$$R_{\mathcal{A}}(W; c) = R_{\mathcal{P}}(W; c).$$

Hidden records, shared endpoints, zero prices and price ties are allowed. The statement concerns worst-input expected actual purchased cost at zero error. It does not assert equality under every specified input prior.

Proof We give the reduction, full-policy premise and continuous-price certificate obligations. The accompanying packet supplies the finite assignments and complete analytic case arguments. Its implementation checks the stated obligations using exact arithmetic; the analytic arguments are not replaced by a formal proof kernel.

Complete source cover. Collapse all prior SAME components, including paths through hidden records. The eight marked endpoint occurrences induce a partition π. For an answer word $w \in \{0,1\}^4$, merge the positive query edges. The word is possible exactly when every zero-answer edge and every prior DIFFERENT pair still has separate ends. Necessity follows from transitivity. For sufficiency assign each remaining component a fresh class, and assign fresh classes to hidden-only components. The number of classes is unrestricted, so there is no further extension condition.

For a fixed π, let B_π be its answer support without forbidden pairs. A forbidden pair between endpoint-containing blocks defines a filter $H_j \subseteq B_\pi$. Starting with $\{B_\pi\}$, the iteration

$$\mathcal{L}_j = \mathcal{L}_{j-1} \cup \{S \cap H_j : S \in \mathcal{L}_{j-1}\}$$

covers every finite forbidden-pair conjunction, including redundant filters. All 4140 restricted-growth strings on eight positions give 465 distinct nonempty supports. Every disjoint pair is realized on the same eight physical records with its two label-specific priors. There are 42864 ordered tasks, partitioned into 1356 orbits under simultaneous query-coordinate permutations

and label exchange. Answer-bit complementation is not a physical symmetry used in this cover.

Full policy cover. Repeated, publicly fixed-answer and post-pure queries can be erased without increasing a nonnegative additive cost. In $\mathcal{P}$ erasure is a legal permanent skip. At a mixed feasible state, the adaptive recursion chooses any remaining informative query and combines both answer continuations. For a fixed permutation, the priority recursion unites SKIP on the same feasible set with TAKE on both answer children, using the same remaining suffix on both branches. Pure states stop. Taking the union over all 24 orders covers the full priority class. The finite union of input-specific null sets of erroneous seeds can be removed from any zero-error randomized policy. Seedwise normalization therefore reduces it to a distribution on these finite families. A fresh recursion reconstructs all 39113 adaptive and 33781 priority paid-set vectors. All 629694 original-input paths are checked for correctness, purchased sets, no repeated queries and the shared global order. No chosen-order or restricted-input baseline is substituted for $\mathcal{P}$.

The majority of sources. For any normalized monotone fee F, put

$$K_F = \max_{w \in W} \min_{S \text{ certifies } w} F(S).$$

Every correct path is a certificate, so $K_F \le R_{\mathcal{A}}(W;F)$. The set $I_F = \{S : F(S) \le K_F\}$ is a nonempty downset in $2^{[4]}$ and contains a certificate for every input. There are exactly 167 nonempty downsets. For 1300 orbits, each relevant downset admits a legal original priority policy whose path sets all belong to it. Checking all 155995 relevant source/downset cells yields $R_{\mathcal{P}} \le K_F \le R_{\mathcal{A}}$. For another 31 orbits, each complete adaptive vector has a legal priority vector contained in it on every original input. Replacing each seed's policy proves the same inequality. The reverse inequality follows throughout from $\mathcal{P} \subseteq \mathcal{A}$.

Structural and additive-only sources. Table 3 gives the disjoint remainder. For the 14 structural orbits, complete two-column, three-column and chain saddle arguments, or common-input retractions, imply equality for all normalized monotone fees. The packet includes their complete ordered case splits, generator signs and zero-boundary arguments. Polynomial identities and nonnegative-product decompositions are checked for 64, 105 and 52 sign obligations in the three scalar saddle lemmas, respectively. The physical source bindings are separately checked on the full original families (2336 input obligations). Auxiliary response encodings in a structural template are individually bound between its lower core and upper envelope; they are not added to the physical orbit group. The two cost-preorder regions use opposite scalar inequalities and include their shared boundary.

For clarity, the remaining 11 orbits require only additive costs. At a fixed nonnegative price vector, a sufficient certificate is a probability vector λ on original inputs and a probability vector p on legal priority columns with common value L, satisfying

$$\sum_{w} \lambda_w \, c(P_A(w)) \ge L \qquad \text{for every full adaptive column } A, \tag{2}$$

$$\sum_{j} p_j \, c(P_{P_j}(w)) \le L \qquad \text{for every original input } w. \tag{3}$$

These imply $R_{\mathcal{A}} \ge L \ge R_{\mathcal{P}}$ directly. For constant rational λ, p, the inequalities are homogeneous linear forms in c. On a closed price-order cone $c_{\pi_0} \le \cdots \le c_{\pi_3}$, write

$$c_{\pi_j} = \sum_{k=0}^{j} t_k, \qquad t_k \ge 0.$$

Nonnegative coefficients in the t variables prove an inequality on the entire cone, with ties and zero prices included.

The retained-library orbit uses all 29 unchanged rational prior/priority-mixture certificates. Every one of the 24 order cones lies in a certified region, as checked on its four suffix-indicator generators. The exact normals are reconstructed from all original adaptive columns and all original-input priority upper inequalities; the library is not justified by its old sampled prices.

Nine more orbits use 24 complete closed order cones each. Each cone has a pure saddle certificate or a sequence of row and column dominance inequalities with nonnegative coefficients. Deleting a column replaces it by a no-more-expensive active column on every active row. Deleting a row maps it to a no-cheaper active row simultaneously for all active columns. Following the row chains gives one common map for every mixture of terminal columns. Each terminal column then lifts to an original legal priority column on all hard rows. This preserves the full worst-input randomized value. In the one orbit with a preliminary deleted input, its paid set is contained in another input's paid set for *every* original adaptive and priority column. Its objective constraint is therefore redundant for both classes, while it remains a correctness obligation. All 16634 dominance steps, 556560 scalar signs and 3008 pure-saddle scalar signs are replayed from the original paid sets.

Proof obligation	Source orbits
Certificate-threshold downsets	1300
Same-input full-policy subset replacement	31
Two-column structural saddle	2
Three-column saddle and signed template bindings	$1+4$
Common retraction; two preorder regions; AND/hint; prefix	$1+1+1+1$
Chain structural saddle	3
Retained rational library, all 29 regions	1
Pure saddles and common-row dominance, 24 cones each	$1+8$
Balanced two-input certificate and 22 old closed cones	1
Total, with disjoint memberships	1356

Table 3 Complete four-query cover. The first six rows prove equality for the full monotone fee class on their stated sources. The last three rows complete the nonnegative additive theorem only. Each orbit is bound to its full original policy families.

The final source and its boundary faces. To make the remaining rational dependence explicit, one orbit has negative words $\{0,3,4,7,8,11\}$ and positive words $\{1,15\}$, where $w=\sum_{q=0}^{3}2^q w_q$. Write its prices as (a,b,c,d). Twenty-two order cones are closed by the preceding dominance method. The two remaining orders are interchanged by swapping c,d; this involution preserves the source and both complete policy families. It suffices to treat $0\le a\le b\le c\le d$. Two legal original priority policies, denoted U,V, have paid masks $(3,14)$ and $(7,12)$ respectively on positive inputs $(1,15)$. Their full original-input traces and global orders are checked. Put

$$D=b+c,\quad S=a+b-d,\quad N=b(a+b+c)+c(c+d).$$

On $S\ge 0$ and $D>0$, use weights S/D and $(c+d-a)/D$ on U,V, and lower prior $b/D, c/D$ on inputs $1,15$. All weights are nonnegative and sum to one. Both positive inputs have upper value N/D. The common row map

$$\tau(0)=\tau(1)=\tau(8)=1,\qquad \tau(3)=\tau(4)=\tau(7)=\tau(11)=\tau(15)=15$$

dominates every original input for both upper policies on this cone.

For the lower bound, every correct tree must buy queries a, b on input 1 and c, d on input 15, using the opposite-label single-coordinate witnesses $0, 3$ and $11, 7$ respectively. These paths share their prefix up to the first differing queried answer. If that first differing query is b, their costs are at least $(a+b, b+c+d)$; if it is c, they are at least $(a+b+c, c+d)$. The prior-weighted numerators in both cases are N. If it is d, the latter lower numerator increases by $b(d-c) \geq 0$. Extra purchases cannot reduce cost. Thus (2) holds for every correct adaptive tree, not just the retained finite columns.

On $S \leq 0$, pure policy V has worst cost $c+d$; the mandatory c, d purchases on input 15 give the matching lower bound. The packet checks all original-input upper inequalities. Closed nonnegative parameterizations for the three regions are

$$\begin{aligned} S \geq 0 &: (S, b-a, c-b, d-c), \\ S \leq 0,\ c \leq a+b &: (a+b-c, b-a, c-b, d-a-b), \\ S \leq 0,\ c \geq a+b &: (a, b-a, c-a-b, d-c). \end{aligned}$$

Their invertible linear maps and all numerator coefficients are checked exactly (1576 scalar coefficients, including zeros). In the first region D has coefficient vector $(2, 2, 3, 2)$, so $D = 0$ is only the all-zero price vector. Use legal pure V there at cost zero, without division. On the switching face, $N - (c+d)D = bS = 0$, and the mixed weight on U vanishes. This includes the shared boundaries and closes both remaining orders.

All 1356 orbits are now covered for the whole nonnegative real additive orthant. Pulling prices back along a physical coordinate permutation preserves the classes and costs. For fewer than four queries, append fresh known-SAME zero-price dummy pairs to both fibres. Their insertion and erasure preserve the two policy classes and all actual costs, proving the stated at-most-four scope. □

Corollary 8 (Randomized additive first-occurrence interval) *Let m_{add} and m_{pos} be the least menu sizes with $R_{\mathcal{A}} < R_{\mathcal{P}}$ in this same complete-source model, for nonnegative additive and strictly positive additive prices, respectively. Then*

$$5 \leq m_{\mathrm{add}} \leq m_{\mathrm{pos}} \leq 7.$$

The exact additive minima remain undetermined. The corresponding submodular minimum is four by Theorem 26.

Proof Theorem 7 excludes all smaller menus. On the complete seven-query source in Section 3.3, use the old prices $(2, 2, 3, 1/3, 1, 1, 1)$. A correct adaptive tree has worst cost $13/3$. The original priority mixture has upper $227/51$ and the fixed prior supported on world indices $(0, 4, 5, 13, 15)$, with weights $(9, 1, 1, 1, 5)/17$, gives a matching lower bound over all 5040 global orders with causal skips. Worlds with prior zero are still included in every correctness/purity test. Therefore

$$R_{\mathcal{P}} - R_{\mathcal{A}} \geq \frac{227}{51} - \frac{13}{3} = \frac{2}{17} > 0.$$

The retained exact curve in Appendix O also supplies the adaptive optimum. Only its legal adaptive upper and matched full-priority value are needed for this corollary; the upper alone is not a new proof of an exact adaptive optimum. □

6 Fee-class bounds on the fixed seven-query source

The following results concern exactly the complete source in Section 3.3. They compare deterministic values and do not give universal ratio bounds for all menus of size seven.

Theorem 9 (Fixed-source price bound) *For exactly the source in Section 3.3,*

$$\max_{\kappa \geq 0 : \mathsf{A}^*(\kappa) > 0} \frac{\mathsf{P}^*(\kappa)}{\mathsf{A}^*(\kappa)} = \frac{5}{4}.$$

Equality holds precisely on $\kappa = t(2, 2, 3, 0, 1, 1, 1)$, $t > 0$, in the stated numbering. Over strictly positive prices the supremum remains $5/4$, but is not attained. This is not an upper bound for every seven-query source.

Theorem 10 (Fixed-source submodular bound) *For every normalized nonnegative monotone submodular F on this fixed source, $\mathsf{A}_F = 0$ implies $\mathsf{P}_F = 0$. Whenever $\mathsf{P}_F > \mathsf{A}_F$,*

$$3\mathsf{P}_F + F(\{q_3\}) \leq 4\mathsf{A}_F. \tag{4}$$

The maximum of $\mathsf{P}_F/\mathsf{A}_F$ over costs with $\mathsf{A}_F > 0$ is $4/3$, already attained by a coverage cost. If F must increase strictly under every proper inclusion, the supremum is $4/3$ and is not attained, both for submodular and for coverage costs.

Appendix B gives the full frontiers and unchanged proofs, including attainment and strict-cost limits. For unrestricted strictly monotone fees the ratio is unbounded on that same source (Appendix D). The nonadditive construction is outside the submodular class, so it supplies no unbounded additive or submodular ratio. The specified eleven-query extension has its own fixed-source quantifier (Appendix J).

7 Scope, reproducibility and open questions

The unresolved additive first-occurrence size is five, six or seven. The submodular and coverage minima are four; a classification of all four-query submodular sources and the largest randomized ratio remain open. The pair-resource minimum in Theorem 28 is restricted to its stated source and two-query shared groups. We claim neither an arbitrary-size extremal ratio nor polynomial-time strategy synthesis. The GS-restricted pair-support minimum three is also universal over complete four-query sources, within the quadratic pair-coverage class only. It does not classify all three-pair instances, general GS functions or larger menus; its adaptive upper witness need not be optimal. The universal source results use unrestricted class count and exact, reliable equality queries. A specified correlated population, bounded error, noisy tests or a different information unit requires another analysis.

The complete six-query proof and fixed-seven supplements are available in the public review archive. The four-query all-additive proof has a separate self-contained packet and retained actual Linux execution evidence. Appendix H maps the older proof dependencies; Appendix T states the newer packet's full scope. The new GS extension has a separate private combined packet and an actual relocated Windows replay: seventeen mathematical children, eighteen whole output comparisons,

with the global source and full policy enumeration explicitly inherited. Its point and relative-neighborhood priority lower bounds do not require cached priority-profile exhaustiveness. Appendix Y states the freshly executed modules and provenance binding. Replaying exact code across systems is not independent implementation, human review or formal proof-kernel certification.

Applicability controls are retained, rather than interpreted as positive application results. Changing pair tests to record reads removes the gap on a fixed control, and an all-size even-path family has a priority normal form (Appendix I). The selected Wine execution and offline VPS3 audit have same-information fixed verifiers and establish no acquisition advantage (Appendix L). Controlled recorded identities and byte budgets have their own engineered contracts (Appendix P). These results identify conditions a future academic-evidence application must satisfy.

The contribution under consideration is the exact source- and fee-dependent boundary, with specified fixed-source bounds. The cited minimax, completion and ordered-evaluation methods are established. The bounded prior-work comparison does not certify exact literature priority, and the paper establishes no improvement in academic rankings, semantic judgement or historical authentication.

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Research process and generative AI.

Generative AI tools, including Codex agents, were used in research exploration, proof development, implementation of search and verification programs, automated cross-checking, manuscript drafting and formatting. Programmatic certificate replay and agent cross-checks are not human peer review. Technical checking is performed with the stated programmatic and AI-assisted methods. Actual author information, final disclosures and informed approval by the responsible human authors remain pending. Before submission those authors must understand the final claims, approve the manuscript and accept responsibility for it; this statement does not impose a separate manual replay of every technical check. Combinatorial cases are not independent scientific studies or human gold labels.

Data and code availability.

The public core packets and actual execution records are identified in Section 7. Appendix S retains the additional source-specific deposit and version correspondence. Public availability does not grant a blanket repository licence; source-specific notices

state applicable terms. This private manuscript has not been submitted or publicly deposited.

Other declarations.

Human-participant, consent and other ethical declarations must be assessed and completed by the responsible authors as applicable; this preparation text does not certify those determinations.

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A Completion and direct order comparisons

A.1 Compatible completions

For a labeled promise W, let $\mathrm{Comp}(W)$ contain its compatible total Boolean functions. Let $O_F(g,\pi)$ be the minimum full-domain worst purchased-set cost of a π-ordered decision diagram computing g, and $R_F(g)$ the corresponding minimum over read-once branching programs, without diagram-size restrictions. Then

$$\mathsf{P}_F(W) = \min_{g\in\mathrm{Comp}(W)} \min_{\pi} O_F(g,\pi), \qquad \mathsf{A}_F(W) = \min_{g\in\mathrm{Comp}(W)} R_F(g).$$

Unfolding and completing promise policies proves these identities; Appendix E.1 gives the cost-preserving argument.

The minima may use different completions; answer-equivalent coordinates retain separate cost identities. The appendix addresses paid-context costs, without asserting scalar Bellman optimality.

A.2 Certificate competition and minimax transfer

Fix nonnegative additive prices and write $c(S) = \sum_{q\in S} c_q$, with $F = c$. For $w \in W$, let $C^c_W(w)$ be the minimum cost of a coordinate set whose answers at w determine its label within W, and put $K^c_W = \max_{w\in W} C^c_W(w)$. Every terminating correct policy path is a certificate. Hence

$$K^c_W \le \mathsf{A}_c(W) \le \mathsf{P}_c(W).$$

If a legal priority policy T satisfies $c(P_T(w)) \le \rho C^c_W(w)$ for every $w \in W$, with finite $\rho \ge 1$, then

$$\mathsf{P}_c(W) \le C_c(T) \le \rho K^c_W \le \rho \mathsf{A}_c(W).$$

The multiplicative formulation includes zero certificate costs. A certificate-competitive lower bound alone gives no adaptivity separation: with unit prices, Boolean OR has optimal certificate ratio m, but both minimax values equal m.

For an arbitrary compatible total completion g, every full-domain certificate is a promise certificate, so $C^c_g(w) \ge C^c_W(w)$ on W. The reverse inequality need not hold: for $W = \{00 \mapsto 0, 11 \mapsto 1\}$ with unit prices, both promise certificate costs are one, whereas the completion $g(x) = x_1 \wedge x_2$ has $C^c_g(11) = 2$. One sufficient transfer route is $C^c_g(w) \le \alpha C^c_W(w)$ on W, together with a legal priority implementation of a ρ-certificate-competitive policy for g on the same menu and prices; restriction to W then gives $\mathsf{P}_c(W) \le \rho\alpha \mathsf{A}_c(W)$.

Pointwise certificate preservation is not necessary. Complete an optimal adaptive promise tree as in Appendix E.1, obtaining g_T and a full-domain tree of worst cost $\mathsf{A}_c(W)$. Each of its paths certifies g_T, so $\max_{x\in\{0,1\}^m} C^c_{g_T}(x) \le \mathsf{A}_c(W)$. Thus a legal priority policy for this particular g_T with certificate factor ρ would also give $\mathsf{P}_c(W) \le \rho \mathsf{A}_c(W)$. The remaining issue is whether g_T belongs to a function family with such a uniform guarantee and an admissible global-order implementation. The completion identities do not supply that membership or implementation. Applying

the priced-information results to our boundary therefore still requires a menu- and cost-preserving bridge; additive guarantees do not automatically cover shared submodular costs. These transfer observations are supporting comparisons, not claims of new competitive-analysis results.

The two certificate comparisons are quantitatively different on our fixed source. Appendix K proves that the least uniform pointwise inflation over *all* compatible completions is exactly 2 at prices $(2, 2, 3, 0, 1, 1, 1)$, although a suitable completion has maximum full-domain certificate cost 4. It also gives a strict positive-price pointwise obstruction and an exact maximum-certificate bound for read-once AND/OR completions. These are specific transfer limitations, not exclusions of every classical algorithm or reduction.

A.3 Compatible formulas and full-domain cost

The fixed source has compatible read-once AND/OR formulas, but this representation need not preserve optimal query cost. List bits in coordinate order $q_0, \dots, q_6$. The opposite-label pairs

$$(0100000, 1100000), \quad (0100000, 0000000), \quad (1111111, 1101111)$$

have their first member in A and second in B, and differ only at q_0, q_1, q_2, respectively. These three coordinates are therefore essential in every compatible total function. With A positive, monotonicity after answer flips forces q_0 negative and q_1, q_2 positive. Conversely,

$$g(q) = q_2 \vee (q_1 \wedge \neg q_0)$$

is compatible: on A, either $\alpha = 1$ or $(q_1, q_0) = (1, 0)$; on B, $q_2 = 0$ and $q_0 = q_1$. It ignores the last four coordinates. Thus exactly 16 answer orientations admit a monotone completion for this label polarity.

Let $\mathcal{R}(W)$ be the compatible total functions admitting an AND/OR formula in which each physical coordinate appears at most once as a literal; unused coordinates are allowed. This formula restriction is stronger than querying each coordinate at most once per decision-tree path. A read-once AND/OR function is evasive on its essential variables [1]. Directly, answer a queried literal by its parent's neutral value, delete it and suppress unary gates. The residual formula stays nonconstant until its final essential variable is queried. Negated leaves only reverse the returned bit. Hence for nonnegative additive prices, $R_c(g)$ is the sum of all essential-coordinate prices, including zero-price cases. The three necessary coordinates and the displayed formula give

$$\min_{g \in \mathcal{R}(W)} R_c(g) = c_0 + c_1 + c_2.$$

At $(2, 2, 3, 1/2, 1, 1, 1)$ this minimum is 7, whereas the adaptive promise value is $9/2$. An adaptive-optimal completion from Section A.2 therefore lies outside this formula family at these prices. Other, nonoptimal completions may still support useful promise-restricted certificate bounds. This cost-preservation obstruction concerns the stated family on the original menu; it establishes neither exclusion of every prior reduction nor

a practical application advantage. Proposition 16 separately determines the best maximum full-domain certificate cost in this family; Proposition 17 addresses pointwise inflation without a formula restriction.

A.4 A direct order obstruction

We prove directly that the source in Section 3.3, at prices $(2, 2, 3, \varepsilon, 1, 1, 1)$ with $0 < \varepsilon < 1$, satisfies

$$\mathsf{A}^* \le 4 + \varepsilon < 5 = \mathsf{P}^*.$$

The complete-fibre realization and the adaptive and priority upper bounds are already given there. It remains to prove the priority lower bound. This argument does not use the enumerated frontiers and does not replace the six-query classification or the all-price theorem.

Disclose $q_3 = \beta$ for free at the outset, deleting its position but keeping *one common permutation* of all other coordinates for both values of β. An original policy can ignore this information until its former q_3 position and simulate its original decisions without paying for that query. This is therefore a relaxation; earlier pure stopping can only lower cost. Put

$$(x, y, z, \ell, m, r) = (q_0, q_1, q_2, q_4, q_5, q_6), \qquad (c_x, c_y, c_z, c_\ell, c_m, c_r) = (2, 2, 3, 1, 1, 1).$$

The exact conditional supports in this notation are

$$\mathcal{A}_\beta = \{(a, 1, a, a, c, c) : a, c \in \{0, 1\}\}, \qquad \mathcal{B}_\beta = \{(u, u, 0, w, w, \beta) : u, w \in \{0, 1\}\}.$$

Suppose every path costs at most four in this relaxed game. Both labels are initially feasible for either β. Preliminary SKIPs reveal nothing. Opposite-label worlds agreeing on all available coordinates except one force its purchase on their common transcript; if that coordinate was skipped, correctness is impossible. Granting previously skipped coordinates back can only make a lower-bound subproblem easier.

The branch $\beta = 0$ ***forces*** $m \prec \ell$***.***

Use the three legal worlds

$$A_0 = (0, 1, 0, 0, 0, 0), \quad B_0 = (0, 0, 0, 0, 0, 0), \quad B_1 = (1, 1, 0, 0, 0, 0).$$

On A_0, both y and x are necessary to exclude B_0 and B_1, respectively. The following cases exhaust the first actual purchase.

First query	A feasible response defeating budget four
ℓ, m, r	Answer 0. The three displayed worlds remain; the additional x, y cost is four after a unit purchase.
z	Answer 0. The same three worlds require an additional four after paying three.
y	Answer 1. Worlds A_0, B_1 force x, using the entire remaining budget two; any other positive purchase first makes x unaffordable. Yet $x = 1$ still leaves both $(1, 1, 1, 1, 0, 0) \in \mathcal{A}_\beta$ and B_1 feasible, so buying only x cannot finish.

Thus the first purchase must be x. On its feasible answer 1, y is constant and z is unaffordable. The informative unit-coordinate supports, with budget two, are exactly

$$A' = \{100, 111\}, \qquad B' = \{000, 110\} \quad \text{on } (\ell, m, r).$$

No unit coordinate can have been skipped before the first unit purchase: the opposite-label pairs 100/000, 100/110 and 111/110 differ only at ℓ, m, r, respectively. Nor can an endpoint be purchased first. After $\ell = 1$, the surviving worlds are $100, 111$ versus 110; neither remaining coordinate alone separates their labels. After $r = 0$, they are 100 versus $000, 110$, again not separable by either single remaining coordinate. A constant-query purchase only consumes budget. Consequently m must be purchased first among the units, which requires $m \prec \ell$ in the fixed order.

The branch $\beta = 1$ ***forces*** $\ell \prec m$***.***

The legal pair

$$A_+ = (1, 1, 1, 1, 1, 1), \qquad B_+ = (1, 1, 0, 1, 1, 1)$$

differs only at z. Again consider every first purchase other than ℓ.

First query	A feasible response defeating budget four
x, y	Answer 1. The displayed pair forces z, whose price three exceeds the remaining budget two.
z	Answer 0. Budget one remains. Each affordable query has a mixed answer: $\ell = 0$, $m = 0$, or $r = 1$, respectively, as follows directly from the conditional templates. Hence no one-query continuation works.
m, r	Answer 1. The displayed pair forces z, consuming the entire remaining budget three. Any other purchase first makes this common transcript unaffordable. Buying z next also fails: its answer 0 remains mixed, with $A = (0, 1, 0, 0, 1, 1)$ and a matching B (choose $u = 0, w = 1$ after $m = 1$, or $u = 0, w = 0$ after $r = 1$).

Only ℓ can be the first purchase. If $m \prec \ell$, then m must already have been permanently skipped. On the feasible answer $\ell = 0$, take

$$\widehat{A} = (0, 1, 0, 0, 1, 1), \quad \widehat{B}_0 = (0, 0, 0, 0, 0, 1), \quad \widehat{B}_1 = (1, 1, 0, 0, 0, 1).$$

After omitting the unavailable m, these worlds agree outside x, y. On $\widehat{A}$ both heavy queries are necessary, costing four while only three remain. The coordinates z, r supply no substitute; additional prior skips cannot help. Thus m cannot precede ℓ, proving $\ell \prec m$. This step is what rules out using a fixed order with history-dependent permanent SKIPs, rather than merely comparing two particular decision trees.

The two necessary precedences contradict the common remaining permutation. Finally, all six nonprobe prices are integers. An original priority policy of total worst cost strictly below five would induce, after free disclosure, a policy whose every nonprobe path costs at most four. The contradiction gives $\mathsf{P}^* \geq 5$. The priority policy in Section 3.3 costs exactly five and ignores q_3; its adaptive policy costs at most $4 + \varepsilon$. This proves the displayed strict separation without an exact adaptive lower bound or a permutation scan.

B Complete fixed-source fee-class proofs

Source quantifier	Cost class	Conclusion
Every complete source, $m \leq 6$	Normalized monotone	$\mathsf{A}_F = \mathsf{P}_F$
One complete source, $m = 7$	Strictly positive additive	$\mathsf{P}_F > \mathsf{A}_F$ is possible
That same fixed source	Nonnegative additive	Maximum ratio 5/4
That same fixed source	Monotone submodular	Maximum ratio 4/3
That same fixed source	Strictly monotone	Ratios unbounded
One specified source, $m = 11$, and its identifiable projections	Nonnegative additive	Upper 5/4

Table 4 All set costs are normalized and nonnegative; ratios require $\mathsf{A}_F > 0$. On the seven-query source, strictly positive additive and strictly increasing submodular costs retain the respective suprema without attainment. The eleven-query row concerns one specified source; Appendix J gives sixteen matching menus. No all-seven-source or all-eleven-source upper bound is asserted.

B.1 Additive bound

Proof of Theorem 9.

B.2 Complete frontiers

An integer mask encodes a purchased set, with bit q denoting coordinate q. The complete adaptive and priority frontiers are shown in Table 5. Display rows do not pair policies; five envelopes are shared. The producer used feasible-world masks. Its independent checker used 2,187 ternary partial assignments, saturated copy clauses and frozenset families. It compared every feasibility flag against direct support enumeration, every order's frontier, and all 13 saved witnesses.

Adaptive	Priority
$\{3, 5\}$	$\{3, 5\}$
$\{11, 28, 57, 105, 120\}$	$\{11, 28, 121\}$
$\{11, 57, 60, 105\}$	$\{11, 57, 60, 105\}$
$\{19, 20\}$	$\{19, 20\}$
$\{28, 43, 57, 120\}$	—
$\{35, 49, 52\}$	$\{35, 49, 52\}$
$\{67, 84, 113\}$	$\{67, 84, 113\}$

Table 5 Undominated envelopes, with TAKE/SKIP optimization in all $5,040$ orders.

Write $\kappa = (a, b, c, z, d, e, f)$ for prices, not source bits. The priority costs are

$$\begin{aligned}
G_0 &= \max(a + b, a + c),\\
G_1 &= z + \max(a + b, c + d, a + d + e + f),\\
G_2 &= z + \max(a + b, a + d + e, c + d + e, a + e + f),\\
G_3 &= \max(a + b + d, c + d),\\
G_4 &= \max(a + b + e, a + d + e, c + d + e),\\
G_5 &= \max(a + b + f, c + d + f, a + d + e + f).
\end{aligned}$$

The two adaptive-only costs are

$$\begin{aligned}
H_1 &= z + \max(a + b, c + d, a + d + e, a + e + f, d + e + f),\\
H_2 &= z + \max(c + d, a + b + e, a + d + e, d + e + f).
\end{aligned}$$

Thus $\mathsf{P}^* = \min_{0 \le i \le 5} G_i$ and

$$\mathsf{A}^* = \min(G_0, G_2, G_3, G_4, G_5, H_1, H_2).$$

The formulas hold at all real nonnegative prices because the finite frontiers are complete, not because prices have been sampled.

B.3 Algebraic proof

First eliminate the probe price z. For fixed other prices, write $\mathsf{P}^*(z) = \min(U, z + V)$ with $U = \min(G_0, G_3, G_4, G_5)$ and $V = \min(G_1 - z, G_2 - z)$; write $H_i = z + W_i$. If $V \le W_i$, the ratio P^*/H_i is at most one whenever defined. If $V > W_i > 0$, both $U/(z + W_i)$ and $(z + V)/(z + W_i)$ decrease with z whenever their minimum exceeds one. If $W_i = 0$, the path sets cover all six non-probe coordinates, so those prices vanish and $G_0 = 0$. It suffices to bound each P^*/H_i at $z = 0$. This does not assume that H_i remains adaptive-optimal when z changes. A shared adaptive-optimal envelope gives ratio one.

For H_1, normalize $W_1 = 1$. Then

$$a + b,\ c + d,\ a + d + e,\ a + e + f,\ d + e + f \le 1.$$

If G_0, G_2, G_3 all exceed $r > 1$, their potentially large terms imply $a + c > r$, $e > r - 1$ and $d > r - 1$, respectively. Adding $c + d \le 1$ and $a + d + e \le 1$ yields

$$2 \ge a + c + 2d + e > 4r - 3.$$

At $r = 5/4$ this is impossible. Hence $\mathsf{P}^*/H_1 \le 5/4$.

For H_2, normalize $W_2 = 1$, so

$$c + d,\ a + b + e,\ a + d + e,\ d + e + f \le 1.$$

If G_0, G_4, G_3 all exceed $r > 1$, their terms imply $a + c > r$, $e > r - 1$, and $d - e > r - 1$. Consequently

$$2 \ge a + c + 2d + e > 6r - 5.$$

Setting $r = 7/6$ is impossible, giving $\mathsf{P}^*/H_2 \le 7/6$ and therefore $\mathsf{P}^*/\mathsf{A}^* \le 5/4$.

At $(a, b, c, z, d, e, f) = (1/2, 1/2, 3/4, 0, 1/4, 1/4, 1/4)$, the formulas give $\mathsf{A}^* = 1$ and every priority envelope costs $5/4$. For global equality, only H_1 can be optimal and strict probe-price decrease forces $z = 0$. Normalize $W_1 = 1$. All priority costs are at least $5/4$, forcing $a + c \ge 5/4$, $d, e \ge 1/4$, and $a + c + 2d + e \le 2$. Equality is necessary throughout. Thus $d = e = 1/4$, $c = 3/4$ and $a = 1/2$. From $G_3 \ge 5/4$ and $a + b \le 1$ obtain $b = 1/2$. The inequality $a + e + f \le 1$ gives $f \le 1/4$, and $G_1 = \max(1, 1, 1 + f) \ge 5/4$ gives the reverse bound. Scaling proves the unique ray.

Every strictly positive vector has $z > 0$. When its ratio exceeds one, removing z strictly increases it, ruling out attainment of $5/4$. The vectors $(2, 2, 3, \varepsilon, 1, 1, 1)$ approach ratio $5/(4 + \varepsilon) \to 5/4$ as $\varepsilon \downarrow 0$. If $\mathsf{A}^* = 0$, all zero-price coordinates together separate the supports and can also be queried in a fixed order at zero cost. Hence $\mathsf{P}^* = 0$ too; such vectors are excluded from the ratio. This completes Theorem 9.

A specified eleven-query extension.

Appendix J gives an explicit complete twenty-two-record realization of an eleven-query extension. Its additive upper bound is $5/4$, inherited by every identifiable coordinate projection; the sixteen menus containing the original seven queries have matching maximum $5/4$ over nonnegative prices. This is a specified-source result. Its separate direct certificate packet covers every price cone, without an old seven/eight-query upper-bound premise.

C Submodular fixed-source proof

Throughout this section the source is exactly the one in Section 3.3. A normalized monotone cost F is submodular if

$$F(S) + F(T) \ge F(S \cup T) + F(S \cap T) \qquad (S, T \subseteq Q).$$

This includes nonnegative query charges with shared setup fees. The complete monotone-cost frontiers in Table 5 remain the finite premise.

Proof of Theorem 10.

Proof For this proof only, write $F(0345)$ for $F(\{q_0, q_3, q_4, q_5\})$ and similarly for other coordinate strings; these strings are not decimal masks. Let H_1, H_2 denote the costs of the two adaptive-only envelopes:

$$H_1 = \max\{F(013), F(234), F(0345), F(0356), F(3456)\},$$
$$H_2 = \max\{F(234), F(0135), F(0345), F(3456)\}.$$

Set $Y = 03456$, $X = 2345$, $Z = 014$, $I = 345$ and $J = 34$. Submodularity gives

$$F(0345) + F(3456) \geq F(Y) + F(I),$$
$$F(234) + F(I) \geq F(X) + F(J).$$

For H_1, a third application and monotonicity give

$$F(013) + F(J) \geq F(0134) + F(3) \geq F(Z) + F(3).$$

For H_2, replace this third line by

$$F(0135) + F(J) \geq F(01345) + F(3) \geq F(Z) + F(3).$$

Adding the relevant three lines cancels $F(I)$ and $F(J)$, proving for each $i = 1, 2$ that

$$F(Y) + F(X) + F(Z) + F(3) \leq 4H_i. \tag{5}$$

Suppose $\mathsf{P}_F > H_1$. The priority envelopes with masks $\{11, 28, 121\}$ and $\{11, 57, 60, 105\}$ respectively force $F(Y) \geq \mathsf{P}_F$ and $F(X) \geq \mathsf{P}_F$: all their other path costs are at most H_1, and the cost of any valid priority policy is at least its optimum. The envelope $\{19, 20\}$ forces $F(Z) \geq \mathsf{P}_F$, because $24 \subseteq 234$ implies $F(24) \leq H_1$.

If instead $\mathsf{P}_F > H_2$, the envelope $\{11, 28, 121\}$ again forces $F(Y) \geq \mathsf{P}_F$, since $013 \subseteq 0135$. In $\{35, 49, 52\}$, the paths $015 \subseteq 0135$ and $045 \subseteq 0345$ cost at most H_2. Thus $F(245) \geq \mathsf{P}_F$, and $245 \subseteq X$ gives $F(X) \geq \mathsf{P}_F$. The envelope $\{19, 20\}$ gives the same bound for $F(Z)$. Hence (5) yields $3\mathsf{P}_F + F(3) \leq 4H_i$ whenever $\mathsf{P}_F > H_i$.

If a shared envelope attains A_F, then $\mathsf{P}_F = \mathsf{A}_F$. Otherwise an H_i attains A_F, by completeness of the adaptive frontier. In particular, when $\mathsf{P}_F > \mathsf{A}_F$, selecting such an i proves (4). If $\mathsf{A}_F = 0 < \mathsf{P}_F$, this inequality contradicts nonnegativity. For positive A_F, the ratio is either one or at most 4/3.

For attainment, take the coverage cost

$$F_0(S) = |S \cap \{q_0, q_1, q_4, q_6\}| + 2\mathbf{1}\{q_2 \in S\} + \mathbf{1}\{S \cap \{q_0, q_5\} \neq \varnothing\}. \tag{6}$$

It has independent query charges and one unit setup charge shared by q_0, q_5; q_3 is free. The seven adaptive envelope costs in Table 5 are $4, 3, 4, 4, 3, 4, 4$, and all six priority envelopes cost four. Thus $\mathsf{A}_{F_0} = 3$ and $\mathsf{P}_{F_0} = 4$.

Strict increase implies $F(\{q_3\}) > 0$ and $\mathsf{A}_F > 0$, since every correct policy makes a nonempty purchase. Equation (4) then rules out ratio 4/3. Conversely, $F_\delta(S) = F_0(S) + \delta|S|$ is strictly increasing coverage for $\delta > 0$, and the same finite frontiers give

$$\mathsf{P}_{F_\delta} = 4 + 2\delta, \qquad \mathsf{A}_{F_\delta} = \min(4 + 2\delta, 3 + 4\delta).$$

For $0 < \delta \leq 1/2$ the ratio tends to 4/3 as $\delta \downarrow 0$, proving both suprema. □

D Unrestricted monotone fixed-source fees

The finite bounds in Theorems 9 and 10 depend on their respective cost classes. Let $E = \{11, 28, 57, 105, 120\}$ and, for $\varepsilon > 0$, define

$$F_\varepsilon(S) = \varepsilon|S| + \mathbf{1}\{S \notin \downarrow E\}.$$

It is normalized and strictly increasing under proper inclusion. The six priority envelopes have costs

$$1 + 2\varepsilon,\ 1 + 5\varepsilon,\ 1 + 4\varepsilon,\ 1 + 3\varepsilon,\ 1 + 3\varepsilon,\ 1 + 4\varepsilon.$$

Envelope E costs 4ε, and the other adaptive-only envelope costs $1 + 4\varepsilon$. Therefore

$$\mathsf{P}_{F_\varepsilon} = 1 + 2\varepsilon, \qquad \mathsf{A}_{F_\varepsilon} = \min(4\varepsilon, 1 + 2\varepsilon).$$

For $0 < \varepsilon \le 1/2$ the ratio is $1/(4\varepsilon) + 1/2$, unbounded as $\varepsilon \downarrow 0$, with positive denominators. The finite inside/outside cardinality profiles prove these formulas for all real $\varepsilon \ge 0$; rational software controls are not a sampling argument.

These costs are not submodular. Masks $3, 4$ and their intersection 0 belong to $\downarrow E$, but their union 7 does not. The modular term cancels and the submodular inequality fails by one. The construction thus establishes no unbounded additive or submodular ratio. It applies the standard downset separation in Lemma 2, rather than introducing a new general lower-bound technique.

E Explicit policy recurrence

For precision, write $X = (X_A, X_B)$ for the current labeled feasible worlds and U for available coordinates. Let I_q contain the answers with nonempty feasible children. For a selected child envelope E_b at each $b \in I_q$, define

$$\mathsf{Join}_q((E_b)_{b\in I_q}) = \mathrm{Max}_{\subseteq} \bigcup_{b\in I_q} \{S \cup \{q\} : S \in E_b\}.$$

Here $\mathrm{Max}_{\subseteq}$ retains maximal purchased sets inside one envelope, whereas $\mathrm{Min}_{\preceq}$ retains minimal envelopes. At a nonempty mixed state,

$$\mathcal{A}(X, U) = \mathrm{Min}_{\preceq} \bigcup_{q\in U} \{\mathsf{Join}_q((E_b)) : E_b \in \mathcal{A}(X|_{q=b}, U \setminus \{q\})\ (b \in I_q)\}.$$

The pure-state value is $\{\{\varnothing\}\}$; the value of a mixed state with $U = \varnothing$ is the empty family. Thus an impossible policy combination contributes nothing. Empty feasible children are omitted. For a fixed order, at position i take the minimal union of (i) the family from skipping to $i + 1$ at the same X and (ii) these joined feasible-child families for taking $q_{\pi(i)}$ and continuing at $i+1$. These rules also specify zero-price and constant-coordinate cases.

At a common paid context P, future set S is evaluated by $F(P \cup S)$. Equivalently, set $G_P(S) = F(P \cup S) - F(P)$ for $S \subseteq Q \setminus P$. This cost is normalized, nonnegative and monotone, and remains submodular when F is submodular. Subtracting the common constant preserves continuation comparisons for fixed P, but does not remove dependence on the paid context. Downward-closure inclusion is preserved under adjoining a common query and uniting branch families, so dominance pruning remains valid. General F cannot be replaced by an additive remaining-cost recursion. Empty answer children contribute no legal path.

E.1 Compatible-completion proof

To extend a promise policy, unfold its tree and terminate a branch when no promised world remains. At the first such branch, a feasible sibling world purchases a superset of that branch's purchased coordinates. Monotonicity bounds its cost. The completed tree computes a compatible function, and preserves the order or read-once property. Conversely, restrict a diagram to W, unfold it and stop at pure states. Cost cannot increase. These transformations prove both identities.

F Relation to 2-CNF promises

The source class is not contained in 2-CNF relations: a three-edge path with DIFFERENT endpoints allows every assignment except 111. The majority of legal $110, 101, 011$ is illegal. All 2-CNF relations, including existential projections, are majority-closed. Conversely, $x_0 \Rightarrow x_1$ is 2-CNF but violates the atomic condition. These obstructions concern representations preserving visible query coordinates; encodings with new paid queries require a separate menu and cost analysis.

G Exact rational corroboration

G.1 Independent rational certificates

For the additive costs H_i of Theorem 9, normalize every path cost to at most one. Choose one path per priority envelope and bound a ratio variable by its cost. The 432 combinations per H_i yield 864 linear programs. Their maximal optima equal $\max_{\kappa \geq 0, H_i \leq 1} \mathsf{P}^*$ and, by homogeneity, $\sup_{\kappa \geq 0, H_i > 0} \mathsf{P}^*/H_i$; $H_i = 0$ implies $\mathsf{P}^* = 0$. An individual branch optimum is not the global adaptive ratio.

The original floating-point solver proposed candidates. Acceptance required exact rational primal and nonnegative dual feasibility, exact column equations and equal objectives. The independent checker rebuilt every program and verified all 864 certificates using rational arithmetic without an LP solver. The two maxima were $5/4$ and $7/6$. These checks support the algebraic proof without enlarging its source scope.

Independent coverage certificates.

A separate bounded check represents coverage costs as $F(S) = \sum_{\varnothing \neq R \subseteq Q} w_R \mathbf{1}\{S \cap R \neq \varnothing\}$, with 127 nonnegative resource weights. The same 432 path choices for each adaptive-only envelope produce 864 LP branches; their maxima are both $4/3$. Every

branch has an exact rational primal/dual certificate. A separate checker reconstructs all intersection columns and verifies these certificates without importing the producer or using an LP solver. This certifies the coverage subclass. The three submodular inequalities above, not the coverage LP, prove the upper bound for all submodular costs. Both arguments depend on the already established fixed-source frontiers.

H A map of the computational proof

The six-query catalogue splits into $112,971$ analytic and $2,747,725$ explicit cases. The analytic cases are $51,048$ singleton-side pairs and $61,923$ remaining shared-constant pairs; the latter depend on the complete five-query foundation. The explicit cases are $129,250$ remaining duplicate/complement pairs and $2,618,475$ residual pairs. These disjoint scopes must not be replaced by a sample, a count-only test or a selected-policy comparison.

Obligation	Required evidence
Source completeness	Enumerate endpoint identifications and all distinct forbidden-filter intersections; compare the full support family.
Pair completeness	Genuine source membership, canonical representatives, uniqueness, orbit weights and independent Burnside count.
Five-query foundation	Complete supports and pair orbits; rebuild every adaptive frontier and verify all priority witnesses.
Analytic cases	Singleton proof and constant-coordinate deletion, with monotone paid-context costs.
Explicit six-query cases	Reconstruct full adaptive frontiers and replay all increasing-order TAKE/SKIP certificates for $2,747,725$ cases.
Coverage alignment	Match each physical pair index to its assigned case group and certificate; require no omissions or duplicate accepted ranges.
Fixed-source cost bounds	Rebuild the seven-query frontiers; verify additive and coverage certificates; use algebra and submodularity for the general bounds and nonattainment.

The original V03 portable bundle completed one fresh aggregate run on 11 October 2026. It executed the complete source catalogue, pair-orbit census, five-query foundation and all $2,747,725$ explicit six-query certificates in one invocation. Each stage exited successfully and met its fixed result contract. The explicit layer replayed $4,074,031$ priority witnesses. The terminal status is conditional on the written analytic arguments: neither hashing those arguments nor checking the executable layers constitutes a formal derivation of them. The accompanying iteration-272 closure ledger binds the original bundle, stage receipts, inputs and precise scope. It records an actual local process observation, not an operating-system-signed attestation.

The later frozen candidate retains identical scientific data, mathematical kernels and stage programs. Its aggregate contract changes the textual proof bindings to V08

and omits the historical-index binding; its driver changes only the corresponding contract hash. Its acceptance covers byte inventory, a boundary plan and the fixed-seven supplement. It has not completed a fresh aggregate six-query run under its new contract. The completed original V03 run is evidence about the common computational inputs, not fresh execution of the candidate. This journal-preparation source preserves the V08 mathematical statements and proofs while updating front matter, declarations and this reproduction account; it does not retroactively change either artifact's contract.

The review archive preserves all 5,569 candidate files and adds an outer source notice, version map and execution-evidence index. Original receipt bytes and derived summaries are labelled separately. The fixed delivery record, at commit `57ed5b6`, binds the four public assets by size and SHA256. The release tag points to source commit `a8f60ce`. An anonymous download and extraction checked all 5,636 archive members; a relocated plan-only invocation exited successfully. These distribution checks do not add a mathematical replay. The outer correspondence includes journal draft V02, whose mathematical blocks were unchanged in V03. Journal V04 added three analytic explanations (common filter witnesses, a common physical menu, and continuation costs). Journal V05 explained certificate-to-minimax transfer and cited priced-information work. Journal V06 added a self-contained comparison with compatible read-once AND/OR formulas. Journal V07 added the separately checked fourteen-record acquisition control. Journal V08 appends an analytic normal form for a different even-path family and an additive construction in Appendix I, with explicit positioning against classical path search. These revisions do not change the original boundary or pair-menu finite certificate premise, or the inner executable contract binding V08.

For the boundary, the entries are `verify_catalogue_v02.py`, `verify_orbits_v01.py`, `verify_five.py` and `verify_six.py`. From the extracted archive, the aggregate entry is `candidate/boundary/verify_bundle.py`, with `--mode boundary-full --seconds 14400` and a fresh output directory outside `candidate/`. This explicit cap overrides the wrapper's 900-second default. The original bundle V03 run took about 11,199.1 seconds on its recorded environment; four hours is a cap, not a completion guarantee on another machine. The archive reviewer guide gives the complete command and interpreter requirements. Complete acceptance requires every layer to finish; files-only, planning, smoke or partial results do not suffice. Strategy and rational LP checks use exact integer/rational arithmetic. The orbit census additionally uses NumPy for exact combinatorial operations. A proposed floating-point LP solution is insufficient. Proof inputs and failed executions are preserved. Byte hashes identify versions; they do not establish authorship, human review, historical authenticity or an error-free proof kernel.

The fixed-seven supplement has the entry `verify.py --mode all`. In that invocation, `reproduce_seven.py` rebuilds all 5,040 order frontiers, checks 13 witnesses and verifies 864 additive certificates; `check_coverage.py` then verifies 864 coverage certificates against the same frontier bytes. It needs Python 3.10+ and its standard library, without an LP solver or model API. Coverage enumeration certifies the coverage subclass; the three-inequality argument in the paper supplies the upper bound for

general submodular costs. Exact hashes, executable commands, dependency requirements and accepted versions belong in the accompanying versioned ledger. The public fixed-seven record retains the executed source and results of hosted run 38134598264, attempt 1. That run used source commit `d4903be`; the linked persistent archive is the later commit `5ca862c`. It completed the fixed-seven checks just described, not the universal six-query layers or a machine proof of the general submodular argument.

I Changing the acquisition unit

I.1 A fixed fourteen-record control

Retain the complete 14-record realization in Section 3.3, including every legal cross-block merge. Now an action acquires one previously unread record. Its answer identifies its equivalence class relative to all previously acquired records, with one possible fresh class. The first acquisition conveys no class-name or label information. Only this induced partition is observable: absolute token names, content semantics, sizes and timing are excluded. This is a declared oracle contract, not a measurement of actual file contents or access time. Both policy classes receive the same responses. A priority policy scans one global record order with causal permanent skips and stopping.

Proposition 11 (Finite record-acquisition control) *On this source, give records* $2i, 2i+1$ *the prices* $c_{2i} = c_{2i+1} = \kappa_i/2$*, where* $\kappa = (2, 2, 3, 1/2, 1, 1, 1)$*. The adaptive and strong-priority worst costs are both* $15/4$*. With unit prices on all records they are both* 4*.*

Proof The A prior has seven SAME components and no required inequality, hence $B_7 = 877$ complete partitions, where B_k is the kth Bell number. The B prior has eight SAME components and requires two specified components to remain different, giving $B_8 - B_7 = 3263$. These full partitions, rather than the eight pair-answer templates per label, define the control.

For the matched prices use integer weights

$$w = 4c = (4, 4, 4, 4, 6, 6, 1, 1, 2, 2, 2, 2, 2, 2).$$

Here the weighted conclusion has a finite computational premise. Let h be a partial induced partition and let $H(h, b)$ denote feasibility within additional integer budget b. It is true at a nonempty pure state. At a mixed state it is true exactly when some unread record r with $w_r \le b$ has

$$H(h \cup \{r \mapsto t\}, b - w_r) = \text{true} \quad \text{for every feasible next class } t.$$

If there is no affordable successful action it is false. Every existing observed class and one fresh class are considered. A class assignment is feasible for a label exactly when unioning its observed equalities with that label's SAME constraints violates neither a required DIFFERENT pair nor an observed inequality. Distinct remaining components provide a completion, proving sufficiency of this test. The recursion terminates because each action adds a record.

Exact evaluation gives $H(\varnothing, 14) = \text{false}$. Integer path costs therefore imply an adaptive lower bound of $15/4$. A saved tree attains integer budget 15; all its paths increase in the single order

$$(8, 7, 5, 4, 1, 2, 0, 3, 9, 6, 10, 12, 11, 13).$$

Scanning this order and skipping until the next node of the tree implements the same observations, leaves and costs. This proves the priority upper bound and hence equality. The independent constraint-extension checker verifies all 83 nodes and 53 leaves of this tree, including exhaustive response coverage and label purity, and evaluates the lower recurrence in 3233 states. A separate full-world implementation corroborates the result on all 4140 partitions. No enumeration of 14! orders is needed. These are program-checked finite claims, not proof-assistant-kernel theorems.

For unit prices, first acquire $0, 1$. If equal, acquire $4, 5$; otherwise acquire $2, 3$. The selected pair is equal under A and different under B. The order $(0, 1, 4, 5, 2, 3, 6, 7, 8, 9, 10, 11, 12, 13)$ implements this four-read policy. For the lower bound, answer the first pair DIFFERENT precisely when it is one of $\{4, 5\}, \{4, 1\}, \{4, 9\}, \{5, 0\}, \{5, 8\}$, and SAME otherwise. For every third record there remains a common induced partition under both labels. The checker verifies this extension assertion for all $91 \cdot 12 = 1092$ first-pair/third-record cases. Thus three reads cannot suffice. □

The accompanying record-acquisition supplement contains the source constraints, integer prices, the saved tree and a standard-library verifier. It checks the weighted lower recurrence, all tree branches and their common order, and the unit-price control. This is a separate finite check from the six-query proof and fixed-seven pair-menu supplement. The source predicates and isolated-pair prices are preserved, but the action set and available information change. The result neither contradicts the pair-menu theorem nor establishes equality for all record prices, all complete sources or any natural academic-evidence task.

I.2 An all-size record-path family

Consider a separate family on $V = \{0, \dots, n-1\}$, where $n = 2k \geq 2$. The fibre Ω_A contains every partition satisfying $2r \sim 2r + 1$ for $0 \leq r < k$. The fibre Ω_B contains every partition satisfying $2r + 1 \sim 2r + 2$ for $0 \leq r < k - 1$, and $0 \not\sim n - 1$. All other merges are allowed. The fibres are disjoint: their combined equalities contradict the endpoint inequality. This family differs from the preceding 14-record source; its conclusions do not extend that finite control to all prices.

Use the same relative-record oracle: buying an unread record reveals its equivalences to all previously acquired records. Absolute class names are unavailable. Policies are deterministic and zero-error. A priority policy fixes a record permutation and permits causal permanent skips. Write $S_T(w)$ for the purchased set and $\mathsf{A}_F^{\mathrm{rec}}, \mathsf{P}_F^{\mathrm{rec}}$ for the two optimal worst costs under normalized nonnegative monotone $F : 2^V \to \mathbb{R}_{\geq 0}$. These symbols concern this record model. The first acquisition is uninformative but remains in the charged set.

Theorem 12 (Record-path containment) *For every correct adaptive policy T there is a correct priority policy P such that*

$$\forall w \in \Omega_A \cup \Omega_B \quad \exists h \in \Omega_A \cup \Omega_B : \quad S_P(w) \subseteq S_T(h).$$

Consequently $\mathsf{A}_F^{\mathrm{rec}} = \mathsf{P}_F^{\mathrm{rec}}$ for every such F. The order may depend on T and hence on the chosen cost. The comparison world h need not equal w.

Proof For $j < n-1$, let h_j have two classes $\{0,\dots,j\}$ and $\{j+1,\dots,n-1\}$; let h_{n-1} be all equal. These worlds are legal, with label B for even j and A for odd j. Since n is even, labels alternate cyclically. Retain T's actual first record i, which is fixed before informative observations. No endpoint-first restriction is imposed.

Put $v_t = (i+t) \bmod n$ for $0 \le t \le n$, treating $v_n = i$ as a ghost, and define

$$b_t = \mathbf{1}\{v_t \not\sim i\}, \qquad \epsilon_t = \mathbf{1}\{t \ge n-i\}, \qquad g_t = b_t \oplus \epsilon_t.$$

Thus $g_0 = 0, g_n = 1$; the ghost costs no second acquisition. On h_j, writing $p = (j-i) \bmod n$ gives $g_t = \mathbf{1}\{t > p\}$. On this at-most-two-class subfamily, the anchor bits determine all acquired equivalences. The resulting transition search is closely related to inconsistent chain 2-CNF clause search [14, Section 4]: with endpoints fixed, a 0-to-1 edge violates $g_t \vee \neg g_{t+1}$. The classical search structure and the minimax recurrence below are not claimed as new general algorithms.

Restrict T to these n hard worlds. Surviving cuts form an interval $[L,U]$. A read at position t splits it exactly when $L < t \le U$. Otherwise its bit and complete response are constant on the surviving hard worlds: simulate that response and delete the purchase. Relations involving deleted records remain simulable from their inferred anchor bits on this subfamily. Stop once the interval is singleton. A correct original leaf cannot retain a larger interval, since adjacent cuts have opposite labels. This produces an alphabetic binary tree with n leaves; normalization only deletes purchases from each original hard-world path.

A split at t leaves possible internal positions $L+1,\dots,t-1$ on the left and $t+1,\dots,U$ on the right. These sets are disjoint. Inductively, each nonanchor record occurs at most once in the whole tree, and therefore exactly once among its $n-1$ internal nodes. The anchor followed by internal-node preorder is a permutation. Every tree path increases in this order; off-path nodes can be skipped once the relevant branch is observed.

Run this tree on any legal, possibly multicolour, world, using only equality to the original anchor. Its g sequence need not be monotone. Maintain $g_L = 0, g_{U+1} = 1$. A response $g_t = 1$ selects $[L, t-1]$, and $g_t = 0$ selects $[t,U]$. The invariant persists, so a leaf p gives adjacent bits $g_p = 0, g_{p+1} = 1$. Both boundary records were acquired, or are the anchor/ghost.

For an ordinary physical edge $(j, j+1)$ the biases agree, including an ordinary edge entering the ghost. Thus the raw bits differ: exactly one endpoint equals the anchor, so the endpoints differ. This excludes A when j is even and B when j is odd. At the seam $(n-1, 0)$ the bias changes from zero to one; a 0-to-1 transition forces both raw bits to be zero. The endpoints then equal the anchor and each other, excluding B. These certificates give precisely the hard-world leaf labels, including endpoint anchors.

Every lifted path reaches the same structural leaf as some h_j, and purchases exactly that normalized leaf set. Hence $S_P(w) \subseteq S_T(h_j)$. Monotonicity bounds the worst cost of P by that of T. Minimizing, and using inclusion of priority policies among adaptive policies, proves equality, including zero and nonadditive costs. □

Corollary 13 (Unit prices) *For $F(S) = |S|$, the common optimum is $1 + \lceil \log_2 n \rceil$.*

Proof After the paid uninformative anchor, the normalized binary tree must distinguish n cuts, requiring height at least $\lceil \log_2 n \rceil$. A balanced alphabetic tree has this height and admits the same full-world lift and priority order. □

Corollary 14 (Nonnegative additive prices) *For $F(S) = \sum_{r\in S} c_r$, $c_r \geq 0$, an optimal priority policy can be constructed in $O(n^3)$ exact arithmetic operations using $O(n^2)$ stored values and splits.*

Proof Let $D[a, \ell]$ be the minimum worst additional price for ℓ consecutive cyclic cuts beginning at a, excluding already acquired boundary records. With all record indices modulo n,

$$D[a, 1] = 0,$$
$$D[a, \ell] = \min_{1\leq s<\ell} \big(c_{a+s} + \max\{D[a, s], D[a + s, \ell - s]\}\big).$$

The root comparison costs c_{a+s} on both branches; either child tree can be chosen independently. This proves the recurrence by induction. Every alphabetic tree has the preceding lift, and each leaf is attained by its hard world. Containment therefore gives the exact optimum

$$\mathsf{A}_F^{\mathrm{rec}} = \mathsf{P}_F^{\mathrm{rec}} = \min_{0\leq i<n} \big(c_i + D[i, n]\big).$$

One table filled by increasing ℓ serves every anchor: a continuation depends only on its cyclic interval and internal prices. There are n^2 states and $n^2(n - 1)/2$ split candidates. Saved minimizing splits reconstruct the tree in $O(n)$ time; its anchor and preorder give the causal-skip witness. Execution always compares to the original anchor, even when a subinterval starts elsewhere. Strictly shorter child intervals handle zero-price ties without further assumptions. Exact rational arithmetic additionally depends on input bit length. This is minimax node-priced search, not expected weighted-leaf optimization. □

J A complete eleven-query extension

This appendix concerns one specified complete source and its coordinate projections. All prices here are finite, nonnegative and additive. It does not bound every eleven-query complete source or a source of arbitrary size. Append four coordinates to the templates of Section 3.3:

$$T_A = (\alpha, 1, \alpha, \beta, \alpha, \gamma, \gamma, \alpha, \beta, \beta, \gamma),$$
$$T_B = (\mu, \mu, 0, \nu, \omega, \omega, \nu, \nu, \mu, \omega, \mu).$$

Each label retains all eight assignments to its three free bits.

Complete equivalence realization.

Take records $0, \ldots, 21$ and the common physical queries $q_i = (2i, 2i + 1)$ for $i = 0, \ldots, 10$. Under label A, impose the SAME blocks

$$\{0, 4, 8, 14\}, \quad \{1, 5, 9, 15\}, \quad \{2, 3\},$$
$$\{6, 16, 18\}, \quad \{7, 17, 19\}, \quad \{10, 12, 20\}, \quad \{11, 13, 21\},$$

with no DIFFERENT constraints. Under label B, impose

$$\{0, 2, 16, 20\}, \quad \{1, 3, 17, 21\},$$
$$\{6, 12, 14\}, \quad \{7, 13, 15\}, \quad \{8, 10, 18\}, \quad \{9, 11, 19\},$$

and $\mathsf{DIFFERENT}(4,5)$, leaving $4,5$ as the remaining singleton blocks. The source contains every partition satisfying these constraints. All unlisted cross-block merges remain legal.

For each free bit, all left endpoints form one SAME block and all right endpoints form another. Their equality fixes every occurrence of that bit. The fixed-one pair is SAME and the fixed-zero pair is DIFFERENT. Thus every satisfying partition has its label's displayed template. Conversely, start with exactly the displayed blocks, merge the two blocks of each free bit set to one, and give all remaining blocks distinct fresh classes. The groups and pin blocks are disjoint, so this realizes every assignment without violating the DIFFERENT constraint. This proves exact complete-fibre projection. The construction also extends every original fourteen-record partition; new records are only attached to the corresponding existing group blocks.

Write A^*_{11} and P^*_{11} for the adaptive and causal-skip global-order optima on this source.

Theorem 15 (Specified eleven-query additive bound) *For all admissible prices κ,*

$$\mathsf{P}^*_{11}(\kappa) \le \frac{5}{4}\mathsf{A}^*_{11}(\kappa).$$

The maximum ratio over prices with $\mathsf{A}^*_{11} > 0$ is $5/4$. Every identifiable coordinate projection inherits this upper bound. For each of the sixteen menus containing the original seven coordinates, the nonnegative-price maximum and strictly-positive-price supremum are $5/4$.

Proof The upper uses a computer-assisted finite premise. Complete adaptive recursion gives $2,193$ minimal envelopes. Fix one envelope E, let $U = \bigcup_{S\in E} S$, and normalize its worst path cost to at most one. Introduce $\rho \ge 0$ and nonnegative prices on U, with

$$\sum_{q\in S} \kappa_q \le 1 \quad (S \in E), \qquad \rho \le \sum_{q\in U} \kappa_q.$$

The union U separates labels, so buying all U is a correct priority policy. At a branch use a correct U-only ordered policy F and create one child for each of its maximal purchased sets T, adding $\rho \le \sum_{q\in T} \kappa_q$. For the actual optimum $\rho = \mathsf{P}^*_U$, at least one such child contains the point, since P^*_U is at most the worst path cost of F. Thus a complete finite branch tree whose every leaf has a rational dual upper bound $5/4$ proves $\mathsf{P}^*_U \le 5/4$ throughout the normalized domain. The full priority optimum is at most this U-only optimum. Selecting an optimal adaptive envelope and scaling gives the claimed full-menu bound. If $\mathsf{A}^*_{11} = 0$, the union of coordinates used by a zero-cost adaptive tree is free and separates labels, so $\mathsf{P}^*_{11} = 0$.

The delivered direct proof contains 293 legal priority policies and $130,683$ exact rational nodes, covering all $2,193$ envelope roots and all required children. Of these, $5,395$ nodes directly replace 166 roots previously justified by old seven/eight-query results. The self-contained replay rebuilds the complete adaptive frontier from ternary group states, replays all policies on all legal worlds and checks every primal/dual equation and branch disjunction. It imports neither the producer's mathematical functions nor an LP solver. Completeness of the priority-policy library is not a premise of this upper proof.

For the matching lower, take

$$\kappa = (2, 2, 3, 0, 1, 1, 1, 10, 10, 10, 10).$$

The complete full-menu adaptive frontier evaluates to 4. Scalar fixed-order recursion on the seven-coordinate projection checks all $5,040$ permutations and obtains minimum 5, allowing causal permanent SKIP at every observed history. The legal policy $N_0(N_1(B, A), N_2(B, A))$ costs 5. Any policy purchasing an added coordinate on a legal world costs at least 10, whereas an original-seven policy already costs at most 5. Erasing unobserved outside skips preserves a seven-coordinate policy. Hence the full priority optimum is exactly 5 and the ratio is $5/4$.

For an identifiable submenu M, assign every omitted coordinate a price $H > \sum_{q\in M} \kappa_q$. Buying all M is a finite-cost priority policy. A reachable outside purchase cannot improve either optimum; policies avoiding such purchases have precisely the projected histories. Within-label duplicate projections do not alter correctness or worst cost. Both optima therefore equal their M-only values, and the full-source upper transfers. For menus containing all seven original coordinates, restricting the displayed witness gives the same 4 versus 5 lower. Replacing the zero price by $\varepsilon > 0$ and taking $\varepsilon \downarrow 0$ gives the positive-price supremum, by continuity of finite minima of finite maxima of linear policy costs. No positive-price attainment or uniqueness claim is needed. □

The packet `full11_direct_proof_v01.zip` uses Python's standard library. Its scalar recursion compares SKIP with a purchase followed by the worst feasible child,

for one precommitted order at a time. The exact certificates concern every nonnegative real price vector, not a sampled price grid. The relocated Windows replay is accompanied by an actual Linux run: GitHub Actions run 38812275187, attempt 1, executed source commit `37b10299` with Python 3.12.3 and exited with code zero. Its parsed mathematical fields agree with the Windows result; only runtime metadata and elapsed time are excluded. The retained receipt is archived at commit `cba5d748`, which is distinct from the executed source. This is a cross-platform replay of the same checker, not a new independent mathematical implementation. Neither programmatic replay nor this finite-source extension establishes literature priority, human author review, or application advantage.

K Certificate costs under compatible completions

This appendix concerns the complete seven-query source of Section 3.3. All its eight assignments on each side are retained. It quantifies two different certificate comparisons used in Section A.2; it does not change the deterministic policy objective. Write

$$K_c(g) = \max_{x \in \{0,1\}^7} C_g^c(x)$$

for a compatible total function g. Let $\mathcal{R}(W)$ be the read-once AND/OR formula family of Section A.3, including signed literals and permitting unused coordinates.

Proposition 16 (Maximum full-domain certificates) *For every nonnegative additive price vector c,*

$$\min_{g \in \mathcal{R}(W)} K_c(g) = \max\{c_0 + c_1, c_0 + c_2, c_1 + c_2\}.$$

At $c^0 = (2, 2, 3, 0, 1, 1, 1)$, the minimum over unrestricted compatible total functions is instead 4. For $c^\varepsilon = (2, 2, 3, \varepsilon, 1, 1, 1)$, where $0 \le \varepsilon < 1$, that unrestricted minimum is at most $4+\varepsilon < 5$, while the displayed read-once minimum is 5. No exact unrestricted positive-ε minimum is asserted.

Proof The three opposite-label Hamming pairs in Section A.3 force coordinates $0, 1, 2$ to be essential in every completion. Any nominated set of essential leaves in a reduced read-once AND/OR formula can survive a restriction of all other leaves. At a gate whose two children contain nominated leaves, recurse in both. If only one does, set the other child to the gate's neutral value and suppress the gate. A nonconstant child can be set to either value by setting all its signed literals alike. Read-onceness makes these assignments consistent, and every nominated leaf remains essential.

Maximum certificate cost cannot increase under restriction. A certificate at an extended full input, intersected with the surviving coordinates, certifies the restricted function. Its nonnegative cost is no larger. It therefore suffices to consider three-essential-leaf read-once functions on coordinates $0, 1, 2$.

A pure AND or OR has an input forcing all three literals, so its maximum certificate cost is $c_0 + c_1 + c_2$. A mixed three-leaf formula has maximum certificate cost equal to the largest pair sum. For $a \wedge (b \vee d)$, every input has a certificate using at most two leaves, while the

single-success positive inputs force $\{a, b\}$ and $\{a, d\}$, and the double-failure negative input forces $\{b, d\}$. OR duality and literal flips preserve the statement. These cases exhaust the three-leaf formulas. The compatible function $q_1 \wedge (\neg q_0 \vee q_2)$ attains the largest pair sum on the original full cube, proving the first assertion even with zero prices.

For the unrestricted free-probe minimum, interpret the displayed adaptive tree T of Section 3.3 on every Boolean input. Its full-cube worst purchased cost is 4, so each of its paths is a full certificate of cost at most 4. Conversely, the promised input 0100000 has opposite promised Hamming neighbours at coordinates 0 and 1. Every full certificate there buys both, at cost 4. For c^{ε}, the same tree buys coordinate 3 exactly once on each path and gives the upper bound $4 + \varepsilon$. □

For the remainder put $c = c^0$, and abbreviate C_W^c and C_g^c by C_W and C_g. Every promised certificate cost is positive. Define

$$\alpha(g) = \max_{w \in W} \frac{C_g(w)}{C_W(w)}.$$

This maximizes ratios at promised inputs, rather than maximizing full-cube certificate costs before comparison. Its minimizer need not minimize K_c.

Proposition 17 (Sharp pointwise certificate inflation) *Over all compatible total Boolean functions, without a read-once restriction,*

$$\min_{g \in \mathrm{Comp}(W)} \alpha(g) = 2.$$

At prices c^{ε}, for $0 \le \varepsilon < 1$, every compatible total function has pointwise inflation at least $2/(1 + \varepsilon) > 1$. The latter bound is not claimed sharp for positive ε.

Proof Use the promised inputs, in coordinate order,

$$a = 1110111 \in W_A, \quad b = 0000000 \in W_B, \quad d = 1110100 \in W_A.$$

Their promise certificate costs are $1, 2, 2$. At a, the only promise certificate of cost below 2 is $\{3, 6\}$, fixing $(q_3, q_6) = (0, 1)$. Any cheaper single unit-price coordinate, even with the free probe, still agrees with some B input.

Suppose $C_g(a) < 2$, $C_g(b) < 4$ and $C_g(d) < 4$. The certificate at a forces $g = 1$ throughout the full-cube cylinder $q_3 = 0, q_6 = 1$. A certificate at b must contain q_1: the opposite promised input 0100000 differs only there. It must also contain $q_6 = 0$ to avoid that positive cylinder. These two queries cost 3; under a strict budget 4, only free q_3 may be added. The possible coordinate sets are therefore $\{1, 6\}$ and $\{1, 3, 6\}$, fixing $q_1 = q_6 = 0$ and optionally $q_3 = 0$.

The input d agrees at $3, 6$, so its certificate must fix $q_1 = 1$ to avoid this negative cylinder. No promise certificate at d of cost below 4 contains q_1. With that cost-2 query bought, at most one unit-price query among $4, 5, 6$ and the free probe remain affordable; a B input agrees in each case. This contradiction establishes, for every compatible g,

$$C_g(a) \ge 2 \quad \text{or} \quad C_g(b) \ge 4 \quad \text{or} \quad C_g(d) \ge 4.$$

Dividing by the respective promise costs gives $\alpha(g) \ge 2$. This is a three-input cylinder argument over every completion, not an enumeration of all 2^{112} total completions.

For a matching upper bound define the total function

$$g_*(q) = \begin{cases} 1, & q_3 \neq q_6, \\ 0, & q_3 = q_6,\ q_5 \neq q_6, \\ 0, & q_3 = q_5 = q_6,\ q_1 = 0, \\ \neg q_0 \vee q_4, & q_3 = q_5 = q_6 = 0,\ q_1 = 1, \\ \neg q_4 \vee q_2, & q_3 = q_5 = q_6 = 1,\ q_1 = 1. \end{cases}$$

Substituting either source template proves compatibility. Table 6 gives C_W and sufficient full certificates for g_*. Each listed certificate costs at most $2C_W$, so $\alpha(g_*) \leq 2$.

Table 6 Pointwise comparison at c^0. The source parameters are those in Section 3.3; rows with overlapping conditions for B may use either indicated certificate. Coordinate 3 is free. The last column is a certificate-cost upper bound, not a query-policy execution cost.

Label	Parameter condition	C_W	Certificate	Cost
A	$\beta \neq \gamma$	1	$\{3, 6\}$	1
A	$\alpha \neq \beta = \gamma$	2	$\{3, 5, 1, 4\}$	4
A	$\alpha = \beta = \gamma = 0$	4	$\{3, 5, 1, 0\}$	5
A	$\alpha = \beta = \gamma = 1$	3	$\{3, 5, 1, 2\}$	6
B	$\mu = 0$	2	$\{3, 6, 1\}$	3
B	$\nu \neq \omega$	2	$\{3, 5, 6\}$	2
B	$\mu = 1,\ \nu = \omega = 0$	3	$\{3, 6, 0, 4\}$	4
B	$\mu = 1,\ \nu = \omega = 1$	4	$\{3, 6, 4, 2\}$	5

For completeness, the promise-cost entries follow directly from the templates. On A, $\beta \neq \gamma$ is witnessed by $\{3, 6\}$, and $\alpha \neq \gamma$ by $\{4, 5\}$. The free probe alone is never pure. When $\beta = \gamma$, every one unit-price coordinate together with the probe still agrees with some B input, giving the cost-2 lower bound in the second A row. When neither inequality occurs, the all-zero parameter input has Hamming neighbours forcing $0, 1$, and the all-one parameter input has a neighbour forcing 2. On B, $q_1 = 0$ or $q_5 \neq q_6$ yields cost 2. Otherwise the two remaining inputs force, respectively, 0 and one of $2, 4$, or 2 and one of $0, 4$; the certificates $\{0, 4\}$ and $\{2, 4\}$ attain costs $3, 4$. No single unit-price query with the free probe separates these B inputs from all A inputs. This also establishes the cost-2 lower bounds in the preceding cases.

Finally let $0 \leq \varepsilon < 1$. Prices only increase from c^0, so the certificate families affordable below thresholds $2, 4, 4$ in the lower proof only shrink. The same three-way disjunction therefore holds. The promise costs at a, b, d are now $1 + \varepsilon, 2, 2$: at a the unique base certificate below 2 is $\{3, 6\}$, and the other two costs are attained without 3. Dividing the disjunction by these costs proves the asserted lower bound $2/(1 + \varepsilon)$. □

Transfer scope and finite corroboration.

At c^0, a uniform pointwise comparison $C_g(w) \leq \alpha C_W(w)$ requires $\alpha \geq 2$. Composing it with a full-domain certificate-competitive factor $\rho \geq 1$ cannot by itself recover the fixed-source bound $5/4$ through the product $\rho\alpha$. For $0 < \varepsilon < 3/5$, the positive-price lower bound similarly exceeds $5/4$. This does not exclude an input-dependent analysis,

source-aware algorithms, other reductions, or the different maximum-full-certificate bridge with a suitable unrestricted completion and legal priority implementation.

General completion complexity and distinctions between promise-valid and domain-certifying certificates are established context [15]. Our C_W is a priced version of the promise-valid convention. The exact constants above concern this specified complete source; no new general framework or literature priority is asserted.

Separate implementations corroborate the argument by classifying all 64 three-essential-variable read-once truth functions through formula generation and a 256-table cofactor scan, checking 48 affordable certificate triples for opposite-label cylinder conflicts, and checking all 128 outputs of each explicit total witness. The original bounded formula search stopped at a resource limit and is not a proof premise. Independent programmatic replay is neither a proof-assistant kernel nor human author review.

Public replay provenance.

The separately deposited certificate packet was executed once in Linux, Python 3.12.3, by GitHub Actions run 38833749833, attempt 1, using source commit `a420cd488`. The actual checker process exited with code zero. Every parsed mathematical output field agreed with the frozen Windows report; no mathematical field was excluded. The original service-digest-matched artifact and provider metadata are retained at archive commit `f68a04d13`, distinct from the executed source. Attribution to the hosted run is conditional on the stated trust in provider metadata and the artifact digest. This is a cross-system replay of the same checker, not another independent implementation, human author review, or historical scientific authentication.

L Case evidence and applicability limits

These development cases concern research-artifact auditing. They are separate from the complete-source query theorems: neither supplies a complete possible-world model, an independently justified priced physical menu, or a measured adaptive-policy advantage. The counts below are evidence components of selected workflows, not independent scientific studies or human gold labels.

L.1 A selected prospective Wine execution

A fixed author-code configuration was executed in GitHub Actions, run 38136338509, attempt 1, using source commit `6f64dd2`. Across the prescribed 25 seeds, the saved CSV has mean 0.904 and population standard deviation 0.04606811918168868, compatible with the target three-decimal Wine cell. The retained receipt binds the selected execution version, output and hosted-run provenance. Its inspected syscall trace records a successful 10,782-byte read from the designated Wine path, and the retained signature bundle passes the fixed verification policy.

A fixed direct verifier accepts the same artifact. This supports a selected prospective execution under its observation contract; it does not authenticate the historical experiment or establish the complete set of consumed bytes. Earlier failed attempts remain archived. The persistent Wine record at archive commit `ae2ba2d` links the

actual source and retained evidence. The archive commit is not the executed source. No separate baseline training run or policy-cost benefit is claimed.

L.2 An offline VPS3 artifact audit

The selected workflow comes from the Materials Cloud deposit *AMaRaNTA: automated first-principles exchange parameters in 2D magnets [Results]* by Aditya Putatunda, Giuseppe Cuono, Federico Orlando and Silvia Picozzi (DOI: 10.24435/materialscloud:he-3k). The fixed target comprises twelve parameters, 48 calculation jobs and 93 distinct repository objects in one VPS3 workflow. Prespecified raw-input spin states and final sigma-to-zero energies, combined with the declared four-state conventions, reproduce all twelve native parameter values within 10^{-7} meV and agree with the deposited text at its displayed precision. The declared $S = 1$ convention is not an inferred initialization-moment denominator.

All 48 raw energies equal their corresponding stored List-slot finite-decimal values exactly under the statically identified source slot mapping. None of the twelve results agrees with the separate small-JSON central value under the declared comparison. This discrepancy remains undiagnosed; it is not classified as twelve newly discovered physical errors. Since J_{xx} had already been checked from raw files, completion added eleven parameter coverages. Prior retrieval failures and cache reuse remain in the scoped acquisition record, which is not the whole research cost.

The portable verifier re-extracts text and small-JSON values from digest-bound retained source bytes, checks selected typed database ports and statically checks the archived source. It does not execute that author source. The relocated Windows result and the Linux result from hosted run 38140070343, attempt 1, agree apart from their timestamps. The latter uses source commit `3f39c20`; the persistent VPS3 record is at the later archive commit `7ab1b1d`.

This hosted execution runs the offline audit, not VASP or the original workflow. Numerical agreement does not supply the missing CREATE edges, authenticate historical raw-to-List conversion or parser semantics, establish physical convergence, or resolve the small-JSON discrepancy. The package preserves source attribution, the deposit's stated CC BY 4.0 terms and its original public database metadata; it is not an anonymized archive. Ordinary fixed retrieval, relational lookup and arithmetic suffice for these observations. They do not verify the complete-fibre hypothesis for academic evidence or demonstrate practical gains from the query-policy theorems.

M A sharp randomized small-menu boundary

This appendix extends the policy objective explicitly. It does not change the deterministic six/seven comparison. Here labels are written $0, 1$, with complete-fibre projections W_0, W_1 and $W = W_0 \cup W_1$. All physical query identities and the unrestricted number of equivalence classes are retained. Costs $F : 2^Q \to \mathbb{R}_{\geq 0}$ are finite-valued, normalized and monotone.

M.1 Private randomness and a common completion

For $\mathcal{C} \in \{\mathcal{A}, \mathcal{P}\}$, let $\mathsf{D}_{\mathcal{C}}(W; F)$ denote optimal deterministic worst-input cost. Thus $\mathsf{D}_{\mathcal{A}} = \mathsf{A}_F$ and $\mathsf{D}_{\mathcal{P}} = \mathsf{P}_F$. A zero-error randomized policy is an input-independent distribution over correct deterministic policies, with value

$$\mathsf{R}_{\mathcal{C}}(W; F) = \min_{\mu} \max_{w \in W} \mathbb{E}_{T \sim \mu} F(P_T(w)).$$

For priority policies the private seed may sample the permutation before execution; each supported deterministic policy still has a single global order and causal permanent skips. The input is independent of the private seed. We do not use the cost of an expected purchased set, an average over inputs, or a bounded-error criterion.

A set $S \subseteq Q$ is a promise certificate at w if every promised input agreeing with w on S has its label. Put

$$K_F(W) = \max_{w \in W} \min_{S \text{ a promise certificate at } w} F(S).$$

Every correct terminating path is a certificate. Consequently, for both classes,

$$K_F(W) \le \mathsf{R}_{\mathcal{C}}(W; F) \le \mathsf{D}_{\mathcal{C}}(W; F). \tag{7}$$

For a fixed $g \in \mathrm{Comp}(W)$, let $\mathsf{R}^{\mathrm{full}}_{\mathcal{C}}(g; F)$ require correctness and charge worst-input expected cost on the full Boolean cube. Let $\mathsf{R}^{W}_{\mathcal{C}}(g; F)$ retain full-cube correctness but charge only promised inputs. In both definitions all supported trees compute the same g. Then

$$\mathsf{R}_{\mathcal{C}}(W; F) \le \min_{g \in \mathrm{Comp}(W)} \mathsf{R}^{W}_{\mathcal{C}}(g; F) \le \min_{g \in \mathrm{Comp}(W)} \mathsf{R}^{\mathrm{full}}_{\mathcal{C}}(g; F). \tag{8}$$

The cube includes every indexed answer string, even if some strings are not physically realizable for a particular menu. A compatible completion may depend on the source and on F; no single completion is asserted to optimize all costs at once.

M.2 Certificate ideals remove the price quantifier

Theorem 18 (Three-query certificate optimality, computer-assisted) *For every complete equivalence source on at most three physical queries and every finite-valued normalized nonnegative monotone F,*

$$\begin{aligned} K_F(W) &= \mathsf{D}_{\mathcal{P}}(W; F) = \mathsf{D}_{\mathcal{A}}(W; F) \\ &= \mathsf{R}_{\mathcal{P}}(W; F) = \mathsf{R}_{\mathcal{A}}(W; F). \end{aligned}$$

For each source and each such F, one compatible total function and one deterministic priority tree attain $K_F(W)$ on the entire cube. Thus both completion minima in (8) also equal $K_F(W)$, for either policy class. Zero costs are included.

Proof The finite premise is a certificate-ideal cover, which we specify and reduce to complete checked objects below. For every complete source on $m \le 3$ queries and every nonempty downset $I \subseteq 2^Q$, suppose every promised input has a certificate in I. The premise asserts

that there is a total priority tree correct on the promise whose purchased set lies in I on *every full-cube input.*

Complete source reduction. Duplicate shared query endpoints and impose their identities as SAME constraints in both labels. Collapse all original SAME blocks. Endpoint blocks retain every DIFFERENT edge between them. Any endpoint partition satisfying these induced constraints extends to all original records: assign every remaining block without endpoints its own fresh equivalence class. All remaining DIFFERENT constraints are then satisfied. This uses an unrestricted class count, not transitivity of DIFFERENT. Thus arbitrary hidden records impose no extra restriction on endpoint responses.

For three indexed queries this reduces to six endpoints. Their $B_6 = 203$ SAME partitions, each with an arbitrary DIFFERENT graph on its blocks, give

$$\sum_{k=1}^{6} \left\{ {6 \atop k} \right\} 2^{\binom{k}{2}} = 53{,}071$$

satisfiable specifications. Exact projection gives 55 distinct nonempty supports and 924 ordered disjoint two-label source pairs. Query coordinates are never identified or repriced by this reduction.

Exhaustive ideal cover. For dimensions one and two the check covers all partial Boolean tasks with both labels, a superset of the required sources. At dimension three it covers the complete-source catalogue just described. Enumerating every nonempty downset gives the entire case table in Table 7. Each case supplies either an input with no certificate in I, or a global permutation and an explicit total tree. The producer uses quotient-partition enumeration and order recursion. A separate checker reconstructs all 203 closed equivalence graphs from the 2^{15} undirected edge masks, intersects allowed-world bitsets for all 53,071 specifications, and recovers exactly the same 55 supports and 924 source pairs. It independently enumerates downsets and tests certificate availability.

For every applicable case the checker traces the supplied tree on every full-cube input, checks labels on the promise, checks membership of its purchased set in I, and verifies the same global order along all paths. For an inapplicable case it checks the supplied input against all opposite-label agreements to exclude every allowed certificate. All 17,810 source/ideal pairs are covered exactly once. The 10,974 applicable trees pass 87,308 full-cube path checks. Constant tasks, if allowed, use a zero-purchase leaf. No optimizer or floating-point certificate is used in this finite premise.

Queries	Tasks	Nonempty downsets	Cases	Total trees
1	2	2	4	2
2	50	5	250	118
3	924	19	17,556	10,854

Table 7 Complete finite certificate-ideal cover. Dimensions one and two include all partial tasks with two nonempty labels; dimension three includes all complete equivalence sources after projection. These are exhaustive combinatorial cases, not independently sampled studies.

All-cost deduction. Set $t = K_F(W)$ and $I = \{S \subseteq Q : F(S) \leq t\}$. Normalization, nonnegativity and monotonicity make I a nonempty downset. Every promised input has a certificate in I by the definition of K_F. The finite cover supplies a deterministic priority tree

whose every cube path costs at most t. Its leaf function is a compatible total completion. This gives upper bounds t for all stated quantities, while (7) and (8) give the lower bounds. The quantifiers are $\forall W\,\forall F\,\exists(g,T)$, not $\exists g\,\forall F$. The universal price statement follows from this threshold argument, not from extrapolating a finite price grid. □

M.3 Four positive-priced physical queries attain the gap

Corollary 19 (Sharp four-query minimum) *For complete equivalence sources, four is the smallest menu allowing either $\mathsf{R}_{\mathcal{C}} < \mathsf{D}_{\mathcal{C}}$ or a strict randomized promise/common-completion gap, for each $\mathcal{C} \in \{\mathcal{A}, \mathcal{P}\}$. This holds with strictly positive additive prices, and in the containing normalized monotone cost classes, including monotone submodular costs. Both full-cube and promise-charged common-completion minima have the same minimum menu size.*

Proof Theorem 18 excludes the discrepancies at size at most three. We give the existing four-query witness and its exact analytical values to prove attainability.

Coordinates are listed in order $0, 1, 2, 3$, with positive prices $c = (2, 2, 3, 1)$ and supports

$$W_1 = \{A_0 = 0100, A_1 = 1111\},$$
$$W_0 = \{B_{00} = 0000, B_{01} = 0001, B_{10} = 1100, B_{11} = 1101\}.$$

Use records $r_0, \ldots, r_7$ and disjoint pair query $q_i = [r_{2i} \sim r_{2i+1}]$. For label one the SAME blocks are $\{0, 4, 6\}$, $\{1, 5, 7\}$ and $\{2, 3\}$, with no DIFFERENT constraints. The exact answer form is $(a, 1, a, a)$. For label zero impose SAME blocks $\{0, 2\}$ and $\{1, 3\}$ and DIFFERENT$(4, 5)$, giving exactly $(u, u, 0, w)$. All indicated free bits are realizable. Without the label promise all sixteen strings are physically realizable.

Write $N_i(L, R)$ for query i with zero/one children L, R. Mix the following promise-correct priority trees with masses $1/2, 1/4, 1/4$. Each is total on the cube, but their completions need not agree outside W:

$$U_1 = N_0(N_1(0, 1), N_2(0, 1)),$$
$$U_2 = N_3(N_0(N_1(0, 1), 0), N_2(0, 1)),$$
$$U_3 = N_3(N_1(0, N_0(1, 0)), N_2(0, 1)).$$

Their global orders are $(0, 1, 2, 3)$, $(3, 2, 0, 1)$ and $(3, 2, 1, 0)$ respectively. The expected costs on $(A_0, A_1, B_{00}, B_{01}, B_{10}, B_{11})$ are $(9/2, 9/2, 4, 4, 9/2, 9/2)$. Thus $\mathsf{R}_{\mathcal{P}} \le 9/2$.

For the lower bound assign masses $1/2, 1/6, 1/3$ to A_0, A_1, B_{11}. Opposite-label single-bit neighbours force mandatory query sets $M_{A_0} = \{0, 1\}$ and $M_{A_1} = M_{B_{11}} = \{2\}$. Including the first actual query i in each mandatory set gives average-cost lower bounds $9/2, 9/2, 5, 9/2$ for the four choices of i. That first query is independent of the unknown input. Every correct randomized mixture therefore has worst expected cost at least $9/2$, even in the adaptive class. Deterministic costs are integral, and U_1 attains worst cost 5. Hence, for both classes,

$$\mathsf{R}_{\mathcal{C}}(W; c) = 9/2, \qquad \mathsf{D}_{\mathcal{C}}(W; c) = 5.$$

Now fix a compatible total g and let $z = 0101$. If $g(z) = 0$, opposite-label single-bit neighbours of A_0 force coordinates $0, 1, 3$, costing 5. If $g(z) = 1$, neighbours z, A_1 of B_{11} force coordinates $0, 2$, again costing 5. The input incurring this cost is promised. Every supported deterministic tree computing this fixed g must pay at least 5 on that same input, so randomizing trees cannot remove the lower bound. Finally U_1 computes the compatible total function equal to x_1 when $x_0 = 0$ and to x_2 when $x_0 = 1$, at full-cube worst cost 5. Both common-completion minima therefore equal 5, proving all claimed four-query gaps. □

M.4 Proof provenance, literature and limits

The three-query finite premise has separately generated proof objects and an independent standard-library checker. The original portable packet was extracted and actually replayed at a different local path. The later public derivative retains six raw proof inputs and the complete finite mathematical checker core; its declared packaging changes omit the old path-bearing four-query files. The four-query witness is inherited from the earlier immutable companion result, with its separate certificate report; its sixty closed files were hash-checked when accepting the small-menu boundary. The analytical proof above gives its source, mixture, lower bound and completion collision explicitly. The old four-query computation was not re-executed for this revision.

The public derivative was executed once on Linux with Python 3.12.3, by GitHub Actions run 38844940996, attempt 1, using source commit `aade35d`. A retained checker process handle records PID 2403, exit zero. All parsed result fields equal the Windows control except `wall_seconds`; no mathematical field is excluded. The original 3,928-byte archive matches the fetched service SHA256. It is retained with provider metadata and an offline receipt verifier at archive commit `1f70f51`, distinct from the executed source. Acceptance depends on authentic provider metadata, the executed source and original artifact digest; local copies alone do not authenticate the platform. This is the same mathematical checker on another system, not an additional independent implementation, proof-assistant kernel or human review. The ledger distinguishes the new three-query execution from inherited four-query evidence.

The earlier exploratory scan used 125 partial-task symmetry classes and 64 price triples. Its failure to find a gap is not a proof premise. Theorem 18 instead uses the complete source/ideal cover and the threshold deduction for every monotone F. The certificate and common-completion identity in Theorem 18 is not asserted for every arbitrary partial three-bit function. The weaker randomized adaptive/priority equality for arbitrary partial tasks is proved separately in Theorem 5.

Consistent outputs and compatible total completions are established context in pseudodeterministic search and communication [16–18]. The inspected journal Theorem 5.5 of Chattopadhyay et al. is a randomized decision-tree-size result (Theorem 4.5 in its earlier conference version). The current STOC paper includes a bounded-error query-depth warmup, and Watson gives a zero-sided-error communication separation. These are important prior results. Our objective fixes the complete equivalence source and physical pair menu, charges expected purchased-set cost at zero error, and distinguishes whole-cube from promise-only completion charging. A direct transfer needs these hypotheses and quantifiers to be preserved; no such exact derivation from the inspected statements is asserted. A separate bounded source-expressivity audit supports that comparison and is retained with the preparation record. It is not an additional numbered contribution of this manuscript or a proof of bibliographic priority. The candidate result here is the exact minimum physical menu within complete equivalence fibres and the all-monotone certificate identity below that minimum.

The preceding corollary concerns randomization benefit and a common-completion discrepancy. Subsection M.5 proves the four-query randomized adaptive/priority minimum for general monotone final-set fees, including strictly positive strictly monotone fees. Its witness is nonadditive and nonsubmodular. For additive fees, Theorem 7 now

excludes all complete sources through four queries and Corollary 8 gives a five-to-seven interval. The exact additive minimum remains undetermined. Theorem 26 separately determines the submodular and coverage minima as four. The deterministic adaptive/order boundary remains six versus seven, and no largest randomized four-query ratio is claimed. No practical academic-evaluation gain, new model experiment, human gold label, or new historical execution certification follows from this appendix.

M.5 A four-query randomized adaptive–priority boundary

Write $\mathcal{A}$ for correct adaptive policies and $\mathcal{P}$ for correct policies that sample a global query order with the private seed, permit causal permanent skips, and may stop once the output is determined. For a normalized monotone final purchased-set fee F, put

$$R_{\mathcal{C}}(F) = \min_{\mu \in \Delta(\mathcal{C})} \max_{w \in W} \mathbb{E}_{T \sim \mu} F(S_T(w)), \qquad \mathcal{C} \in \{\mathcal{A}, \mathcal{P}\}.$$

Here $R_{\mathcal{C}}(F)$ abbreviates the previously defined $\mathsf{R}_{\mathcal{C}}(W; F)$ for a fixed promise W. The input is chosen before the private seed. Both policy classes receive the same complete source promise, query menu and fee function.

Proof of Theorem 6.

Proof Theorem 18 gives equality for every complete source with at most three queries and every normalized monotone fee. We describe the four-query upper witness and matching bounds explicitly.

Use records $0, \dots, 7$, with $q_i = (2i, 2i+1)$ for $i = 0, \dots, 3$; an answer bit is one for equality. The negative source has required equality blocks $\{0,1,2,4,6\}$, $\{3,5\}$ and $\{7\}$, with $3 \neq 7$. The positive source has required equality blocks $\{0,2,4,6\}$, $\{1,3,7\}$ and $\{5\}$, with no required inequality. Each source contains all coarsenings consistent with its requirements. Their projected supports, using $w = \sum_i 2^i q_i$, are

$$W_0 = \{1, 7, 9\}, \qquad W_1 = \{0, 4, 11, 15\}.$$

These supports are disjoint. The three compatible negative and five positive record partitions produce exactly these projected supports.

For a query set with mask s, define

$$(F_0(s))_{s=0}^{15} = (0,0,0,0,0,0,0,4,0,0,1,4,0,4,2,4).$$

For $d \geq 0$, let $G_d(S) = |S| + dF_0(S)$. This fee is normalized, positive on nonempty sets, and strictly increasing under proper inclusion, including at $d = 0$.

Here are complete matching bounds for every $d \geq 0$. The following three adaptive paid-set profiles list their masks on inputs $(0, 1, 4, 7, 9, 11, 15)$:

$$a_1 = (3,3,3,10,3,10,10),$$
$$a_2 = (9,9,9,9,10,10,10),$$
$$a_3 = (15,15,14,14,14,6,12).$$

Use their certified tree representatives with weights

$$\frac{(2+4d,\ 2+4d,\ 2d)}{4+10d}.$$

For the priority upper bound, the four certified profiles are a_1, a_2, p_3, p_4, where

$$p_3 = (7, 7, 14, 14, 7, 6, 12),$$
$$p_4 = (15, 15, 13, 13, 14, 6, 12).$$

with weights $(2 + 6d, 2 + 6d, d, d)/(4 + 14d)$. To specify the representatives, let $T(q, u, v)$ query q, continuing with u after zero and v after one; constants denote output leaves. Put $u = T(0, 1, 0)$, $v = T(1, 1, 0)$ and $z = T(3, u, 0)$. The five trees are

$$a_1 = T(1, u, T(3, 0, 1)), \qquad a_2 = T(3, u, T(1, 0, 1)),$$
$$a_3 = T(2, T(1, z, 1), T(3, v, 1)), \quad p_3 = T(2, T(1, u, 1), T(3, v, 1)),$$
$$p_4 = T(2, T(1, z, 1), T(3, u, 1)).$$

The four priority trees a_1, a_2, p_3, p_4 admit orders $(1, 0, 2, 3)$, $(2, 3, 0, 1)$, $(2, 3, 1, 0)$ and $(2, 1, 3, 0)$, respectively. All five trees are correct on the original seven inputs. The respective dual input distributions are supported on $(7, 9, 11, 15)$, with weights

$$\frac{(2d, 2d, 2 + 3d, 2 + 3d)}{4 + 10d}, \qquad \frac{(2d, 2d, 2 + 5d, 2 + 5d)}{4 + 14d}.$$

The first mixture has cost at most $2 + d - d^2/(5d + 2)$ on every original input, while every correct adaptive tree has at least this expected cost under the first dual distribution. The second mixture and second distribution give the corresponding matching value $2 + d - d^2/(7d + 2)$ for all priority policies. The complete families contain forty adaptive and thirty-two priority paid-set vectors, independently reconstructed from all four-level tree skeletons. Every column inequality, every original input upper bound, normalization and positive denominator is checked symbolically for the entire interval $d \geq 0$. Hence

$$R_{\mathcal{A}}(G_d) = 2 + d - \frac{d^2}{5d + 2}, \qquad R_{\mathcal{P}}(G_d) = 2 + d - \frac{d^2}{7d + 2},$$

and

$$R_{\mathcal{P}}(G_d) - R_{\mathcal{A}}(G_d) = \frac{2d^3}{(5d + 2)(7d + 2)} > 0 \quad (d > 0).$$

This proves the four-query witness within the stated strictly positive, strictly monotone subclass, and thus in the containing monotone class. Together with the complete-source lower bound, the minimum is four. □

Deterministic and randomized costs on the same source.

For this source, the previously defined deterministic optima satisfy

$$D_{\mathcal{A}}(G_d) = D_{\mathcal{P}}(G_d) = 2 + d \qquad (d \geq 0).$$

Here $D_{\mathcal{C}}(G_d)$ abbreviates $\mathsf{D}_{\mathcal{C}}(W; G_d)$. The priority tree a_1 uses only purchased masks 3 and 10, whose fees are 2 and $2 + d$, proving the upper bound. For a direct lower bound, any path buying at least three coordinates costs at least $3 + 2d > 2 + d$. A correct tree of depth at most two cannot start with q_0: after $q_0 = 1$, no single remaining coordinate separates all opposite labels. It cannot start with q_2 either, by the branch $q_2 = 1$. If it starts with q_1, the branch $q_1 = 1$ forces q_3 as the second purchase; if it starts with q_3, the branch $q_3 = 1$ forces q_1. Either case purchases mask 10 on some input and costs $2 + d$. This also proves the unit-fee endpoint $d = 0$.

The same values can be expressed using the existing certificate family $H_e = F_0 + e|\cdot|$. That family has $D_{\mathcal{A}}(H_e) = D_{\mathcal{P}}(H_e) = 1 + 2e$, $R_{\mathcal{A}}(H_e) = 1 + 2e - 1/(5 + 2e)$

and $R_{\mathcal{P}}(H_e) = 1+2e-1/(7+2e)$. For $d > 0$, $G_d = dH_{1/d}$, and all four optimum values are positively homogeneous in the fee: multiplication by d commutes with the policy minimum, the input maximum and the private-seed expectation. This yields the displayed deterministic value and re-expresses the randomized formulas above without changing the source or policy class. The contrast therefore combines the already established boundaries and exact fee families, rather than asserting another independent separation contribution.

Deterministic envelope domination need not supply randomized domination. For each priority path it permits a comparison with an adaptive path on another input; that comparison input may depend on the sampled policy. It gives no same-input coupling that would compare the two worst-input expectations.

Fees arbitrarily close to unit additive fees.

For all sets,

$$|S| \le G_d(S) \le (1 + 4d/3)|S|, \qquad \|G_d - |\cdot|\|_\infty = 4d.$$

Thus for every $\eta > 0$, choosing $d = 3\eta/4$ gives a strict adaptive–priority gap while every set fee lies between its unit additive value and $(1 + \eta)$ times that value. All singleton fees are exactly one. For every $d > 0$,

$$G_d(\{q_1, q_3\}) + G_d(\varnothing) - G_d(\{q_1\}) - G_d(\{q_3\}) = d > 0.$$

The witness therefore remains nonadditive, nonsubadditive and nonsubmodular for every positive perturbation. It proves no separation inside additive or submodular fee classes. Its advantage is cubic:

$$\lim_{d\downarrow 0} \frac{R_{\mathcal{P}}(G_d) - R_{\mathcal{A}}(G_d)}{d^3} = \frac{1}{2}, \qquad \lim_{d\downarrow 0} \frac{R_{\mathcal{P}}(G_d)/R_{\mathcal{A}}(G_d) - 1}{d^3} = \frac{1}{4}.$$

For $d = 3/40$, the fee factor is at most $11/10$, and the exact values are $1969/950$ and $4187/2020$, with gap $27/191900$.

Perturbation robustness.

For any fixed policy family, its optimal randomized worst-input value is 1-Lipschitz in the supremum norm of final-set fees. Each fixed policy's expected cost changes by at most that norm on each input; taking maximum inputs and then minimum mixtures preserves the bound. Consequently, every legal fee G' satisfying $\|G' - G_d\|_\infty < (R_{\mathcal{P}}(G_d) - R_{\mathcal{A}}(G_d))/2$ retains the strict gap. This is a fee-model guarantee, not a measured API-billing or academic-workflow improvement.

Comparator conventions.

Amano's recursive directional class [4] first fixes child-subformula orders. For each seed, depth-first leaf expansion gives a global order; causal skips implement completed subformulas. This class is contained in $\mathcal{P}$. A fixed-permutation prefix class without

skips is also contained in $\mathcal{P}$; the explicit permutation-prefix convention appears in [6]. Their lower bounds alone do not give a lower bound against $\mathcal{P}$. As a finite check on the distinction, for $(x_0 \wedge x_1) \vee (x_2 \wedge x_3)$ on the full cube at unit prices, adaptive, priority and recursive-directional randomized worst-input values are all 3, while the no-skip prefix value is 10/3. The full-cube example is not an added complete-source classification result. These comparator translations and generic minimax arguments are supporting context; exact model-specific literature priority remains under review.

Exact replay scope.

The separate private companion packet `randomized_gap_v01.zip` provides three explicit verification components: the inherited three-query certificate-ideal checker, a portable adaptation of the independent full-family four-query saddle checker, and a portable adaptation of the near-additive/comparator checker. Each component reconstructs its stated finite objects and checks exact rational or symbolic inequalities. The packet contains the complete forty and thirty-two profiles for this source. It does not rely on a selected price grid, a reduced priority comparator, model responses or human labels. The release ledger distinguishes original independent checks from replay of the same checker kernels at a different path. It is a local replay packet; no new hosted or historical execution is claimed.

N A generic randomized three-query boundary

N.1 Model and scope

Let Q be a finite coordinate set, $D \subseteq \{0,1\}^Q$ a promise, and $f : D \to \{0,1\}$. We take $D \neq \varnothing$; the empty promise has value zero by convention. A purchased-set fee is a finite function $F : 2^Q \to \mathbb{R}_{\geq 0}$ with $F(\varnothing) = 0$ and $F(S) \leq F(T)$ for $S \subseteq T$. A deterministic policy must return the correct label on every promised input. The class $\mathcal{A}$ allows causal query choices. In $\mathcal{P}$, each private seed fixes one global order; queries may be permanently skipped using the observed history. Different seeds may use different orders. The value is

$$R_{\mathcal{C}}(f,F) = \min_{\pi \in \Delta(\mathcal{C})} \max_{w \in D} \mathbb{E}_{T \sim \pi} F(P_T(w)), \qquad \mathcal{C} \in \{\mathcal{A}, \mathcal{P}\}.$$

Each supported tree is promise-correct. No single common total completion across all seeds is required. Here $R_{\mathcal{C}}(f,F)$ extends the notation $\mathsf{R}_{\mathcal{C}}(W;F)$ from Appendix M to an arbitrary labeled promise $f : D \to \{0,1\}$.

Theorem 5 is stated in Section 4. The following reductions and finite cover prove it.

The theorem concerns the worst-input expectation over private randomness. It asserts neither equality for each fixed input prior nor a universal identity with the cheapest-certificate lower bound. The stronger certificate identity for complete relational sources remains a separate, narrower result. In particular, the three-query randomized equality itself needs no relational-source completeness assumption.

N.2 Analytical reductions

Stop a tree as soon as its label is determined and remove any query whose answer is fixed by the current promise and history. These operations shrink purchased sets pointwise and preserve global-order feasibility. Every resulting tree on three coordinates is obtained by pruning one of the twelve complete query skeletons. Independent recursion on the remaining input subset gives the same finite tree family. A tree belongs to $\mathcal{P}$ precisely when the union of its path precedence constraints is acyclic; a topological order implements the tree with permanent skips.

Lemma 20 (Translation and subset replacement) *Assume $|Q| \le 3$. Let G be the group of pairs (v, k) satisfying $D \oplus v = D$ and $f(w \oplus v) = f(w) \oplus k$. Suppose every normalized $\mathcal{A}$ tree either has a $\mathcal{P}$ tree with a pointwise subset of purchased coordinates, or has a root coordinate flipped by some element of G. Then $R_{\mathcal{A}} = R_{\mathcal{P}}$ for every stated fee.*

Proof Swapping a tree's branches at coordinates indicated by v and flipping its leaves by k maps its purchased set at w to its original purchased set at $w \oplus v$. Coordinates are not permuted, so F need not be symmetric. Uniformly averaging an arbitrary tree mixture over G cannot increase its worst cost.

Fix a root-flipping element h. Pair group elements as $\{g, g \oplus h\}$. If a root branch is label-constant, its paired branch is constant as well. Otherwise let a, b be the next queries in the root's two branches. After this choice, at most one coordinate remains, so the normalized continuation is forced. The two orders $(q, a, \text{remaining})$ and $(q, b, \text{remaining})$ reproduce, for each input, the purchased-set multiset of the paired trees. Each has weight $1/|G|$. For roots that are not flipped, use the supplied pointwise subset replacements. Applying these operations to the same full-group average produces a $\mathcal{P}$ mixture of no larger worst cost. The inclusion $\mathcal{P} \subseteq \mathcal{A}$ completes the proof. □

For a promised input w, call S a certificate if no opposite-label promised input agrees with w on S, and set

$$K_F = \max_{w \in D} \min_{S \text{ certifies } w} F(S).$$

Every correct path is a certificate. For the sublevel downset $I_F = \{S : F(S) \le K_F\}$, every input has a certificate in I_F. Thus, if every such downset has a priority tree with all promised path sets inside it, then

$$K_F \le R_{\mathcal{A}} \le R_{\mathcal{P}} \le \mathsf{D}_{\mathcal{P}}(f; F) \le K_F.$$

There are exactly nineteen nonempty downsets on three coordinates. This is the existing certificate-threshold reduction, rather than a new minimax framework.

For $H \subseteq D$, write $R_{\mathcal{C}}(H, F)$ for the value of the restricted labeled task $f|_H$; the notation for D has the same meaning.

Lemma 21 (Promise extension preserving worst cost) *Write $D = H \cup \{z\}$. Suppose every normalized H-correct adaptive tree also classifies z correctly, and one fixed $u \in H$ satisfies*

$P_T(z) \subseteq P_T(u)$ for every such tree T. Then $R_{\mathcal{C}}(D,F) = R_{\mathcal{C}}(H,F)$ for $\mathcal{C} \in \{\mathcal{A}, \mathcal{P}\}$ and every stated fee.

Proof Restriction gives $R_{\mathcal{C}}(H,F) \le R_{\mathcal{C}}(D,F)$. Normalize a mixture on H without increasing its cost. Each tree already extends correctly to D, and the same u works for every seed. Hence

$$\mathbb{E}F(P_T(z)) \le \mathbb{E}F(P_T(u)) \le \max_{w \in H} \mathbb{E}F(P_T(w)).$$

This proves the reverse inequality. The common choice of u is essential to this argument; a seed-dependent choice would not justify it. □

N.3 Exhaustive certificate coverage

Proof of Theorem 5 There are $3^8 - 2 \cdot 2^8 + 1 = 6050$ nonconstant three-bit partial tasks. They are covered by three disjoint verified classes.

1. For 4498 tasks, Lemma 20 applies. The complete structural reconstruction contains 43302 adaptive and 38598 priority normal trees over all tasks. The covered class uses 24264 direct subset obligations and 23376 pair/input identities.
2. For 1344 additional tasks, all applicable downsets have the certificate-threshold priority witnesses. The full residual check covers $1552 \cdot 19 = 29488$ task/downset pairs, retaining 25040 inapplicable pairs, 4240 successful priority witnesses, and 208 refutations of the stronger universal K_F conjecture.
3. All final 208 tasks satisfy Lemma 21 after deleting one input. Dependencies strictly decrease the promise-domain size and terminate at already proved tasks. There are 1536 tree-level label/subset obligations; 420 earlier deletion options with incorrect labels remain in the ledger.

The independent checker verifies the disjoint exhaustive union $4498 + 1344 + 208 = 6050$. Constants cost zero. Fewer coordinates embed by fixing the added bits and extending the fee by intersection with the original coordinate set; fixed-answer queries can be removed. □

Corollary 22 (Minimum randomized adaptive/priority menu) *For arbitrary partial Boolean tasks, four is the minimum number of coordinates permitting $R_{\mathcal{A}} < R_{\mathcal{P}}$ under general monotone fees, and also under strictly positive strictly increasing fees.*

Proof Theorem 5 excludes a gap below four. The complete four-query source in Theorem 6 is also a partial Boolean task. With its fee F_0, the exact randomized adaptive and priority values are $4/5$ and $6/7$; with $14F_0(S) + |S|$, they are $239/18$ and $351/25$. The latter fee is positive on every nonempty set and strictly increases under proper inclusion. The complete $40\mathcal{A}/32\mathcal{P}$ policy coverage and exact primal/dual certificates underlying that theorem establish the upper witness. Together the lower and upper bounds prove both minima. □

This historical witness is nonadditive and nonsubmodular. The separate Theorem 26 determines the coverage/submodular minimum as four. The additive exclusion in Theorem 7 leaves the five-to-seven interval. Randomization itself can help on three variables: the classical unit-price majority example has $K = 2$, deterministic

cost 3, and randomized cost 8/3 [4]. The 208 certificate-threshold refutations remain valid even though their randomized adaptive/priority values are equal.

N.4 Primary-source comparison

The present contribution candidate is the exact finite boundary under the stated quantifiers. Ordered skipping, partial-function consistency, randomized worst-input expectation, and nonadditive testing costs are established concepts.

The following comparisons use the actual source witnesses and unit leaf-query costs. They clarify the comparator and objective; they are not additional separation contributions. Write $\mathsf{R}_{\mathcal{C}}$ for the full-cube, worst-input expected value and $B_{\mathcal{C}}(\nu)$ for the optimal expected cost under a fixed input prior ν, with every supported policy still correct on the full cube. For either class,

$$B_{\mathcal{C}}(\nu) \le \mathsf{R}_{\mathcal{C}}.$$

The fixed-prior objective is linear in the policy mixture. Thus a valid lower bound covering every deterministic ordered-skip policy also covers private mixtures of different orders and gives a minimax lower bound. A prior-versus-minimax distinction alone does not invalidate it.

Recursive directional strategies.

Amano's directional policies recursively finish a selected subformula before entering another [4]. After fixing the coins, a depth-first leaf expansion embeds them in $\mathcal{P}$. The source's Eq. (3.2) has 154 variables. Let f_i be disjunctions on disjoint sets of sizes $(11, 12, 12, 39, 79)$ and let z be separate; the formula is

$$f = ((f_1 \wedge f_2 \wedge f_3 \wedge f_5) \vee z) \wedge f_4.$$

The algorithm used for the general upper bound in its Theorem 3.8 also has a global-order implementation. Pre-sample each disjunction's leaf order uniformly and a uniform permutation of the three small blocks. Fix the global order as z, f_5, f_4, then those small blocks. If $z = 1$, skip f_5 and the small blocks; otherwise evaluate the conjunction in that order, skipping the remainder after any false block. Within a disjunction stop at the first true leaf. These skips use only purchased answers. The small-block conjunction has expected cost at most 73/4 on false inputs and 19 on true inputs; the conjunction $f_5 \wedge f_4$ has bounds 79 and 60, respectively. Including z gives a worst-input expected cost at most 80. Independent pre-sampling in disjoint recursive copies preserves the global order by leaf-block concatenation, so

$$\mathsf{R}_{\mathcal{P}}(f^{(k)}) \le 80^k.$$

In contrast, the published directional Claim 3.9, first row of Eq. (3.4) at the base case, gives $d_0(f) \ge 536183/6341 > 80$. We adopt that published directional lower bound rather than reprove it. It cannot be relabeled as an all-$\mathcal{P}$ lower bound in view of the

displayed feasible policy. The value 80^k is an upper bound, not a computation of the exact $\mathsf{R}_{\mathcal{P}}$ or $\mathsf{R}_{\mathcal{A}}$ for this family.

Linear strategies and fixed-prior lower bounds.

Greiner et al.'s Definition 16 already permits irrelevant-test skipping [3]. On the full cube, a correct policy cannot permanently skip a still-relevant variable: two completions differing only in that variable would remain indistinguishable. Pruning redundant queries therefore identifies that linear class with our full-cube comparator. Numbering here follows the author manuscript dated 9 September 2005. Its Theorem 18 uses a unit-cost formula of the form

$$h = \bigvee_{i=1}^{r} \left[\left(X_i \vee \bigwedge_{j=1}^{D} Y_{ij} \right) \wedge \bigwedge_{k=1}^{r} B_{ik} \right].$$

Under the displayed prior ν_0, the X_i are independent Bernoulli(q) and all Y, B leaves have probability one of being true. Correctness still concerns the full cube, and those leaves must be purchased to confirm them. Put $K = D + r$ and $N = r(D + r + 1)$, the number of query leaves. The source's adaptive strategy scans X_i until a true one and then its B block; after all false X_i, it uses one Y block and its B block. Extend it by querying remaining leaves after an unexpected Y/B outcome. Its expected cost under ν_0 is

$$U = \frac{1 - (1 - q)^r}{q} + r + D(1 - q)^r.$$

The source's canonical-block lower-bound argument yields the conservative finite expression $B_{\mathcal{P}}(\nu_0) \geq \min\{r/q, (1 - q)D\}$. This is a legitimate linear-class Bayes lower bound; we adopt the source's exchange argument and do not claim a new full proof of it.

The obstacle to transferring the displayed average-cost ratio is instead a concrete upper/lower comparison. A fixed order tests all Y_{1j}, then all B_{1k}, with the other leaves afterwards. It costs K on every ν_0-supported input. On the supported input w_* with all $X_i = 0$ and all $Y, B = 1$, every positive certificate contains a complete Y_i block and its B_i block. Hence every correct adaptive seed costs at least K at w_*, giving

$$B_{\mathcal{P}}(\nu_0) \leq K \leq \mathsf{R}_{\mathcal{A}}.$$

Thus this particular prior cannot certify a strict minimax $\mathcal{P}/\mathcal{A}$ gap. Restricting correctness to its constant-label support would be a different promise problem.

The source also prescribes replacing the probability-one leaves by probability $1 - \varepsilon$, for example $\varepsilon = 2^{-2^n}$ with its total-node parameter n. Keep the X probabilities unchanged and call the prior ν_ε. Couple the same X bits and strategy coins; any of the $N - r$ other leaves changes with probability at most $(N - r)\varepsilon$. Since every nonrepeating policy costs at most N, every fixed-policy mean, and hence each optimal Bayes value, changes by at most $N(N - r)\varepsilon$. The source's mean bounds therefore remain meaningful. More sharply, the preceding fixed order costs K when its first K

leaves are true and at most N otherwise:

$$B_{\mathcal{P}}(\nu_\varepsilon) \le K + (N-K)\big(1-(1-\varepsilon)^K\big) \le K + NK\varepsilon,$$
$$\frac{B_{\mathcal{P}}(\nu_\varepsilon)}{\mathsf{R}_{\mathcal{A}}} \le 1 + N\varepsilon.$$

The last inequality uses the unchanged full-cube lower bound at w_*. It caps what this prior can certify by averaging; it is not an upper bound on the actual minimax ratio $\mathsf{R}_{\mathcal{P}}/\mathsf{R}_{\mathcal{A}}$. A tiny gap, another prior, or another reduction is not excluded.

Permutation prefixes versus causal skips.

Hellerstein et al. define a non-adaptive strategy as a test permutation whose prefix is purchased until the function is determined [6]. We use arXiv:2208.03810v1, 7 August 2022, its first-page definition and Theorem 7. Denote that prefix class by $\mathcal{N}_{\mathrm{pre}}$. Its address-function witness has d selector bits and 2^d dedicated bits, all independent uniform and unit-priced, and returns the selected dedicated bit. Put all selectors before all dedicated bits in one global order. After paying for the selectors, skip the now irrelevant unselected bits and buy the selected bit. This $\mathcal{P}$ strategy costs $d+1$ on every input, implementing the source's adaptive upper bound. The source's $\Omega(2^d)$ prefix-search lower bound consequently is not an all-$\mathcal{P}$ lower bound.

The smallest member $f(a, z_0, z_1) = z_a$ calibrates the distinction with no change of costs or prior. The order a, z_0, z_1 with causal skips always buys two bits; no single bit determines this full-cube function. Thus $\mathsf{D}_{\mathcal{A}} = \mathsf{D}_{\mathcal{P}} = \mathsf{R}_{\mathcal{A}} = \mathsf{R}_{\mathcal{P}} = 2$. In contrast, every prefix permutation has uniform mean $5/2$ and worst cost 3: each possible first pair determines the value on exactly half the inputs. Averaging gives a $5/2$ lower bound for any private mixture of prefix orders. A half-half mixture of (a, z_0, z_1) and (a, z_1, z_0) attains $5/2$ on each input. Therefore $\mathsf{D}_{\mathcal{N}_{\mathrm{pre}}} = 3$ and $\mathsf{R}_{\mathcal{N}_{\mathrm{pre}}} = 5/2$ under the same minimax convention. These exact small-member values are our calibration of a classical source function, not a quoted new theorem of that paper. They establish no exact values for the larger address-function family.

The Las Vegas OBDD class equivalence in [19] concerns polynomial representation size. We use no representation-size bound. The partial-table model and compatible completions in [20] already formalize promise consistency. The inspected statements therefore do not directly supply the exact finite cost boundary stated here.

Batch-history charging.

The batch-testing objective in [9] adds the cost of each performed batch. A direct substitution of its set-cost symbol for our terminal-set fee need not preserve path charges. For nonempty batches $c(B) = \rho + \sum_{i\in B} a_i$ with $\rho > 0$, one batch $\{a, b\}$ costs $\rho + a_a + a_b$, whereas two singleton batches cost $2\rho + a_a + a_b$. The final tested union is the same. A single $F(\{a, b\})$ cannot represent both charges. This elementary control addresses that substitution, without excluding other reductions.

Primary result	Inspected model and measure	Transfer issue for the present theorem
Karpinski–Mubarakzjanov [19]	Polynomial-size OBDD classes; bounded probability of an unknown answer; the proof keeps a common variable order.	Exact expected purchased-set cost and seed-specific orders require additional arguments.
Kristensen–Miltersen [20]	Fixed-order diagrams consistent with a partial truth table; decision-node minimization.	Compatible completion is prior; the node-count objective differs from the present path-cost objective.
Harris–Nagarajan–Tan [9]	Known independent probabilities, series diagnosis, and subadditive batch costs accumulated along performed batches.	Promise class, prior, and charging convention require an explicit cost-preserving translation.

Table 8 A bounded comparison of inspected statements. It is not an exhaustive exclusion of earlier theorems or reductions.

The same paper assumes subadditive costs. In our existing strict gap family $G_d(S) = |S| + dF_0(S)$, $d > 0$, one has

$$G_d(\{1,3\}) = 2 + d > G_d(\{1\}) + G_d(\{3\}) = 2.$$

That particular near-additive witness lies outside the subadditive class. The new witness in Theorem 26, however, is a coverage cost and does belong to that class. Harris et al.'s coverage-machine setting [9, Section 4] is relevant prior cost modeling; a cost-class mismatch does not distinguish the new result. Transferring a theorem still requires preserving the batch-history versus final-union charge, source, objective and full comparator.

Source assumptions and priority.

Because Theorem 5 now concerns arbitrary partial functions, exclusion of an earlier witness from complete relational sources does not establish distinctness of this generic lower bound. The complete-source K_F identity needs its own comparison. This bounded primary audit establishes no bibliographic priority and rules out no arbitrary recoding. The full 1998 Vereshchagin text remains an access limitation; the later directional construction and attribution were inspected in Amano [4].

N.5 Execution evidence and manuscript status

The unified private packet has 35 files and 379013 compressed bytes. Its ZIP was actually extracted into another directory and its four components were executed on Windows Python 3.12.14. All exited with code zero, and every non-runtime mathematical field matched the closed originals. The run preserves the 208 stronger-K counterexamples and all 420 rejected label extensions. Three lower checking cores

retain their original text except four path-bearing content bindings; the four-query upper component is byte-identical to its earlier portable source.

This was a relocation replay of existing checking implementations, with no model requests, numerical minimax solves, or new price/prior grids. It adds no human review or practical academic-evaluation advantage. Other operating systems were not tested.

The earlier V24 extended the sourcewise pair-resource result to all complete sources in Theorem 30. At most one pairwise setup gives equality through four queries, and the over-source four-query pair-support minimum is two. The combined proof was actually replayed at a different directory: 25 mathematical child processes exited zero and 26 whole output objects matched. The 1345 finite monotone certificates and all residual readers were freshly executed; the global catalogue cover remains an inherited accepted premise. All 29 earlier statement bodies and 26 earlier proof bodies are unchanged, including the corrected retained/minimal family distinction. Earlier public, source and executed versions retain their identities. Literature priority, natural application advantage and final author approval remain unestablished.

O An exact private-coin curve on the seven-query source

This appendix returns to the complete source in Section 3.3, with all sixteen template assignments and the original seven physical coordinates. Put

$$c_\varepsilon = (2, 2, 3, \varepsilon, 1, 1, 1), \qquad \varepsilon \geq 0,$$

with additive fee $F_\varepsilon(S) = \sum_{i \in S} c_{\varepsilon,i}$. Abbreviate $R_{\mathcal{C}}(\varepsilon) = \mathsf{R}_{\mathcal{C}}(W; F_\varepsilon)$, using the zero-error, worst-input expected-cost definition of Appendix M. Priority policies may randomize their global order and causal permanent skips; the input remains independent of the private seed.

Theorem 23 (Exact curve in one additive price family, computer-assisted) *Let* $\eta = \sqrt{19} - 4$. *On this fixed complete source,*

$$R_{\mathcal{A}}(\varepsilon) = \min\{4 + \varepsilon, 9/2\}, \tag{9}$$

$$R_{\mathcal{P}}(\varepsilon) = \begin{cases} (\varepsilon^2 + 13\varepsilon + 46)/(\varepsilon + 11), & 0 \leq \varepsilon \leq \eta, \\ (17 - 2\varepsilon)/(4 - \varepsilon), & \eta \leq \varepsilon \leq 2/5, \\ 9/2, & \varepsilon \geq 2/5. \end{cases} \tag{10}$$

Consequently, $R_{\mathcal{P}} > R_{\mathcal{A}}$ *exactly when* $0 \leq \varepsilon < 1/2$. *All seven prices are strictly positive when* $\varepsilon > 0$. *The priority value changes rational branch at* η *and reaches its constant value at* $2/5$. *Its cost gap above the adaptive value closes only at* $1/2$.

Proof We give explicit upper policies and the finite, full-class premises for matching lower bounds. Indices $0, \ldots, 7$ enumerate the A worlds lexicographically by (α, β, γ); indices $8, \ldots, 15$ enumerate B by (μ, ν, ω). The priors below certify lower bounds; none is an assumed population distribution.

Upper policies. Retain the notation $N_i(L,R)$ for a query with zero and one continuations. Set $U_i = N_i(B,A)$, $V_i = N_i(A,B)$ and

$$H = N_4\big(N_5(V_6,A),U_2\big).$$

Five correct priority trees are

$$\begin{aligned}
T_0 &= N_4\big(N_0(U_1,B),N_3(N_5(A,U_6),U_2)\big),\\
T_1 &= N_3\big(N_0(U_1,N_5(U_2,U_6)),H\big),\\
T_2 &= N_3\big(N_0(U_1,N_4(B,N_5(A,U_6))),H\big),\\
T_3 &= N_0\big(U_1,N_3(N_5(U_4,U_6),U_2)\big),\\
T_4 &= N_0(U_1,U_2).
\end{aligned}$$

Table 9 supplies one shared physical order for each entire tree, including both answer branches. Unused coordinates are permanently skipped.

Tree	Shared order of physical coordinates
T_0	$(4,0,1,3,2,5,6)$
T_1	$(3,0,1,4,5,2,6)$
T_2	$(3,0,1,4,2,5,6)$
T_3	$(0,1,3,2,5,4,6)$
T_4	$(0,1,2,3,4,5,6)$

Table 9 Legal global orders for the explicit mixed-policy upper certificates. They are witnesses, not a restriction of the lower-bound comparison class.

For the first branch, mix T_0,T_1,T_2,T_3 with probabilities

$$\frac{1}{\varepsilon+11}\big(2+2\varepsilon,\ 3+\varepsilon,\ 4,\ 2-2\varepsilon\big).$$

For every world, its expected-cost numerator is at most $\varepsilon^2+13\varepsilon+46$. The possible slacks are

$$\begin{gathered}
0,\quad 7+\varepsilon,\quad 14+2\varepsilon,\quad 4+2\varepsilon-2\varepsilon^2,\\
2-2\varepsilon^2,\quad 2+2\varepsilon^2,\quad 9+5\varepsilon+2\varepsilon^2.
\end{gathered}$$

They are nonnegative on $[0,1/2]$, and some worlds attain the target. For the second branch, mix T_0,T_1,T_2,T_4 with probabilities

$$\frac{1}{4-\varepsilon}(1,1,1,1-\varepsilon).$$

The target numerator is $17-2\varepsilon$; the possible slacks are $0,2,4,2-\varepsilon,1-\varepsilon,\varepsilon,3+\varepsilon$. Thus this is a legal upper bound on $[\eta,2/5]$.

A probe-free upper bound $9/2$, valid for every ε, mixes T_4 with weight $1/2$ and

$$S_1 = N_4(N_0(U_1,B),U_2),\qquad S_2 = N_4(N_1(B,V_0),U_2)$$

with weights $1/4$ each. Shared orders for S_1,S_2 are $(3,4,0,1,2,5,6)$ and $(3,4,1,0,2,5,6)$; both skip coordinate 3 without purchasing it. Direct evaluation of the sixteen worlds proves the bound. An adaptive witness buys q_3, with zero child $N_0(U_1,N_5(U_4,U_6))$ and one child

$N_4(N_6(A, U_5), U_2)$. Its worst cost is at most $4 + \varepsilon$. These supply all upper bounds needed below.

Full-class lower certificates. For any prior, every correct deterministic policy has average cost at least its exact Bayes optimum. The same lower bound holds for any input-independent mixture, and worst-input expectation is at least prior average. Correctness and leaf purity always retain all sixteen worlds, including those of prior mass zero.

One constant prior puts masses $(6, 1, 1, 4)/12$ on worlds $(0, 4, 5, 15)$. Its adaptive root cost is

$$\min\{54, 51 + 6\varepsilon, 48 + 12\varepsilon\}/12 = \min\{9/2, 4 + \varepsilon\}.$$

For the first priority branch, put prior numerators

$$(6, 1 - \varepsilon, 1 - \varepsilon, 1 - \varepsilon, 2 + 4\varepsilon)$$

on worlds $(0, 4, 5, 13, 15)$, with denominator $11 + \varepsilon$. The complete priority cost frontier on $[0, 1/2]$ reduces to

$$\varepsilon^2 + 13\varepsilon + 46, \qquad 49 + 5\varepsilon.$$

Their difference, second minus first, is $3 - 8\varepsilon - \varepsilon^2$, nonnegative on $[0, \eta]$. For the second branch, use the same five worlds with numerators

$$6(1 - \varepsilon), \quad (1 - \varepsilon)^2, \quad (1 - \varepsilon)^2,$$
$$(1 - \varepsilon)^2, \quad -1 + 10\varepsilon - 3\varepsilon^2$$

and denominator $2(4 - \varepsilon)$. The retained root polynomials are

$$31 + 4\varepsilon + \varepsilon^2, \qquad 34 - 4\varepsilon, \qquad 38 - 14\varepsilon.$$

Relative to $34 - 4\varepsilon$, their differences are $\varepsilon^2 + 8\varepsilon - 3$, 0 and $4 - 10\varepsilon$. All are nonnegative on $[\eta, 2/5]$. The priors normalize and are nonnegative on their stated intervals. They give the matching lower bounds.

Finite premises and their coverage. For each fixed physical permutation, a state consists of its remaining position and the feasible world set. A pure state costs zero; an exhausted mixed state is infeasible. Each other state compares permanently skipping the current query with purchasing it, adding prior mass times price and both response continuations under that same remaining order. Taking the minimum over all 5040 permutations covers every causal-skip priority policy. Adaptive recursion instead chooses the next query at each feasible state. Constant-answer and post-purity purchases may be omitted because prices are nonnegative.

The computations retain polynomial costs, pruning only by a domination relation valid on the entire declared parameter interval. Addition of common fees and independent response continuations preserves that relation. The first variable prior produces degree at most two. The second produces degree at most three; the cubic terms happen to cancel in the final three root witnesses. A separately implemented world-tuple recursion rebuilds every order, using different sufficient coefficient-dominance tests. Each per-order verification covers 1,763,124 state instances over all orders. Polynomial nonnegativity is checked exactly at endpoints and internal derivative critical points, with quadratic algebraic endpoints where necessary. These are finite proof premises and exact real-interval sign checks, not a sampled price grid or a new general optimization method.

Finally, at $2/5$ the matching value is $9/2$. Every fixed policy's cost is nondecreasing in ε, so minimization of worst-input expectation is also nondecreasing. The probe-free upper bound then proves the entire constant branch. Writing the first two formulas as g, h,

$$g - h = \frac{(1 - \varepsilon)(\varepsilon^2 + 8\varepsilon - 3)}{(\varepsilon + 11)(4 - \varepsilon)}.$$

Thus η is an actual branch change, not just the failure of a selected prior. The formulas agree at their shared endpoints. Subtracting (9) proves the gap assertion. □

The curved pieces satisfy $g = \varepsilon + 2 + 24/(\varepsilon + 11)$ and $h = 2 + 9/(4 - \varepsilon)$, so each has positive second derivative. Their first derivatives jump down at η, and the derivative drops to zero at $2/5$. Hence this explicit value curve is neither globally convex nor globally concave. This is a consequence of the displayed formulas, not a new general parametric-game principle. The result concerns one price family on one source. It does not determine a randomized minimum within additive or submodular fees, an all-additive-price ratio, or practical academic-evaluation gains.

P A controlled recorded-identity realization with byte costs

We separate a controlled realization of the query model from a natural academic task. A record here points to a locally recorded invocation identifier. Equality means the same recorded source invocation. Distinct identifiers do not establish statistical independence of research results, and the structural labels A, B do not denote paper truth or quality.

Proposition 24 (Controlled byte-cost realization) *The seven-query source admits a controlled recorded-identity realization with a trusted paid-bit interface and positive additive query-phase byte costs*

$$L = (1535, 1535, 2303, 255, 767, 767, 767).$$

In the valid-enrollment model, one executed adaptive tree has worst-input cost 3325 *bytes, while every correct privately randomized global-order policy has worst-input expected cost at least* 57993/17 *bytes. Consequently,*

$$R_{\mathcal{P}}^{\text{byte}} - R_{\mathcal{A}}^{\text{byte}} \geq 1468/17 > 0.$$

The cost is newly returned explicit query-payload read buffers. No device-I/O, latency, dollar, unrestricted end-to-end or natural-task advantage is asserted.

Proof Source construction. Give each block of a valid partition of the fourteen records a distinct registered invocation tag, and point every record in that block to the same tag. At most fourteen tags suffice. Conversely, the kernel of a record-to-tag map is an equivalence relation. A validator imposing only the original SAME blocks and DIFFERENT edges therefore admits exactly the complete source. The general registry and encoder allow up to fourteen roots; they do not impose a two-class promise. The exact sixteen-signature projection follows from Section 3.3. Two tags suffice for representative witnesses, without replacing that complete-source proof by sampling.

The controlled registry uses a unique fixed-length tag per invocation, echoed by a child together with its PID. A trusted parent retains the real process handle and records the actual exit and byte bindings. A query file contains only the two endpoint tags in JSON, followed by whitespace padding to L_q. The encoder has no label argument. The paid interface checks its fixed byte binding and returns only the equality bit. A fresh policy instance receives the same cost metadata for every possible world; it receives no root tags, world index, filenames, content hashes, harness order or unpaid preflight answers. This is an explicit trusted observation contract, not an OS sandbox or a historical author authenticator. Missing, malformed or

changed enrollment would invalidate the source instance and remain unresolved, rather than becoming an opposite structural label.

Actual cost and full-class comparison. At $\varepsilon = 1/3$, multiply the reference prices by three to obtain

$$c = (6, 6, 9, 1, 3, 3, 3), \qquad R_{\mathcal{A}} = 13, \quad R_{\mathcal{P}} = 227/17.$$

Choose $B = 256$ and set $L_q = Bc_q - 1$. A valid fixed-size file read with cap $L_q + 1$ returns exactly L_q bytes. Different queries have different physical files, and a repeated coordinate conveys no new information. For any deterministic policy and world,

$$C_{\text{byte}}(T, w) = BC_c(T, w) - N_T(w), \qquad 0 \le N_T(w) \le 7.$$

The inequalities $BC_c - 7 \le C_{\text{byte}} \le BC_c$ survive private expectation, worst-input maximization and minimization over either full policy class. The preceding theorem therefore gives

$$R_{\mathcal{P}}^{\text{byte}} \ge 256(227/17) - 7 = 57993/17.$$

The fixed adaptive tree was executed on every signature; its largest actual path cost is 3325. Since the valid bit projection is complete and file lengths are fixed, this is an upper bound over the entire source, not just the particular root names used in the capture. It proves the stated difference. The new byte-price optima are not solved exactly. □

Execution and independent checking.

The source map and cost argument were frozen and checked before starting the identity children. Exactly two trivial local identity invocations were captured, both with retained-handle exit zero; neither trained a model or reproduced an author's experiment. Their identifiers generated sixteen signature representatives and 112 fixed-size payload files. One adaptive tree and all four components of the fixed reference priority mixture were actually executed on every signature, giving 80 paths and 259 query-payload reads. Input-independent weights $4/17, 5/17, 6/17, 2/17$ give the exact expected costs in Table 10. No policies were selected or reoptimized after observing these traces.

Quantity in the controlled query phase	Bytes
Executed adaptive tree, worst-input upper bound	3325
Executed reference priority mixture, feasible upper bound	58063/17
All randomized priority policies, analytical lower bound	57993/17
Optimal priority minus optimal adaptive, lower bound	1468/17

Table 10 Measured feasible bounds and a transferred full-class lower bound are different evidence types. Neither feasible policy is asserted optimal at the new byte prices.

A separate program imports none of the source producer, encoder, paid oracle or policy interpreter. It checks the actual root receipts and stdout hashes, all 112 raw

files, their root assignments, source constraints and signatures, then reconstructs all 80 policy paths, public observations and byte ledgers. It reuses the pinned complete 5040-order reference certificate and checks the cost perturbation deduction. This combines an analytical complete-source argument with actual representative executions; it is not a proof-assistant kernel, human review or exhaustive execution of every partition/root count.

Common costs and limits.

Payload creation wrote 126864 bytes and common enrollment read 126864 bytes. These operations are shared preparation, not free work or information supplied to one policy. Evaluating all mixture components consumed 256509 query bytes in total; that study cost is distinct from executing one component drawn from the deployed private mixture. Code/import/metadata readsets, OS caches and device transfers are outside the query-buffer metric. Root differences certify only the trusted local invocation identity relation. The two roots and sixteen representative signatures are one engineered fixture, not independent natural studies.

The whitespace lengths were deliberately chosen to realize the reference fee family up to a bounded perturbation. Thus this proposition validates a controlled model implementation, not a natural pricing distribution or academic-evaluation benefit. It does not overturn the separate native equal-value and fee-transfer obstructions, the frozen model-study negatives, or the absence of human gold and historical paper authentication.

Evidence and manuscript status.

The exact curve and controlled realization have separately frozen local proof objects, actual process receipts and independent checks. Their private evidence ledger distinguishes prior results reused from new executions. They are not contained in the earlier public review archive, and no new public deposit or venue readiness is claimed. Existing numbered statements and proof bodies remain unchanged. Bibliographic priority, natural application advantage, author facts and final author approval remain unresolved or deferred.

Q Additional prior-work positioning

Charikar et al. [1] compare deterministic priced queries with an input-specific cheapest certificate. Such guarantees can bound minimax cost, but require a strategy in the allowed class and a suitable cost comparison. Section A.2 makes this connection explicit; the two objectives cannot be separated merely by their denominators. Section A.3 gives a concrete fixed-source formula-cost comparison.

Positive-rational additive instances fit discrete function evaluation [2], whose logarithmic approximation concerns adaptive trees. Ordered diagrams permit omitted variables; maximum-path objectives [21, 22], compatible completions [23, 24], and average-path cofactoring [25] are established. Greiner et al. [3] separate expected-cost AND/OR strategies under fixed permutations with causal skips. Section A.1 identifies our comparator.

Graphic-flat and multicut representations [12, 13] are standard. Transferring cut-edge cost results or inconsistent 2-CNF clause-search bounds [14] requires preserving our visible menu, purchased-set costs and classification objective; complete fibres need not be 2-CNF.

Partition learning [26, 27] reconstructs the whole partition under different class-count and round assumptions. Matroidal query bounds [28] concern full-cube independence recognition. Neither task is our ordered-skip classification problem.

For independent box values, Berger et al. [29] give an optimal fixed exploration order for expected maximum reward minus submodular set cost. Our worst-cost zero-error relational classification needs a separate argument; no reduction is asserted.

Completion complexity, defined by minimizing a measure over compatible total functions, is established context. Huffstutler et al. [15] distinguish promise-valid from domain-certifying certificates and record preservation of deterministic unit-query complexity under an optimal completion. Our purchased-set completion identities are supporting tree transformations, not a new completion framework.

Appendix M distinguishes three four-query phenomena on complete sources: randomization benefit, a promise/common- completion discrepancy, and adaptive/global-order separation. The first two have a positive-additive witness. For the third, the earlier near-additive witnesses remain nonsubmodular, while Appendix V now supplies a distinct strictly increasing coverage witness and the exact coverage/submodular minimum. Appendix N gives the broader partial-task three-query lower bound.

Worst-input private randomness and restricted-order separations are established [4]. A directional formula strategy chooses a recursive child order and completes a child before entering another. Its deterministic seeds compile to global leaf orders with short-circuit skips. Our full priority comparator may also interleave unfinished children: on $(x_0 \wedge x_1) \vee (x_2 \wedge x_3)$, order $(0, 2, 1, 3)$ has query prefix $(0, 2, 1)$ on the all-one input. A directional lower bound need not bound this larger class. The comparison uses the original paper's public full-text copy; the publisher's bytes were not authenticated in the bounded audit. Permutation-prefix policies [6] are also inside our comparator; Greiner et al.'s linear strategies already permit irrelevant-test skips [3]. Appendix N.4 translates concrete witnesses and retains valid fixed-prior lower bounds. Further prior context includes Las Vegas ordered diagrams [19], consistent partial-table diagrams [20], and nonadditive batch testing [9].

Specified-distribution objectives need a separate quantifier comparison. Hellerstein et al. [6] study expected additive cost under independent inputs. Nielsen et al. [7] give a tight stochastic gap of two for k-of-n evaluation with fixed input probabilities; the inspected full version is v2. A fixed-prior gap does not alone establish a worst-input gap between two separately minimized policy classes. Correlated stochastic certification is also established [8]; its specified correlated distribution and one-sided active-set certificate objective differ from two complete equivalence fibres. These comparisons do not rule out all model-preserving reductions. Neither correlation nor generic adaptivity, minimax or threshold techniques are claimed as new.

Consistent-output and completion phenomena also occur in search and communication complexity [16–18]. The formal journal version of Chattopadhyay et al. distinguishes query depth from decision-tree size; its Theorem 5.5 concerns size of compatible

completions. Transferring such a result requires a cost- and comparator-preserving argument. Complete-source expressivity exclusions do not establish distinctness of our new generic lower bound. We claim no new generic minimax or completion framework; exact literature priority remains under review.

R Applicability and evidence context

The deterministic boundary concerns menu size; ratios concern the specified seven-query source and eleven-query extension. The randomized additive minimum lies between five and seven, while the monotone, submodular and coverage minima are four. The new pair-support minimum of two is over complete four-query sources with pair-activation fees; the one-pair exclusion also covers smaller menus. It is not a bound for larger menus or larger shared groups. We assert no polynomial-time synthesis or practical acquisition advantage. Academic-evidence use requires validated executable queries, a defensible complete promise and same-information reads, joins, batching and caching. No ranking, semantic-judgement or historical-authentication gain is established.

The acquisition unit matters even on this fixed source. Appendix I replaces paid pair answers by paid record reads that reveal every equivalence relation among acquired records. Splitting each price in $(2, 2, 3, 1/2, 1, 1, 1)$ equally between its two endpoints preserves the cost of acquiring that isolated pair, but the new adaptive and priority optima are both $15/4$. Thus the pair-menu gap does not survive this particular change of information and cost contract. This finite control supplies no measured file-system or scientific-workflow benefit. For a separate even-length alternating source family, Theorem 12 gives a priority normal form for every monotone purchased-set cost. The appendix also gives an exact additive-cost construction. These supporting model comparisons use classical path-search ideas and do not strengthen the six/seven pair-query boundary.

Appendix L reports a prospective Wine execution and an offline audit of deposited VPS3 artifacts. These cases distinguish observed execution, numerical compatibility and historical provenance. A same-information fixed verifier also accepts the Wine receipt, and ordinary fixed retrieval and arithmetic suffice for VPS3. Neither case establishes a complete-fibre promise for a natural task or an adaptive acquisition advantage.

Appendix H separates the complete six-query computational layers and their analytic premises from the fixed-seven supplement. The latter rebuilds $5,040$ order frontiers and checks 13 trees plus both 864-certificate families; coverage checks supplement the general submodular proof. The accompanying versioned ledger records the completed original V03 aggregate run and the narrower acceptance of the later frozen candidate. The generic randomized lower bound has a separate finite proof covering all 6050 nonconstant three-bit partial tasks. Appendix N gives its analytical reductions, the disjoint $4498 + 1344 + 208$ coverage, and the scope of an actual relocated replay. The 208 failures of a stronger certificate-cost conjecture are retained. This extension leaves the deterministic six/seven classification and the earlier physical-source claims

unchanged. Section 5 separately adds the all-real-price complete four-query additive theorem, with a portable full-family replay. Neither result establishes a natural application advantage.

S Additional availability and version ledger

The complete boundary evidence and fixed-seven supplement are available in the public review archive. Appendix H identifies the fixed delivery record, version correspondence and verification scope. The separate Wine and VPS3 packages are public through the versioned records in Appendix L; executed source commits and later archive commits are distinguished. The fourteen-record control has a separate public record-acquisition supplement. The full-eleven direct-proof packet has a separate public source record and a completed Linux replay, with retained hosted receipts; these are separate from the frozen original review archive. The certificate-transfer supplement has a separate public source packet and retained Linux replay receipts. They are not part of the frozen original review archive; Appendix K specifies their mathematical and execution scope. Appendix J states its source, direct certificate obligations and execution scope. The optional even-path calibration and additive-solver implementation are prepared separately; they are not included in the frozen review archive. The even-path theorem has an analytic proof independent of this calibration. The randomized small-menu extension in Appendix M has a separate public proof-object and replay packet, with retained original Linux receipts. These are later deposits, not members of the frozen original review archive. The public derivative preserves the six finite-proof inputs and mathematical checker core; it omits the path-bearing old four-query files and identifies that inherited analytical leg separately. The current manuscript evidence ledger binds these versions. Public access does not grant a blanket project licence. The responsible authors must finalize applicable distribution terms and author declarations before submission. Source-specific notices accompany the materials; a deposited dataset's licence is not a blanket repository licence. The generic randomized boundary additionally has a private standard-library proof packet, `generic_randomized_minimum4_v01.zip`. It was extracted and executed locally at another path; Appendix N.5 identifies that replay's scope. It is separate from the public packets above. The four-query additive packet `PORTABLE_PROOF_V01.zip` was subsequently published with retained Linux execution evidence (Appendix T). It includes the complete source/family/continuous-price proof obligations and analytic texts, with an actual extracted relocation replay as described in Section 5. The accepted V22 reading and coverage proof are public in the versioned V22 deposit. The V24 reading manuscript and unrestricted global pair-support proof were subsequently published in the versioned V24 reading deposit and its accompanying global pair-support proof deposit. Earlier descriptions of private packets in the unchanged proof text refer to their construction stage; these outer records distinguish later publication. This material V25 derivative is private and adds the GS minimum-three theorem and relative robustness after an actual combined relocation replay. The GS packet has not been

publicly deposited or executed by the earlier hosted runs. Its new evidence ledger identifies inherited and freshly checked modules. No human author approval or submission is implied.

T Four-query all-additive reproduction and limits

The self-contained packet `PORTABLE_PROOF_V01.zip` contains 52 files (1304852 compressed bytes), 25 compressed exact scientific projections and 16 analytic proof texts. It regenerates the complete source/policy cover and checks the continuous-price obligations, including the positive-price seven-query dual. Two Windows executions, one after extraction at a different directory, produced identical mathematical objects.

The exact packet was subsequently published at source commit `8374985`. One manual Linux execution, run 39003766001, attempt one, exited zero under Python 3.12.3. The entire deterministic scientific object matched Windows; no scientific field was excluded. The service's original output ZIP matched its provider-reported digest and is retained at the later archive commit `468d5a4`. That archive commit was not the executed source. These are cross-system executions of the same implementation, not independent implementations or human review.

All projections were compared with 95 frozen original files using compressed and decoded hashes. An original provenance reader omitted the already frozen packet specification from its allowable origins and failed; a versioned index-only repair passed without changing the ZIP. The older count transcription from 26 to 29 is separately retained. Neither repair changed prices, priors, mixtures, policy families or inequalities.

Extract the packet and, from its root, run `python -B verify.py --out ../result.json`, choosing a new result path outside the extracted directory. Assertions must remain enabled. No solver, model API, key, network or original research directory is required. The restricted file-open check allows packet files, Python runtime imports and the new result file; it is not an operating-system sandbox. The analytic case exhaustion and generator arguments remain mathematical premises supplied as proof texts. Their exact algebraic checks are not formal proof-kernel certification. Specific literature priority and natural academic-evaluation advantage remain unestablished.

U A total-function completion obstruction and AND–OR comparison

These supporting results fix the physical queries and unit prices. We prove an exact minimum over every compatible total Boolean function on one complete source, then examine a specific transfer of Ito–Suzuki's total-tree result [5]. Other reductions and changes of menu or cost remain outside these claims.

U.1 A complete source and a compatible total function

Use eight records and the four indexed queries $q_i = (2i, 2i{+}1)$, $i = 0, 1, 2, 3$. Under label zero impose the SAME partition $(0, 1, 0, 1, 0, 2, 0, 3)$ and $\mathsf{DIFFERENT}(5, 7)$. Under label one impose the SAME partition $(0, 0, 0, 1, 0, 1, 0, 1)$ and no DIFFERENT constraint.

Retain every satisfying equivalence partition. Encode a word by $w = \sum_{i=0}^{3} 2^i x_i$. The full answer supports are

$$W_0 = \{0, 3, 4, 7, 8, 11\}, \qquad W_1 = \{1, 15\}.$$

Indeed, the first prior gives $x_0 = x_1$ and forbids $x_2 = x_3 = 1$; the second gives $x_0 = 1$ and $x_1 = x_2 = x_3$. Assigning each remaining block a distinct fresh class proves sufficiency. This is the existing complete eight-input source in the four-query proof.

The total function

$$g(x) = (x_0 \wedge \neg x_1) \vee (x_2 \wedge x_3)$$

is compatible. In variables $(x_0, 1-x_1, x_2, x_3)$ this is a positive AND–OR tree. Negation decodes the same paid query; it is not a physical symmetry in the source catalogue.

U.2 The forced hard-product distribution leaves the promise

Let S be uniform on the eight original words, and let L, R be the two AND child values of g. Their joint law is

$$\delta(0,0) = \frac{3}{4}, \quad \delta(1,0) = \delta(0,1) = \frac{1}{8}, \quad \delta(1,1) = 0.$$

For a unit-cost AND of two literals, a distribution conditional on output zero assigns probabilities p, q to the two single-zero assignments and r to the double-zero assignment. The two directional orders cost $1 + q$ and $1 + p$, so

$$\min(1 + p, 1 + q) \le 1 + \frac{p+q}{2} = \frac{3}{2} - \frac{r}{2} \le \frac{3}{2}.$$

The unique least-favourable distribution has $r = 0$, $p = q = 1/2$; conditional on output one there is only one assignment. Thus the hard child laws required in the cited Lemma 4.3 are forced in this example: the left zero law is uniform on actual bits $00, 11$, and the right zero law is uniform on $10, 01$. Their one laws are 10 and 11, respectively.

Combining these laws independently conditional on the child values, as in the cited Definition 4.1, gives the following positive masses:

$$S^{\delta}(w) = \begin{cases} 3/16, & w \in \{4, 7, 8, 11\}, \\ 1/16, & w \in \{5, 9, 12, 15\}. \end{cases}$$

Words $5, 9, 12 \notin W_0 \cup W_1$ carry exactly $3/16$ mass outside the original promise. Uniqueness forces this escape for the specified completion, δ and hard-child requirement. The product remains legal on the cited paper's full cube; its preservation of this relational promise would be an extra, false premise.

U.3 The values of this promise and this completion differ

At unit prices, the existing four-query certificate specializes to $R_{\mathcal{A}} = R_{\mathcal{P}} = 5/2$ on this source. This value also has a short direct proof. Opposite-label single-bit neighbours

force queries $0, 1$ on word 1 and $2, 3$ on word 15. The first query of any correct tree is therefore outside the mandatory pair of one of these words. Its cost is at least three there and at least two on the other word. The prior uniform on $1, 15$ gives lower $5/2$, including for mixtures. Zero-prior inputs remain required for correctness.

For the upper bound let $N_i(L, R)$ query i and take L, R on answers zero and one, respectively. Mix the two legal priority trees

$$T_1 = N_1\big(N_0(0,1), N_2(0, N_3(0,1))\big),$$
$$T_2 = N_2\big(N_1(N_0(0,1), 0), N_3(0,1)\big)$$

equally, with orders $(1, 0, 2, 3)$ and $(2, 1, 0, 3)$. On words $(0, 1, 3, 4, 7, 8, 11, 15)$ their costs are $(2, 2, 2, 2, 3, 2, 2, 3)$ and $(3, 3, 2, 2, 2, 3, 2, 2)$, so every expected cost is at most $5/2$. These are the existing policies P_5, P_{14} on a certified tie face; no new minimax solve is used.

For the specified total g, both full-cube values equal 3. For the lower bound give words $4, 7, 8, 11$ equal probability. Independently in each child, the unique zero literal has a uniformly random position. Full-cube correctness requires finding a zero in each AND. The first query in either child costs one, and with probability $1/2$ forces its second query, even after observations in the other independent child. Hence every adaptive algorithm has expected cost at least 3. The eight directional seeds uniformly order the root's children and each child's leaves. When both children are false, each costs at most $3/2$ in expectation; when both are true, cost is 2. When exactly one is true, the uniform root order gives at most $(2 + (3/2 + 2))/2 = 11/4$. Thus the full-cube priority upper bound is 3. Restriction gives a valid but non-sharp promise upper bound. The next subsection proves that the full-cube minimum remains 3 over every compatible total function on this same menu, including nonunate ones.

U.4 The minimum over all compatible total functions

Let $W = W_0 \cup W_1$ be this eight-word promise. Write $\mathrm{Comp}(W)$ for all compatible total Boolean functions on the same four coordinates. For $f \in \mathrm{Comp}(W)$, $R_{\mathcal{C}}(f; \mathbf{1})$ requires correctness on the full cube; every seed computes the same f. The notation $R_{\mathcal{C}}(W; \mathbf{1})$ charges only the original promise.

Proposition 25 (Exact unit-cost common-completion minimum) *On this complete source, with unit query prices and zero-error input-independent private randomness,*

$$\min_{f \in \mathrm{Comp}(W)} R_{\mathcal{A}}(f; \mathbf{1}) = \min_{f \in \mathrm{Comp}(W)} R_{\mathcal{P}}(f; \mathbf{1}) = 3,$$
$$R_{\mathcal{A}}(W; \mathbf{1}) = R_{\mathcal{P}}(W; \mathbf{1}) = \frac{5}{2}.$$

The minima include nonunate functions. Thus the common-total-completion minimum is exactly $6/5$ *times the promise value on this fixed source.*

Proof The promise value and the full-cube upper bound 3 for the displayed g were proved above. We prove that every compatible total f has $R_{\mathcal{A}}(f; \mathbf{1}) \geq 3$.

Let $w^{(q)}$ denote word w with coordinate q toggled, and define $s_f(w) = |\{q : f(w) \neq f(w^{(q)})\}|$. If these labels differ, any deterministic tree computing f must query q on w: otherwise both inputs have the same purchased transcript and output. Therefore every zero-error seed costs at least $s_f(w)$ on that same input, and

$$R_{\mathcal{A}}(f; \mathbf{1}) \geq \max_w s_f(w).$$

For an arbitrary seed probability space, discard the finite union of erroneous null seed sets over all 16 inputs first. This preserves expected costs. The sensitivity lower bound is standard decision-tree reasoning; no new general lower-bound method is proposed.

Suppose all sensitivities are at most two. The eight original labels then force every other value by the following six implications.

1. Word 1 has value one and its neighbours $0, 3$ have value zero. Its other neighbours $5, 9$ must have value one.
2. Word 15 has value one and neighbours $7, 11$ have value zero. Its other neighbours $13, 14$ must have value one.
3. Word 7 has value zero and neighbours $5, 15$ now have value one. Its neighbour 6 must have value zero.
4. Word 11 has value zero and neighbours $9, 15$ now have value one. Its neighbour 10 must have value zero.
5. The four neighbours $0, 3, 6, 10$ of word 2 all have value zero. Value one at 2 would give sensitivity four, so $f(2) = 0$.
6. Word 14 now has value one and neighbours $6, 10$ have value zero. Its neighbour 12 must have value one.

All 16 values are fixed. Their one-set is $\{1, 5, 9, 12, 13, 14, 15\}$, exactly the one-set of g. Thus every compatible $f \neq g$ has an input with at least three mandatory queries and hence randomized value at least three. For g itself, the preceding independent-child hard prior gives lower three even though its maximum sensitivity is only two. The legal priority upper three for g and $R_{\mathcal{A}} \leq R_{\mathcal{P}}$ complete both minima. □

This excludes a value-preserving common-total-function route on the unchanged menu at these prices. Promise-correct seed trees may have different off-promise completions; their value $5/2$ is therefore consistent with the common-completion minimum 3. The off-promise words used in the proof are correctness obligations for f, not added original source inputs. The result gives no unequal-price statement, general $6/5$ bound, or change to the five-to-seven randomized additive adaptive/priority interval. Generic completion phenomena and methods remain established context; exact literature priority is not certified.

U.5 Depth-first and global-order sets are incomparable

For $h = (x_0 \wedge x_1) \vee (x_2 \wedge x_3)$, global order $(0, 2, 1, 3)$ is a legal priority policy with short-circuit skips. Its prefix $(0, 2, 1)$ on the all-one input interleaves unfinished children, violating depth-firstness.

Conversely, query 0 first. If it is zero, evaluate the other child in order $(2, 3)$. If it is one, query 1; on a zero answer evaluate the other child in order $(3, 2)$, and otherwise stop with one. Short-circuit each AND. This policy is depth-first and correct on all 16

words. At words $0, 4$ its paid masks are $5, 13$: purchasing 3 depends on the answer at 2, and these words differ only at 2. Any causal priority policy with this entire paid-set vector must put 2 before 3. At words $1, 9$ the masks are $11, 15$, forcing 3 before 2 by the same argument. No global order has this vector. This statement concerns a specific policy vector; it does not assert unequal class optima for h.

Incomparability does not contradict the valid unit-cost total-domain chain

$$R_{\mathcal{A}} \leq R_{\mathcal{P}} \leq R_{\mathrm{dir}} = R_{\mathrm{DF}},$$

because the last comparison uses the smaller directional embedding and the cited value theorem, rather than a depth-first/priority inclusion. A sufficient transfer route to a promise W would supply a compatible total tree T on the same menu/prices, a valid directional/depth-first equality at those prices, and $R_{\mathrm{DF}}(T) \leq R_{\mathcal{A}}(W)$. The resulting sandwich proves equality on W. On our unit-cost source, Proposition 25 rules out that sharp route through any compatible total tree on the unchanged menu. Other reductions and changes of the information or cost contract remain open.

Verification scope.

Two implementations reconstruct the complete source and check all 16 product masses, the policy witnesses and the precedence cycle. The analytic arguments establish the hard-child uniqueness and lower bounds. The six fixed sensitivity implications were additionally checked without enumerating or optimizing completions. Their universal-policy lower bound is the written proof, rather than a conclusion from fixtures. These checks establish neither human review, formal certification, literature priority nor natural application gain.

V The randomized coverage and submodular boundary

This appendix strengthens the four-query upper witness while preserving the complete source, the information actions and the full causal-skip priority comparator. It concerns the worst-input expectation of the *final purchased-set fee*, not an expected sum of batch charges.

Theorem 26 (Four-query coverage/submodular boundary, computer-assisted) *In the complete equivalence-partition fibre model, the smallest indexed pair-query menu permitting $R_{\mathcal{A}} < R_{\mathcal{P}}$ is four for normalized nonnegative monotone submodular fees, and for nonnegative weighted coverage fees. The same minima hold with positive nonempty fees and strict increase under every proper inclusion. A witness already has strictly positive individual query charges and strictly positive pairwise shared-setup charges, without a global setup charge. It is strictly submodular on every incomparable pair of query sets. Its exact values are*

$$R_{\mathcal{A}} = \frac{125}{7}, \qquad R_{\mathcal{P}} = \frac{161}{9}, \qquad R_{\mathcal{P}} - R_{\mathcal{A}} = \frac{2}{63}.$$

Proof Theorem 18 excludes a gap on complete sources of at most three queries for every normalized nonnegative monotone fee. Each fee class in the statement is contained in that class. We give a matching complete four-query source and exact upper and lower certificates.

Complete source.

Use records $0, \dots, 7$ and queries $q_i = (2i, 2i+1)$, $0 \le i < 4$, with integer words $w = \sum_i 2^i x_i$. Under label zero, the required SAME blocks are $\{0,1,2,4,6\}, \{3,5\}, \{7\}$, with records $3, 7$ required DIFFERENT. Under label one they are $\{0,2,4,6\}, \{1,3,7\}, \{5\}$, with no DIFFERENT requirement. In each hypothesis retain every satisfying partition. Such a partition is a coarsening of its three prior blocks. The five possible coarsenings leave three legal negative partitions and five positive ones. Their distinct projections are

$$W_0 = \{1, 7, 9\}, \qquad W_1 = \{0, 4, 11, 15\}.$$

These are the same source as in Appendix M.5, not selected subsets of its worlds. Write $\mathcal{W} = (0, 1, 4, 7, 9, 11, 15)$ for input order.

Positive charges and pairwise sharing.

For $S \subseteq \{0,1,2,3\}$, let

$$C(S) = \sum_{i \in S} c_i + \sum_{i<j} w_{ij} \mathbf{1}\{S \cap \{i,j\} \neq \varnothing\}, \tag{11}$$

where

$$\begin{aligned}(c_0, c_1, c_2, c_3) &= (3, 1, 1, 1),\\ (w_{01}, w_{02}, w_{03}) &= (2, 1, 2),\\ (w_{12}, w_{13}, w_{23}) &= (4, 4, 4).\end{aligned}$$

A shared setup is charged once when either associated query is bought; activating it reveals no additional answer. This leaves exactly the four original information actions for both policy classes. Every summand is a positive weighted coverage indicator. Consequently C is normalized, nonnegative, monotone and submodular. Its least nonempty fee is 8 and its least proper single-query marginal is 1. For incomparable S, T,

$$C(S) + C(T) - C(S \cup T) - C(S \cap T) = \sum_{\substack{i \in S \setminus T \\ j \in T \setminus S}} w_{ij} \ge 1. \tag{12}$$

Here $w_{ij} = w_{ji}$ when necessary. Each set difference is nonempty and every pair weight is positive. Comparable sets necessarily give equality. Thus the asserted strictness excludes only those identities. In mask order the fee values are

$$\begin{aligned}(C(0), \dots, C(7)) &= (0, 8, 11, 17, 10, 17, 17, 22),\\ (C(8), \dots, C(15)) &= (11, 17, 18, 22, 17, 22, 20, 23).\end{aligned}$$

All-input upper certificates.

Write $T(q, u, v)$ for buying q and following u on zero and v on one; constants denote labels. Set $u = T(0,1,0)$, $v = T(1,1,0)$, $z = T(3,u,0)$. Use

$$\begin{aligned}a_1 &= T(1, u, T(3,0,1)),\\ a_2 &= T(3, u, T(1,0,1)),\\ a_3 &= T(2, T(1,z,1), T(3,v,1)),\end{aligned}$$

$$p_3 = T(2, T(1, u, 1), T(3, v, 1)),$$
$$p_4 = T(2, T(1, z, 1), T(3, u, 1)).$$

Their paid-set masks on $\mathcal{W}$ are respectively

$$a_1 : (3, 3, 3, 10, 3, 10, 10),$$
$$a_2 : (9, 9, 9, 9, 10, 10, 10),$$
$$a_3 : (15, 15, 14, 14, 14, 6, 12),$$
$$p_3 : (7, 7, 14, 14, 7, 6, 12),$$
$$p_4 : (15, 15, 13, 13, 14, 6, 12).$$

All five trees are correct on all seven inputs. The four trees a_1, a_2, p_3, p_4 admit global orders $(1, 0, 2, 3)$, $(2, 3, 0, 1)$, $(2, 3, 1, 0)$ and $(2, 1, 3, 0)$, respectively, with causal skips. Mixing a_1, a_2, a_3 with weights $(3/7, 3/7, 1/7)$ gives input costs

$$(125/7, 125/7, 122/7, 125/7, 125/7, 125/7, 125/7).$$

Mixing a_1, a_2, p_3, p_4 with $(4/9, 4/9, 1/18, 1/18)$ gives

$$(317/18, 317/18, 157/9, 161/9, 161/9, 161/9, 161/9).$$

These prove $R_{\mathcal{A}} \leq 125/7$ and $R_{\mathcal{P}} \leq 161/9$. The priority mixture is an achievable witness, not a restriction of the comparison class to four templates.

Lower certificates for the full classes.

Use priors on $(7, 9, 11, 15)$

$$\pi_A = (1/7, 1/7, 5/14, 5/14), \qquad \pi_P = (1/9, 1/9, 7/18, 7/18).$$

The other three inputs remain correctness obligations with prior zero. Every correct deterministic adaptive tree has π_A mean at least $125/7$, and every correct global-order priority tree has π_P mean at least $161/9$. The complete retained families have 40 adaptive and 32 priority paid-set profiles; pointwise dominated profiles may remain. Their completeness and pointwise paid-set domination are inherited from the source/family proofs used in Section 5; no new catalogue is assumed. Direct feasible-state reconstruction gives 35 adaptive and 27 priority pointwise-minimal profiles, with five redundant profiles in each retained family. Every minimal profile occurs in its retained family, and every retained profile has a reconstructed profile whose paid set is a subset on every input. The original legal paths and global orders were also checked directly. All retained columns are kept in the certificates below; their bounds are unchanged. The reduced costs for all these columns, multiplied by 14 and 18 respectively, are the following concatenated rows:

$$\begin{array}{cccccccccc}
48 & 55 & 58 & 48 & 55 & 65 & 0 & 9 & 58 & 44 \\
33 & 29 & 65 & 51 & 40 & 36 & 33 & 10 & 19 & 8 \\
4 & 0 & 9 & 2 & 11 & 11 & 44 & 30 & 10 & 19 \\
6 & 15 & 29 & 6 & 15 & 4 & 2 & 11 & 11 & 0
\end{array}$$

and

$$\begin{array}{cccccccc}
64 & 73 & 74 & 64 & 73 & 83 & 0 & 11 \\
74 & 56 & 39 & 35 & 83 & 44 & 39 & 10 \\
21 & 4 & 0 & 0 & 11 & 2 & 13 & 13 \\
10 & 21 & 6 & 17 & 6 & 17 & 0 & 2.
\end{array}$$

All entries are nonnegative. Under a monotone fee, a pointwise paid-set-dominated profile cannot improve the lower bound. The exact checker re-traced all 504 original policy/input

paths and their order witnesses, and checked every displayed column inequality. Mixing private seeds preserves a fixed-prior lower bound, and a maximum over inputs dominates that mean. The two priors differ; this is not an assertion of adaptive/priority equality for every specified prior.

There is also a separate full-class check that uses no retained family or first checker result. It reconstructs the physical source above. For a prior π, feasible input set U and already purchased set B, the pure-state value is $\pi(U)C(B)$. At an impure state, the adaptive recurrence minimizes the sum of its two child contributions over every unbought query. At a fixed priority order position it considers both permanent SKIP and TAKE, and advances the same order in both answer children. An impure exhausted state is illegal even if $\pi(U) = 0$. Induction on remaining queries or positions proves that these recurrences cover every legal deterministic policy. Minimizing over all 24 orders covers full $\mathcal{P}$. Randomizing cannot reduce a fixed-prior mean below the best deterministic value.

The two actual checks used the single frozen fee $F(S) = C(S) + 9\mathbf{1}\{S \neq \varnothing\}$. They obtained exact lower and upper values 188/7 and 242/9. Because both labels occur, every correct deterministic seed buys a nonempty set on every input. Removing the global charge therefore subtracts exactly 9 from every policy/input cost, and from each mixed worst-input objective. This proves the claimed C values without a second cost-game optimization. The separate check uses 63 adaptive states and 1602 states over the 24 priority orders. The arbitrary-policy lower justification is the stated recurrence induction, supported by these exact finite calculations. For private seeds defined only almost surely, discard the finite union of erroneous null seed sets over the seven inputs first. □

V.1 How sharing repairs the earlier fee

Let F_0 be the old nonsubmodular fee in Appendix M.5 and put $H_1(S) = F_0(S) + |S|$. Directly on all 16 sets,

$$\begin{aligned} F(S) = H_1(S) &+ 5\mathbf{1}\{S \cap \{0,1,2\} \neq \varnothing\} \\ &+ 6\mathbf{1}\{S \cap \{0,1,3\} \neq \varnothing\} \\ &+ 5\mathbf{1}\{S \cap \{0,2,3\} \neq \varnothing\} \\ &+ 8\mathbf{1}\{S \cap \{1,2,3\} \neq \varnothing\}. \end{aligned}$$

If $|S| \geq 2$, all four triples are met, so $F(S) = H_1(S) + 24$ and $C(S) = H_1(S) + 15$. Every input in the dual support $(7, 9, 11, 15)$ has an opposite-label input agreeing with each possible first query answer. Every correct policy therefore buys at least two queries there. The five explicit upper trees buy at least two on all seven inputs. The old matched H_1 bounds 20/7 and 26/9 consequently transfer by adding 15, giving another explanation of the same gap.

It is false that every original input requires two queries: inputs $0, 4$ have one-query certificates at q_0. They are not deleted. The argument needs two-query necessity only on the dual support, plus the all-input property of the displayed upper trees. In particular it makes no uniform H_1 cost-shift claim for arbitrary policies on the other inputs. The uniform deletion of the *global* charge 9, in contrast, uses only nonempty purchase sets and is valid for all correct policies.

Corollary 27 (Robust submodular and coverage neighborhoods) *For the fixed source and fee C in Theorem 26, let $G(\varnothing) = 0$ and $\max_{S \neq \varnothing} |G(S) - C(S)| \leq 1/126$. Every such G is positive on nonempty sets, strictly increasing, and strictly submodular on incomparable pairs, with*

$$R_{\mathcal{P}}(G) - R_{\mathcal{A}}(G) \geq \frac{1}{63}.$$

There is also a coverage neighborhood: perturb each of the ten positive parameters in (11) by at most $1/1260$. Every resulting fee has these properties and the same gap lower bound.

Proof If the per-set fee change is at most r, each policy's cost on each input changes by at most r. The explicit adaptive mixture and full priority dual from Theorem 26 give

$$R_{\mathcal{P}}(G) - R_{\mathcal{A}}(G) \geq 2/63 - 2r.$$

At $r = 1/126$ this is $1/63$. Nonempty fees remain at least $8 - r > 0$; proper single-query marginals remain at least $1-2r > 0$; incomparable submodular slacks remain at least $1-4r > 0$. Comparable inequalities are identities. Thus the full 15-dimensional normalized real-fee box lies inside the specified submodular class. It need not consist of coverage functions. For the ten-parameter box, every parameter remains positive and every set fee changes by at most $10/1260 = 1/126$. The previous argument applies. These radii are conservative analytic guarantees, not maximum-radius or sampled-neighborhood claims. □

V.2 Evidence, prior costs and remaining scope

The main rational check uses the original 40/32 families and checks all 256 submodular inequalities and 64 query-addition conditions, including the 32 proper additions. A separate implementation uses the physical constraints and frozen priors rather than those policy tables. Both actual processes exited zero. They were written and run by the same AI agent; algorithmic separation is not human or external review. The initial concave-cardinality repair candidate was retained unexecuted. The coverage fee and certificates were fixed before their first check; global-charge removal and the neighborhood bounds are analytic corollaries. No new fee/prior search, LP solver, model API or hosted run was used.

Coverage costs and shared resources are established models. In particular, Harris et al. [9] include coverage machine costs in their batch-testing setting. Their independent series-diagnosis objective charges the sum of performed batch costs in expectation. Here the source is a correlated complete label promise, the objective maximizes over inputs after private expectation, and each path pays its final union fee once. These differences require a cost-, source-, objective- and comparator-preserving reduction before transferring a theorem. No such subsuming reduction is established here, and arbitrary alternative reductions are not ruled out. Exact bibliographic priority remains open.

The minimum four does not classify all four-query submodular tasks. The 1345 source orbits already equal for all monotone fees remain equal for submodular fees. The witness orbit has 48 physical coordinate/label transports; this follows by symmetry, not 48 new runs. Ten other four-query source orbits remain unclassified for general submodular fees. The additive exclusion through four and the interval $5 \leq m_{\mathrm{add}} \leq m_{\mathrm{pos}} \leq 7$ are unchanged. The displayed ratio 1127/1125 is one exact witness value,

not a largest four-query ratio. No natural academic-evaluation advantage, new human gold labels, human proof review, formal-kernel certification or historical execution authentication follows from this theorem.

W A sourcewise minimum for pairwise shared costs

The number of queries and the structure of their fees are different parameters. We now fix the complete four-query source of Theorem 26 and determine a small cost-structure threshold. Throughout this appendix a resource shared by a pair is charged once when either query is purchased. Its activation reveals no additional answer, so both policy classes keep the same four information actions and the same full causal-skip priority comparator.

Theorem 28 (Sourcewise pair-resource minimum, computer-assisted) *Fix the complete source with* $W_0 = \{1, 7, 9\}$, $W_1 = \{0, 4, 11, 15\}$ *and the physical realization in Theorem 26. Consider fees*

$$F(S) = \sum_{i \in S} c_i + \sum_{e \in \binom{\{0,1,2,3\}}{2}} w_e \mathbf{1}\{S \cap e \neq \varnothing\}, \qquad c_i, w_e \geq 0. \tag{13}$$

The minimum number of positive distinct pair weights permitting $R_{\mathcal{A}} < R_{\mathcal{P}}$ *on this source is exactly two. In particular, for every pair* e, *every* $c_i \geq 0$ *and every* $\lambda \geq 0$,

$$F(S) = \sum_{i \in S} c_i + \lambda \mathbf{1}\{S \cap e \neq \varnothing\} \quad \Longrightarrow \quad R_{\mathcal{A}} = R_{\mathcal{P}}.$$

Two positive weights suffice with strictly positive individual charges:

$$(c_0, c_1, c_2, c_3) = (4, 3, 2, 3), \qquad w_{12} = w_{23} = 2,$$

all other pair weights being zero. This strictly increasing coverage fee has exact randomized worst-input values

$$R_{\mathcal{A}} = \frac{286}{29}, \qquad R_{\mathcal{P}} = \frac{79}{8}, \qquad R_{\mathcal{P}} - R_{\mathcal{A}} = \frac{3}{232}.$$

The minimum concerns this source and the pair groups in (13). It does not give an all-source resource lower bound or a minimum for arbitrary larger resource groups. Repeated copies of the same pair combine into a single weight. There is no separate global setup term in the family being minimized.

Proof We first exclude support zero or one over its entire continuous price domain, then give an exact support-two witness. The complete monotone-cost policy families for this unchanged source contain 40 adaptive and 32 priority profiles, as in the preceding appendix. The new proof reuses that finite completeness premise. Every priority profile occurs among the adaptive profiles; every original policy remains correct on all seven inputs, including inputs assigned zero mass by a lower-bound prior.

Universal one-pair exclusion.

Fix a pair e and a permutation o of the four individual prices. Write

$$c_{o(j)} = d_0 + \cdots + d_j \quad (0 \le j < 4), \qquad \lambda = d_4, \qquad d \in \mathbb{R}^5_{\ge 0}.$$

The 24 closed price-order cones cover every nonnegative price vector, including equal and zero prices. A purchased-set fee and every fee difference are linear in d. Thus a nonnegative difference on every generator of a cone is nonnegative throughout the cone. Generator checks are universal linear inequalities; they are not sampled minimax games.

Use the 32 original priority columns followed by the 40 original adaptive columns. Their union has the same adaptive value. The proof certificate closes a cone by either of two arguments.

First, a full-game saddle can use an original hard input w and original priority policy p. If H is p's purchased set on w, verify

$$F(S_a(w)) \ge F(H) \quad \text{for every original adaptive } a, \qquad F(S_p(v)) \le F(H) \quad \text{for every } v \in W.$$

The two optimal values then equal $F(H)$. The certificate uses only the four fixed templates $(w,p) = (1,P_6), (7,P_6), (9,P_{19}), (1,P_{19})$, with original zero-based priority indices. Even at a descendant cone, these inequalities refer to all original policies and inputs.

Second, weak dominance may remove a strategy column when another is no dearer on every surviving row, or an objective row when another is no cheaper on every surviving column. These operations preserve the finite mixed minimax value. At a terminal node, each remaining column t must have an original legal priority column p_t satisfying

$$F(S_{p_t}(v)) \le F(S_t(v)) \qquad \text{for all seven original inputs } v. \tag{14}$$

The all-original-row condition is essential. A deleted row was dominated on a superset of the final columns. Induction through row deletions therefore bounds it by a terminal row on every terminal mixture. An optimal terminal mixture extended by zero to the original game attains the original adaptive value on all inputs. Replacing each column by p_t with the same private probability, using (14), gives $R_{\mathcal{P}} \le R_{\mathcal{A}}$. The reverse inequality follows from policy inclusion. Earlier row inequalities remain valid on the further column subsets of a descendant cone. No claim that separate dominance runs on the two classes reach the same fixed point is needed.

For completeness of the parameter cover, a node may split on a linear form f taking both signs on its generators. The two closed halves $f \ge 0$ and $f \le 0$ cover their parent. Given positive generator p and negative generator n, the boundary vector

$$f(p)n - f(n)p$$

has nonnegative coefficients in p, n and lies on $f = 0$. The nonnegative half is generated by its retained positive and zero generators together with these boundary vectors: cancel negative contributions against positive contributions in any parent conic decomposition. The opposite half follows by negating f. All cones are pointed, since $d \ge 0$, and have full dimension five. For the accumulated complete inequality description, a nonzero feasible ray is extreme exactly when its active normals have rank four. Removing only candidate rays of smaller active rank consequently preserves every extreme ray and the full cone. The reader reconstructs both clipped generator sets and checks full rank five. The rank test uses exact rational Gauss–Jordan elimination, separate from the producer's primitive-integer elimination.

The recorded proof covers all $6 \cdot 24 = 144$ roots with 164 nodes and 10 splits. All 154 leaves close: 73 by full-game saddles and 81 by dominance and full-row lifting. There are 5,533 column deletions, 342 row deletions and 252 full-original-row priority lifts. The separate reader checks all 504 original tree/input paths, 223,769 scalar generator inequalities and 289 exact ranks. Every node is reached exactly once; no root or residual leaf is omitted. Together with the cone-cover argument, these finite certificates prove equality for every nonnegative c, λ, including all boundary faces. The case $\lambda = 0$ also excludes pair support zero.

A two-pair witness.

For the charges in the statement, the complete fee vector in mask order is

$$(F(0),\dots,F(7)) = (0,4,5,9,6,10,9,13),$$
$$(F(8),\dots,F(15)) = (5,9,10,14,9,13,12,16).$$

It is a positive combination of four singleton coverage indicators and two pair indicators, hence is normalized, nonnegative, monotone and submodular. Its least proper query marginal is two. Some incomparable sets have submodularity slack zero; strict increase is the asserted strictness for this sparse witness.

Reuse the five legal trees a_1, a_2, a_3, p_3, p_4 from the preceding appendix and its input order $\mathcal{W} = (0,1,4,7,9,11,15)$. Their costs under the new fee are

$$a_1 : (9,9,9,10,9,10,10),$$
$$a_2 : (9,9,9,9,10,10,10),$$
$$a_3 : (16,16,12,12,12,9,9),$$
$$p_3 : (13,13,12,12,13,9,9),$$
$$p_4 : (16,16,13,13,12,9,9).$$

The adaptive mixture (a_1, a_2, a_3, p_3) with weights $(13,12,3,1)/29$ has input costs

$$(286,286,273,286,286,286,286)/29.$$

The priority mixture (a_1, a_2, p_3, p_4) with weights $(7,7,1,1)/16$ has costs $(155,155,151,158,158,158,158)/16$. These give the two upper bounds on all original inputs. Each mixture chooses its seed before seeing the input; each priority seed has one legal global order.

For the matching lower bounds, use the two priors on $\mathcal{W}$

$$\pi_A = (0,1,0,3,3,11,11)/29, \qquad \pi_P = (0,0,0,1,1,3,3)/8.$$

The expected-cost slacks above $286/29$ for all 40 original adaptive columns, multiplied by 29, are

$$\begin{array}{cccccccccc} 100 & 117 & 91 & 100 & 117 & 133 & 0 & 17 & 91 & 77 \\ 47 & 44 & 133 & 119 & 89 & 86 & 47 & 16 & 33 & 3 \\ 0 & 0 & 17 & 8 & 25 & 25 & 77 & 63 & 16 & 33 \\ 13 & 30 & 47 & 16 & 33 & 3 & 10 & 27 & 27 & 0 \end{array}$$

and the slacks above $79/8$ for all 32 original priority columns, multiplied by 8, are

$$\begin{array}{cccccccc} 28 & 33 & 25 & 28 & 33 & 37 & 0 & 5 \\ 25 & 21 & 13 & 12 & 37 & 24 & 13 & 4 \\ 9 & 1 & 0 & 0 & 5 & 1 & 6 & 6 \\ 4 & 9 & 3 & 8 & 3 & 8 & 0 & 1 \end{array}$$

All are nonnegative. The binding adaptive indices are $6, 20, 21, 39$; the binding priority indices are $6, 18, 19, 30$. These are lower bounds against the full original classes, rather than comparisons with the five displayed templates alone. Any private mixture has prior expectation at least its corresponding bound, and its maximum input expectation is at least that prior expectation. This proves both exact values and the gap $3/232$.

Separate recurrence and evidence scope.

A second point checker reconstructs the physical promise from all coarsenings of the two three-block priors. It reads the frozen candidate, with a research-input read audit excluding the policy catalogue and main result. Under π_A, it minimizes, at each impure feasible-world set, the sum of the two successor values over all unbought queries. A pure state with purchased

set S has value $\mathrm{mass}(X)F(S)$. Under π_P, it considers all 24 global orders, with an explicit permanent SKIP or purchase at each position and the same remaining order in both branches. Every impure state must continue even when its prior mass is zero. The resulting Bayes minima are 286/29 and 79/8; the adaptive recurrence has 63 states. Their lower-bound role, together with the all-input upper mixtures, gives another proof of the same randomized worst-input values without reading the retained matrix.

The point, mixtures and priors were frozen before full-column and recurrence checking. They were obtained by balancing the five displayed profiles; no price/prior grid or LP solver was used. The first cone attempt left 113 leaves unresolved using dominance/lifting alone. The retained versioned repair added the four existing saddle templates without changing the fee family or game. Unresolved certificates were not treated as counterexamples or as proofs. The final certificate and separate reader close the full domain. These are internal exact checks using standard finite-game and polyhedral tools, with actual local exit receipts; they are not external human review or proof-kernel validation. □

The two-pair example has $R_{\mathcal{P}}/R_{\mathcal{A}} = 2291/2288$. Its cost reduction relative to the optimal priority value is $3/2291$, approximately 0.13%. It supplies an exact structural separation, without establishing a large practical saving. Shared-resource and coverage cost modeling are established [9]; the sourcewise support minimum is the claim being examined here. Pandora's fixed-order optimality theorem [29, Theorem 4.1 of the inspected version] uses independent rewards and a different utility objective. Stochastic Boolean evaluation [6] uses its own input distribution, costs and comparator. A theorem transfer requires preserving these contracts. The bounded primary comparison leaves exact literature priority open.

Corollary 29 (A sparse resource-parameter neighborhood) *For the fixed source and two-pair fee of Theorem 28, perturb each of the four individual charges and the two positive pair weights by at most* $1/1856$. *Throughout this six-dimensional parameter box, the fee remains strictly increasing coverage and submodular, has exactly two positive pair groups, and satisfies*

$$R_{\mathcal{P}} - R_{\mathcal{A}} \geq \frac{3}{464}.$$

Proof For fixed input and seed, changing all purchased-set fees by at most r changes cost by at most r. Private expectation, maximum over inputs and infimum over a fixed policy class preserve this bound. Each optimal value is therefore 1-Lipschitz in the fee sup norm, so the gap loses at most $2r$. Each set activates at most six of the listed resources. A coordinate perturbation of size $\delta = 1/1856$ changes its fee by at most $6\delta = 3/928$. The resulting gap is at least

$$\frac{3}{232} - 12\delta = \frac{3}{464}.$$

Every resource weight remains at least $2 - \delta > 0$. The four positive singleton weights give strict increase; the positive coverage representation gives submodularity. This parameter box is restricted to the six stated resources and asserts no strict-submodular neighborhood in all fifteen nonempty fee coordinates. □

X The pair-resource minimum over all complete four-query sources

The sourcewise exclusion in Appendix W can be extended to every complete source. This changes the source quantifier while keeping the same observations, final-set charges and full causal-skip priority class. The finite proof below treats the entire nonnegative price orthant, rather than a price sample.

Theorem 30 (Universal one-pair equality and the four-query resource minimum, computer-assisted) *Let W_0, W_1 be nonempty, disjoint projections of two complete equivalence-partition fibres, with unrestricted class count, on a menu Q of at most four indexed pair-equality queries. For finite weights $c_i, w_e \geq 0$, charge*

$$F(S) = \sum_{i \in S} c_i + \sum_{e \in \binom{Q}{2}} w_e \mathbf{1}\{S \cap e \neq \varnothing\}. \tag{15}$$

If at most one distinct pair has positive weight, then $R_{\mathcal{A}}(W; F) = R_{\mathcal{P}}(W; F)$. The values use input-independent private randomness, zero error on every promised input, and worst-input expected actual final-set cost. Each priority seed may choose its own global order and make causal permanent skips.

Over all complete sources with exactly four indexed queries and fees of the form (15), the least number of positive distinct pair weights permitting $R_{\mathcal{A}} < R_{\mathcal{P}}$ is exactly two. The same minimum holds when all individual charges are strictly positive.

Repeated occurrences of one pair combine into one weight. Each shared group here has size exactly two; its activation supplies no extra information. There is no separate global setup term. The minimum is over complete four-query sources, not over arbitrarily large menus or arbitrary resource groups. In particular, an additive gap already exists at seven queries.

Proof ***Source cover and monotone majority.***

Use the complete four-query source cover and policy normalization in the proof of Theorem 7. Endpoint SAME components and DIFFERENT filters give all 465 nonempty one-sided supports from 4140 endpoint partitions. The 42864 ordered disjoint tasks form 1356 orbits under simultaneous coordinate permutations and label exchange. These are complete physical sources; no literal answer complementation is used as a source symmetry.

For $S \subseteq T$, each coefficient in

$$F(T) - F(S) = \sum_i c_i \big(\mathbf{1}_{i \in T} - \mathbf{1}_{i \in S}\big) + \sum_e w_e \big(\mathbf{1}_{T \cap e \neq \varnothing} - \mathbf{1}_{S \cap e \neq \varnothing}\big)$$

is nonnegative. Thus the fee is finite, normalized, nonnegative and monotone. The earlier 1300 certificate-threshold cases, 31 same-input policy-subset cases and 14 structural cases already prove equality for every fee in this larger monotone class. This closes 1345 disjoint source orbits for all one-pair parameters. It uses the mathematical scope of those certificates, not agreement at sampled prices. Their full finite certificate outputs were also freshly replayed in the combined execution described below.

Physical transport and full-policy premise.

The remaining 11 sources are listed in Table 11. Their physical realizations are retained as label-specific endpoint partitions and DIFFERENT pairs. A direct connectivity check on all 16 answer words reconstructs both nonempty label supports, for 22 supports in total. It verifies the exact catalogue task, not just its number of words.

For each source, recurse on its feasible words. At an impure state an adaptive policy chooses any remaining informative query and combines both continuations. For each of the 24 global permutations, priority recursion allows either SKIP with the same feasible words or TAKE with the same remaining suffix in both branches. Pure states stop. If a query has a known answer, repeats an earlier purchase, or follows a pure state, erasing it cannot raise a monotone final-set fee. Pointwise-subset pruning is valid under every common previously paid context. This proves that every full policy has a no-dearer member in the reconstructed minimal family. The original retained families need not be irredundant: all their legal paths and orders are kept, every minimal vector occurs in them, and every retained vector has a reconstructed pointwise-subset vector.

The audit checks 6957 original policy/input paths. With query permutations and label exchange it explicitly checks 333936 paths. For a permutation σ, transport c_i to $c_{\sigma(i)}$, e to $\sigma(e)$, each tree query and global order likewise, and exchange terminal labels if needed. For every purchased mask, all five coefficients of the one-pair fee agree. The $24 \cdot 6 \cdot 16 = 2304$ coefficient identities therefore preserve costs for every real parameter, the input-independent seed distribution and the maximum over inputs. The inverse transformation gives the reverse policy map. This covers all 408 ordered tasks in the residual orbits without adding an answer-complement symmetry.

Source ID	Inputs	Retained $\mathcal{A}/\mathcal{P}$	Minimal $\mathcal{A}/\mathcal{P}$	Ordered tasks
1209	7	40/32	35/27	48
1217	7	30/28	30/28	24
1221	7	26/25	21/20	48
1258	9	46/36	40/30	24
1265	8	51/42	48/39	48
1292	10	58/40	48/30	48
1312	8	45/33	31/23	48
1322	8	45/31	41/27	24
1328	8	30/26	30/26	24
1343	9	55/37	51/33	24
1344	9	45/35	35/25	48

Table 11 All 11 residual canonical sources in the frozen four-query catalogue. IDs identify exact projected tasks in the packet. Retained families, including redundant columns, are used in the certificates; the freshly reconstructed minimal families establish completeness.

Closed continuous-parameter certificates.

For each source and each of the six pairs, use the 24 closed individual-price orders. If o is an order, write

$$c_{o(j)} = d_0 + \cdots + d_j \quad (0 \le j < 4), \qquad \lambda = d_4, \qquad d \in \mathbb{R}^5_{\ge 0}.$$

Every path fee is a homogeneous linear form in these five variables. The price-order orthants cover all nonnegative c, λ including ties, zero prices and zero setup charge. Each root is closed by the same two finite-game mechanisms proved in Theorem 28.

At a pure-saddle leaf, one original hard input lower-bounds every full adaptive column by an anchor fee, while a legal original priority column is bounded by that anchor on every original input. At a dominance/lifting leaf, removing a weakly dominated minimizer column or maximizer row preserves the finite mixed minimax value. Following the deleted-row chains gives a common no-cheaper surviving row for every terminal mixture. Every remaining column then has an original priority replacement whose cost is no greater on *all original inputs*. This last condition includes removed objective rows and all zero-prior correctness inputs. Replacing the terminal mixture with the same seed weights gives $R_{\mathcal{P}} \le R_{\mathcal{A}}$. Policy inclusion gives equality. Separate dominance runs on the two classes are not assumed to have the same terminal game.

A split on a linear form f uses both closed halves $f \ge 0$ and $f \le 0$. For parent generators p, n with opposite signs, the crossing vector $f(p)n - f(n)p$ lies on their shared boundary and is a nonnegative combination of p, n. Together with the retained same-sign and zero generators, these vectors generate the respective halfcones. All cones lie in the nonnegative orthant and are pointed. The reader checks full dimension five. A feasible nonzero ray is extreme exactly when its active normals in the complete accumulated inequality description have rank four. Removing only nonextreme candidate rays consequently preserves the entire cone. Exact Gauss–Jordan elimination over fractions reconstructs the clipped cones and ranks separately from the producer's primitive-integer elimination. A generator inequality therefore holds on the whole cone, rather than only at tested cost vectors.

Across all $11 \cdot 6 \cdot 24 = 1584$ roots there are 1776 nodes and 96 splits. All 1680 leaves close: 887 by pure saddles and 793 by dominance and full-original-row lifting. The separate readers check 55742 column deletions, 3913 row deletions, 2829 full-original-row priority lifts, 2704249 scalar inequalities and 3275 exact ranks. Every node is reached once, all roots close and there is no unstarted root or residual leaf. No mixed-template leaf is needed. These checks, together with the full-policy and closed cone-cover arguments, prove one-pair equality on every residual source for the full continuous nonnegative domain. The 1345 monotone cases and transport complete the four-query exclusion.

Smaller menus and the upper witness.

For fewer than four queries, add fresh record pairs forced DIFFERENT under both label hypotheses. Give each dummy query zero individual charge and no membership in a positive pair group. Every original physical world extends by fresh classes, so the dummy answers are constant zero. Removing known-answer dummy purchases preserves all original observations and fees. Deleting them from each seed's global order is a legal permanent skip. Thus both minimax values are unchanged by padding, and the four-query exclusion covers every smaller menu. A menu without any eligible pair has additive fees and is also covered by Theorem 7.

On the complete four-query source of Theorem 28, take $(c_0, c_1, c_2, c_3) = (4, 3, 2, 3)$ and $w_{12} = w_{23} = 2$, with all other pair weights zero. That theorem gives the full-class values $R_{\mathcal{A}} = 286/29$ and $R_{\mathcal{P}} = 79/8$, with gap $3/232$. The combined scope check freshly recomputes the 40+32 matching dual inequalities and both all-input upper mixtures; it does not introduce a new fee, prior or numerical solve. At most one positive pair is excluded universally above, and this two-pair witness has strictly positive individual charges. The minimum is therefore exactly two in both the nonnegative and positive individual-charge classes. Corollary 29 still gives a six-resource box of radius $1/1856$ with gap at least $3/464$. □

Combined execution and dependence.

The private combined packet is named `combined_global_four_query_pair_proof_v03.zip`. On Windows it was extracted into a new directory and actually run with a separate work directory and output directory. Its 25 mathematical child processes all exited zero, with retained process handles and separate start/terminal records. They comprise one residual-source audit, one fresh replay of the 1345 finite monotone certificates, ten symbolic producers, eleven fraction-based readers, the global composition and the final scope/upper check. All 26 complete output objects match the pre-execution frozen reference objects under declared provenance-only normalization; no mathematical field is excluded. The ten regenerated candidate proof gzip files are byte-identical to their original files. All replayed mathematical checking cores, including the earlier single-source reader and monotone-certificate kernels, retain their original source bytes. Staging wrappers are declared separately.

The complete global source and full policy-catalogue cover remains an inherited accepted premise from the earlier all-additive packet; it is not re-enumerated in this combined replay. The 1345 finite certificate kernels, all eleven residual physical/completeness checks and all continuous one-pair readers are freshly executed. The old two-pair full-class and physical-recursion results remain declared inherited modules; the additional 72 fixed dual checks and all-input mixtures are freshly recalculated.

Byte-exact original scopes commit to the earlier audit/projection hashes. Those reference snapshots are therefore retained as inputs, and the fresh complete audit objects must match them before dependent readers run. They are not prefilled fresh outputs. The unchanged composer's discovery-era process counts and original proof text remain explicitly origin context; distinct current execution records give the actual replay counts. Two failed Windows launch-audit adapters were retained. Both failed before any mathematical child started; their repairs handle the platform's command-line representation and explicitly pass the same interpreter. No mathematical predicate or input scope was changed. Python audit hooks constrain scientific file access and deny network and nested process launch after bootstrap; they are not operating-system isolation. The optional original storage preflight is a separately declared host-environment dependency, not a mathematical input.

Interpretation and limits.

The theorem strengthens the source quantifier of Theorem 28; it does not change that theorem's sourcewise witness or improve its approximately 0.13% cost reduction. It does not classify every four-query submodular fee, every two-pair instance, larger setup groups, all specified-prior Bayes games, or larger query menus. Cost sharing, coverage modeling, finite minimax certificates and cone clipping are established tools. The candidate contribution is the exact universal pair-support boundary under the stated complete-source and comparator contract. The bounded comparisons with shared-cost testing [9], Pandora's reward-search objective [29] and stochastic Boolean evaluation [6] do not establish literature priority or exclude all possible preserving reductions. This is internal computer-assisted verification using different

programs from one research assistant, without external independent implementation review, human author approval or formal proof-kernel certification. No practical academic-evaluation advantage follows.

Y The pair-support boundary under gross substitutes

The fee restriction in this appendix is stronger than submodularity. It changes the pair-support minimum of Theorem 30 from two to three, without changing the complete-source or full causal-skip comparator contract. Gross substitutes (GS) is applied to the set function used as a fee; no demand, reward or posterior model is imposed on the unknown relational state.

The standard GS condition in the quadratic class.

For sets $X, Y \subseteq Q$ and $i \in X \setminus Y$, the discrete exchange formulation of GS requires

$$F(X) + F(Y) \leq \max\big\{F(X - i) + F(Y + i), \max_{j \in Y \setminus X} [F(X - i + j) + F(Y + i - j)]\big\}. \tag{16}$$

The first option remains when $Y \setminus X$ is empty. Additions and deletions denote set operations. For the nonnegative pair-coverage fee (15), expansion gives

$$F(S) = \sum_i \left(c_i + \sum_{j \neq i} w_{ij} \right) \mathbf{1}_{i \in S} - \sum_{i<j} w_{ij} \mathbf{1}_{\{i,j\} \subseteq S}.$$

Its conditional two-coordinate interaction is w_{ij}, independent of the conditioning set. The standard submodularity/S3 characterization therefore says precisely that the minimum of w_{ij}, w_{ik}, w_{jk} is attained at least twice for each triple. We use this established characterization, as recorded in Hajek [10, Definition 3 and Proposition 4]; neither the characterization nor its hierarchy geometry is a new claim of this paper.

Theorem 31 (GS-restricted pair-support minimum, computer-assisted) *Let the source and private zero-error policy classes be those of Theorem 30, on at most four indexed pair-equality queries, with unrestricted class count. Suppose the nonnegative quadratic pair-coverage fee (15) is GS. If at most two distinct pair weights are positive, then $R_{\mathcal{A}} = R_{\mathcal{P}}$.*

Over all complete sources with exactly four indexed queries, the least positive pair support permitting a strict gap is exactly three. This also holds when all individual charges must be strictly positive. A complete source and fee with three positive pairs satisfy

$$R_{\mathcal{P}} = \frac{1025}{47}, \qquad R_{\mathcal{A}} \leq \frac{324871}{14899}, \qquad R_{\mathcal{P}} - R_{\mathcal{A}} \geq \frac{54}{14899} > 0.$$

No exact adaptive value is asserted for this witness.

The support counts positive weights on named groups of size two. Repeated occurrences of a pair combine into its one weight. It does not count resources in arbitrary

higher-arity representations of the same function. In particular, this is not an extremal statement for all GS set functions or all menu sizes.

Proof ***Why two positive pairs must be disjoint.***

If $w_{ij}, w_{jk} > 0$ and $w_{ik} = 0$, the triple has a unique minimum, contradicting S3. The positive-edge relation is transitive, so its connected components are cliques. With at most two positive edges, there are no edges, one edge, or two disjoint edges. For four labelled vertices the 64 support patterns contain 15 disjoint-clique patterns, of which 10 have at most two edges; these finite counts are also checked by the composer. The transitivity argument, rather than those counts alone, establishes the required exclusion.

For two disjoint pairs, order their weights as $h_1 \geq h_2 \geq 0$. Merge the first pair at height h_1, the second at height h_2, and both pairs at height $h_3 = 0$. The weight on a pair of leaves is the height at which they first meet. The three pair partitions and two relative weight orders give six ranked balanced histories. They also contain the one-edge and empty cases on their zero faces. We verify the larger closed cones $h_1 \geq h_2 \geq h_3 \geq 0$; their $h_3 = 0$ faces suffice for the two-edge lower bound.

For each individual-price order o, use

$$c_{o(j)} = d_0 + \cdots + d_j \quad (0 \leq j < 4), \qquad (h_1, h_2, h_3) = (d_4 + d_5 + d_6, d_5 + d_6, d_6), \quad d \geq 0.$$

Every purchased-set fee is linear in these seven variables. The 24 closed price orders include every nonnegative individual price, with all ties and zeros, and the six balanced histories include every at-most-two-edge GS fee. This is a continuous cover, not a collection of tested price vectors.

The all-source lower cover.

The complete source and policy cover of Theorem 30 supplies 1356 canonical sources and 42864 ordered tasks. Its 1345 monotone-fee equality cases apply here because nonnegative pair-coverage fees are normalized and monotone. The other 11 sources are exactly those of Table 11; none is omitted by assuming a small gap or a convenient policy family.

For each residual source, the six histories and 24 price orders give 144 roots. A separate fraction-based reader rebuilds fees from the merge blocks, rather than importing the producer's fee coefficients. At a pure-saddle leaf an original hard input bounds every adaptive column from below, and one legal priority column achieves that bound on every original input. At a lifting leaf, common weak row/column dominance reductions preserve the finite mixed value. The reader follows removed-row witnesses and checks priority replacements against *all original rows*, so replacing a terminal mixture remains valid even on deleted objective rows and zero-prior correctness inputs. Inclusion gives the reverse inequality, as in the proof of Theorem 30.

Every split retains both closed halves of its linear inequality. The crossing-ray construction there applies in dimension seven. Using the entire accumulated halfspace description, an extreme nonzero ray has active-normal rank six; exact fractional Gauss–Jordan elimination reconstructs that rank and the clipped cones separately from the producer's integer elimination. The reader checks full dimension seven before discarding only nonextreme candidate generators. Consequently generator signs certify the entire cone and its boundary faces.

Across the 1584 roots, the readers check 2074 nodes, 245 splits, 63874 column deletions, 4411 row deletions, 4250 full-original-row priority lifts, 4330508 scalar inequalities and 6005 exact ranks. The 1829 closed leaves comprise 823 pure saddles and 1006 lifting leaves. No fixed-rational template leaf is needed. The source audit reconstructs the 22 physical label supports, 6957 original policy/input paths and their transport to all 408 residual ordered tasks.

Source ID	Closed roots	Nodes	Saddle/lifting leaves	Inequalities
1209	144	288	96/120	366493
1217	144	148	40/106	318037
1221	144	144	60/84	238728
1258	144	144	96/48	289688
1265	144	204	92/82	428417
1292	144	144	96/48	379848
1312	144	144	60/84	394716
1322	144	284	45/169	693161
1328	144	286	106/109	312764
1343	144	144	72/72	456652
1344	144	144	60/84	452004
Total	1584	2074	823/1006	4330508

Table 12 All residual lower certificates for the six balanced GS histories. All roots close, with no unstarted root or residual leaf. The final column counts exact scalar inequalities, not independent experiments. The full global source and policy enumeration is an explicitly inherited premise.

For query permutations the individual and six pair coefficients of every purchased mask rename identically. The composer checks all $24 \cdot 16 = 384$ ten-coefficient identities. Transforming tree queries, global orders and terminal labels therefore preserves every real fee and private input-independent mixture; the inverse transformation gives the reverse map. This transports the lower bound from the canonical representatives. Smaller menus can be padded by fresh pairs forced DIFFERENT under both labels, with zero charge and zero incident pair weights. Such known-answer queries add no information or fee, preserve GS, and can be erased from each seed's order. This proves the equality assertion.

A complete physical three-pair witness.

Let q_i compare records $2i, 2i+1$ for $0 \le i < 4$, returning one on SAME. Label zero has prior SAME partition $(0,0,0,1,0,1,0,2)$ and constraint $3 \not\sim 7$. Label one has prior SAME partition $(0,1,0,1,0,2,0,1)$ and no DIFFERENT constraint. Each label retains every coarsening allowed by its prior, with unrestricted class count. Enumerating the five coarsenings of the three prior blocks on each side gives exactly

$$W_0 = \{1,7,9\}, \qquad W_1 = \{0,4,11,15\}.$$

Integers encode answer words by $w = \sum_i 2^i q_i$. These are the original complete source 1209, rather than a selected subset of its legal worlds.

Set

$$(c_0,c_1,c_2,c_3) = (7,4,3,6), \qquad w_{12} = w_{13} = w_{23} = 4, \qquad w_{0i} = 0.$$

Every triangle either has three equal positive weights or two zero weights, so the fee is GS. All individual charges are positive, and exactly three pair weights are positive. On masks 0 through 15, the fee values are

$$(0,7,12,19,11,18,19,26,14,21,22,29,21,28,25,32).$$

The separate checker also evaluates all set submodularity inequalities and discrete exchange alternatives in (16) directly.

Here are explicit legal witness trees. Write $N_i(L, R)$ for query q_i with zero-child L and one-child R, with terminal labels $0, 1$. The subscripted names are retained column IDs:

$$\begin{aligned}
T_6 &= N_1(N_0(1,0), N_3(0,1)),\\
T_{18} &= N_2(N_1(N_0(1,0),1), N_3(N_1(1,0),1)),\\
T_{19} &= N_3(N_0(1,0), N_1(0,1)),\\
T_{20} &= N_3(N_0(1,0), N_2(N_1(0,1),1)),\\
T_{39}^{\mathcal{A}} &= N_2(N_1(N_3(N_0(1,0),0),1), N_3(N_1(1,0),1)).
\end{aligned}$$

The four T_j are priority trees with orders $(1,0,2,3)$, $(2,3,1,0)$, $(2,3,0,1)$ and $(3,0,2,1)$, respectively. They use only causal permanent skips. Every tree is correct on all seven worlds, regardless of the prior mass below.

A lower bound against the entire priority class.

On $W = (0,1,4,7,9,11,15)$, take the fixed prior

$$\pi = (0, 0, 0, 9/47, 3/47, 13/94, 57/94).$$

Let $B_o(V, k, S)$ be the minimum *unnormalized* prior-weighted future fee for a fixed global order o, current feasible worlds V, suffix position k and already purchased set S. The weights are the original π, not renormalized on V. For pure or empty V put $B_o = 0$. At a mixed state with $k = 4$ put $B_o = +\infty$, even if $\pi(V) = 0$. Otherwise let $q = o(k)$, $V_b = \{w \in V : q(w) = b\}$ and $m(V) = \sum_{w\in V} \pi(w)$. Then

$$B_o(V,k,S) = \min\big\{B_o(V,k+1,S),\\ m(V)\big(F(S+q) - F(S)\big) + \sum_{b=0}^{1} B_o(V_b, k+1, S+q)\big\}. \tag{17}$$

The first option is SKIP. The second is feasible only if both branches are feasible; it includes uninformative purchases. Correctness is imposed on every legal world, including zero-mass ones. A pure state may stop at no greater monotone cost. Every fixed-order causal policy makes precisely one of these choices, and S retains all context needed for a nonadditive final-set fee. Induction on the suffix proves the recurrence. Randomization cannot improve a specified-prior minimum of linear expected costs.

The exact proof and its separate fraction-based reader evaluate all 24 orders and 1602 distinct states: 526 mixed choices, 572 pure states and 504 infeasible mixed terminal states. There are 110 mixed zero-prior states, which retain their correctness obligations. They certify

$$B_{\mathcal{P}}(\pi, F) = \min_o B_o(W, 0, \varnothing) = \frac{1025}{47}.$$

This point lower bound does not assume that the cached 32 priority profiles exhaust the class. It directly covers every global order and causal continuation.

Mix $T_6, T_{18}, T_{19}, T_{20}$ with probabilities $(14, 6, 24, 3)/47$. Its costs on all seven worlds are

$$(989, 989, 983, 1025, 1025, 1025, 1025)/47.$$

The maximum matches the full-class prior lower bound, so $R_{\mathcal{P}} = 1025/47$.

Adaptive upper bound and strictness.

Mix the same four trees and $T_{39}^{\mathcal{A}}$ with probabilities

$$(4240, 1092, 7752, 969, 846)/14899.$$

The all-world cost vector is

$$(319165, 319165, 312151, 324871, 324871, 324871, 324871)/14899.$$

It proves the displayed adaptive upper bound and strict gap. The source checker traces all 504 original retained-policy/input paths, validates their tree/order witnesses and checks both mixtures on the full original seven-world source. Cached policy data are used for those feasibility and cost checks, but not as a premise of the full-class priority lower bound above. The certified at-most-two-edge exclusion and this positive three-edge witness give both claimed minima. □

Corollary 32 (A relative GS neighborhood with a full-class lower bound) *Keep this source, let* $z = (c_0, c_1, c_2, c_3, v)$*, and set* $w_{12} = w_{13} = w_{23} = v$ *and all other pair weights to zero. For every* z *in the closed box*

$$\|z - (7, 4, 3, 6, 4)\|_\infty \le \frac{27}{208586},$$

the fee is GS, all individual charges and exactly three pair weights are positive, and $R_{\mathcal{P}}(z) - R_{\mathcal{A}}(z) \ge 27/14899 > 0$*. This is a relative five-parameter family, not independent perturbations of all ten quadratic coefficients.*

Proof For each purchased set S, the five nonnegative coefficients of $F_z(S)$ sum to at most $4 + 3 = 7$. Thus if $\|z - z^0\|_\infty \le r$, then $|F_z(S) - F_{z^0}(S)| \le 7r$ for *every* S. For any legal priority policy, prior averaging preserves that bound. Taking the minimum over the full policy class gives $B_{\mathcal{P}}(\pi, z) \ge 1025/47 - 7r$. For the explicit adaptive mixture above, the maximum over all worlds is at most $324871/14899 + 7r$. Hence

$$R_{\mathcal{P}}(z) - R_{\mathcal{A}}(z) \ge \frac{54}{14899} - 14r \ge \frac{27}{14899}$$

at the stated radius. Positivity and the fixed equal-weight triangle preserve GS throughout the closed box. This standard Lipschitz argument uses the full-class Bellman center bound, not exhaustiveness of cached priority profiles. The composer also checks all 16 set sensitivities and the 224 older affine interval inequalities as consistency checks; the latter are not a completeness premise for this proof. □

Y.1 Prior objectives, established tools and scope

The same prior supplies a distinct comparison: $T_{39}^{\mathcal{A}}$ has prior expectation $1022/47$, whereas $B_{\mathcal{P}}(\pi, F) = 1025/47$. Its Bayes gap is at least $3/47$; this is not the worst-input gap of the theorem. It also obstructs one specific transfer route to independent-reward submodular-cost Pandora search [29, Theorem 4.1, arXiv version]. If a target simultaneously had optimal adaptive utility $K - B_{\mathcal{A}}$ and optimal fixed-prefix utility $K - B_{\mathcal{P}}$ for one common K, that theorem's utility equality would force Bayes cost equality here, a contradiction. This conditional argument excludes neither other reductions nor other reward-search models. The cited theorem numbering refers to the inspected arXiv v1, without asserting that its body equals the publisher's final article.

Nor is a generic GS adaptivity gap new: additive functions already satisfy GS, and stochastic Boolean evaluation has known additive gaps [6, 7]. Theorem 1 of Nielsen et al. gives a tight gap of two for product-bit k-of-n evaluation. At a mixed feasible state of a full-cube threshold, every unqueried bit is pivotal on some completion; permanently

skipping it cannot give a correct policy. Thus the causal-SKIP convention does not erase that known phenomenon. The candidate increment here is the exact pair-support boundary on complete four-query sources under the stated private minimax contract, compared with unrestricted minimum two. It is not a new GS model, general adaptivity principle or new Bellman method. The bounded primary comparison does not settle priority of that exact boundary or rule out every preserving reduction.

GS of a fee also differs from adaptive submodularity of a posterior utility. Golovin and Krause's definition and named guarantees [11, Definition 3 and Theorem 5] concern conditional expected marginal utility; the cited nonuniform-cost extension uses modular item costs. Those approximation guarantees do not supply an exact equality of the two cost-minimizing classes considered here. This comparison does not rule out a separately proved utility reduction.

Y.2 Actual combined replay and inherited premises

The private packet `combined_GS_minimum3_private_v01.zip` was actually extracted and run in a new Windows directory, with separate work and output directories. All 17 mathematical child processes exited zero; 18 entire JSON output objects matched their frozen pre-execution references, without omitting mathematical fields. The Bellman proof was regenerated byte for byte. The children comprise a residual source audit, the 1345 finite monotone-certificate kernels, eleven balanced-cone readers, the physical/GS/upper checker, a Bellman producer and separate reader, and the composer. Each has an actual start and retained-handle terminal receipt.

The global source and full global policy enumeration remain inherited from the accepted four-query cover; they were not regenerated. The 1345 finite kernels, eleven residual physical and minimal-policy audits, all 1584 lower roots, and the full Bellman point proof were freshly executed. Frozen scopes commit to dependency snapshots; the fresh residual audit and projection must match those whole snapshots before readers run. Original mathematical target files and inputs retain their byte identities. An explicit staging wrapper binds only provenance pin path labels to their original labels, together with actual current byte counts and hashes. All physical reads stay at the new work paths or runtime; separate records identify current paths, PIDs and exits. Original discovery-era hypotheses, scope labels and process data are preserved as origin context, not current verdicts or processes. The Python audit boundary is not operating-system isolation.

The combined proof does read cached policy trees and profiles for legal mixtures, world-cost vectors, path checks and the older affine crosschecks. Only the point and relative-box priority *lower bounds* avoid their exhaustiveness premise. The separate Bellman reader accesses the adaptive profile for the Bayes upper comparison. No claim of avoiding all cached data is intended. This private packet has no new public or hosted replay; the earlier public V24 packet proves unrestricted minimum two. The result is internal computer-assisted verification, without human or external independent review or formal proof-kernel certification. No practical academic-evaluation advantage or publication outcome follows from the constructed strict gap.